The Renormalisation Group for Four-Dimensional Lattice Gauge Theories: A Self-Contained Account of the Results Used for the Continuum Limit of Wilson Loops

Abstract

We give an account of the part of Bałaban's renormalisation group for lattice gauge theories in four dimensions on which the construction of the continuum limit of Wilson loops rests: the averaging operations, the propagators in a background field, the gauge conditions, the variational problem for the background field, the renormalisation step for small fields, the complete step with large-field regions, and the **R**-operation. Each object is defined once, at the generality in which it is used, every constant is named, and no step of a proof is delegated to the original papers, which are cited for attribution only.

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0 Introduction

Bałaban's construction of the effective densities of four-dimensional lattice gauge theories is contained in a series of papers written over six years, from the regularity of lattice Green's functions [1] to the renormalisation of the large fields [12, 13]. A reader who wants to use the construction, and not only its final theorems, meets two difficulties. The first is that the series was not written as one text. The averaging operation which defines the renormalisation transformation is changed between the papers on the linear theory and those on the renormalisation group, because the earlier one "is not symmetric with respect to lattice Euclidean transformations" [9, p. 252], and the symbol $|X|$ denotes the operator norm of a matrix in one paper [4, (19), p. 21] and the Hilbert–Schmidt norm in another [5, p. 390]. The second difficulty is that many steps are taken over by a reference, to an earlier paper of the series, to the three-dimensional theory [8], or to papers on other models, with the remark that the argument is the same.

The paper *The Continuum Limit of the Correlation Functions of Wilson Loops in Four-Dimensional Lattice Yang–Mills Theory*, which we call the main paper, uses the inside of the construction throughout. It inserts loops into the densities and follows them through every operation of a step, it compares the expansions of the propagators term by term on two lattices, and it runs the construction with parameters and test functions of its own. For such a use the original statements do not suffice, since what is needed at each place is a definition, an identity, or the dependence of a term on the fields in a region, and these are found in the proofs. The purpose of the present account is to state and to prove what the main paper and its supplement use, in one notation and in the order of the logic, so that neither of them has to be read with the original papers at hand.

Two principles decide what is included. A statement appears here only if a proof in the main paper, in its supplement, or later in this account uses it. Furthermore, every object is defined once, at the widest generality in which it is used: the dimension is four; there is one averaging operation, that of [9, (0.4), p. 253]; configurations take their values in the complexified group from the start, the gauge fields being the special case; and the parameters, the test functions, and the depth of the **R**-operation are those of the construction of the main paper. As a consequence, a result which the original papers prove for real fields and for the earlier average, and which is then needed for complex fields and for the later one, is proved here once, in the form in which it is needed. We do not aim at reproducing the original statements. What a statement has to deliver is fixed by the proofs of the main paper which use it, and where a definition or a bound of the papers can be replaced by one which is simpler to state or to prove and which serves those proofs equally, we use the simpler one.

The statements are written so that they can be used without their proofs. For each object we give its definition, the identities which it satisfies, the set of variables on which it depends, the domain on which it is analytic, and its bounds, in this order and as separate clauses. Every constant has a name and a place in an order of choices, a constant being allowed to depend only on those chosen before it. A statement which holds "for sufficiently

small" values of a parameter names the threshold.

The state of this version. This version contains Section 1, which fixes the lattices, the fields, the norms, and the geometry of the domains, and Section 2, on the average and the relative average. In Section 1, Lemmata 1.3, 1.8, 1.10, 1.14, and 1.18 are proved, and Lemma 1.2, the first inequality of Lemma 1.15, and Lemma 1.19 are stated without proof. The statements of Section 2 are given without proofs. Section 4 contains the definitions of the operators of a sequence of domains and of the regular pairs, the bounds of the four basic operators, and the walk expansion of the scalar operator with the first approximations of the other two, all without proofs. It also contains the operators of the Landau gauge at a pair, the propagator of the fluctuation, and the dependence of the operators on the sequence of domains, with their bounds and identities. Lemma 4.10, on the composition of bounds between blocks, is proved. The expansions of the block inverse and of the operator G into walks of local factors are stated as an existence theorem, Theorem 4.11(ii); the walks themselves are not yet defined. Section 3 states the momentum representation at the trivial background, without proof. Section 5 defines the regular fields, the two transports, and the axial and the Landau gauge; Lemma 5.4 is proved, and Lemma 5.5 and Theorem 5.8 are stated without proof. Section 6 states the variational problem and its solution: Lemma 6.4, Corollary 6.6, and Lemmata 6.10 and 6.19 are proved from the statements before them, and Lemma 6.3(i), Theorems 6.5, 6.16, 6.17, and 6.20, and Propositions 6.8, 6.11, 6.13, and 6.14 are stated without proof. Not yet stated in that section are the minimiser with a part of the data left free, the bound of the plaquettes of the data by the action, and the bounds of the second order of the solutions at a pair whose second entry is not the current of the first. Section 7 is begun: it defines the spaces of pairs of one scale and states the criterion for membership, Proposition 7.4, without proof. Not yet stated are the walks of the operators of the Landau gauge, the gauge transformation at a complex base, the variants of Theorem 5.8 without the bound of the current and on cubes, and the covariance of the unit lattice, which belongs to Section 7.

1 Lattices, Fields, Norms, and Domains

Nothing in this section is an estimate of the field theory. We fix the group and the norms on its algebra, the lattices of the scales which a renormalisation step relates, the fields on them with the action and the covariant difference operators, and then the two geometric structures in terms of which every later bound is stated: the sequences of domains on which the field lives at different scales, with their weighted distance, and the localisation domains with their linear size.

1.1 The Gauge Group

Throughout, $N \geq 1$ is an integer, $M_N(\mathbb{C})$ is the algebra of the complex $N \times N$ matrices, with unit $\mathbf{1}$, and G is a compact connected Lie subgroup of the unitary group $U(N)$. We follow [4, Sect. A] in writing the elements of the group near the unit as e^{iA} with a Hermitian matrix A, so that the Lie algebra consists of Hermitian matrices and its bracket is $(A, B) \mapsto i[A, B]$.

Definition 1.1 (Algebra, Complexification, and Norms)**.** (i) The *Lie algebra* of G is $\mathfrak{g} \coloneqq \{A \in M_N(\mathbb{C}) \,|\, A^* = A \text{ and } e^{itA} \in G \text{ for all } t \in \mathbb{R}\}$, and $\mathfrak{g}^c \coloneqq \{A + iB \,|\, A, B \in \mathfrak{g}\}$ is its *complexification*. The *complexified group* G^c is the subgroup of $GL(N, \mathbb{C})$ generated by the matrices e^{iA} with $A \in \mathfrak{g}^c$.

(ii) The *normalised trace* is $\operatorname{tr} X \coloneqq N^{-1} \operatorname{Tr} X$. On $M_N(\mathbb{C})$ we use the scalar product

$\langle X, Y\rangle \coloneqq \operatorname{tr} X^*Y$, the *Hilbert norm* $h(X) \coloneqq (\operatorname{tr} X^*X)^{1/2}$, and the operator norm $\|X\|$ with respect to the Euclidean norm of $\mathbb{C}^N$. The *bond norm* $|X|$ is one of these two norms, chosen once and for all.

(iii) π is the orthogonal projection of $M_N(\mathbb{C})$ onto $\mathfrak{g}^c$ with respect to $\langle\cdot,\cdot\rangle$.

(iv) For an invertible matrix $\mathbf{U}$ and $X \in M_N(\mathbb{C})$ we write $R(\mathbf{U})X \coloneqq \mathbf{U}X\mathbf{U}^{-1}$.

The bond norm is the norm in which the fields are compared with their thresholds, and we do not decide between its two candidates: every statement of this account holds for either choice, the constants being allowed to depend on N. What makes this possible is that the two norms differ by the factor $\sqrt{N}$ at most and share the properties of Lemma 1.3, which are the only ones that the proofs use. The Hilbert norm is the norm of the action, by (1.1) below, and of the scalar products of fields. We need the following facts about the algebra.

Lemma 1.2 (The Lie Algebra)**.** *The set $\mathfrak{g}$ is a real linear subspace of $M_N(\mathbb{C})$, and $i[A, B] \in \mathfrak{g}$ for all $A, B \in \mathfrak{g}$. Consequently $\mathfrak{g}^c$ is a complex linear subspace of $M_N(\mathbb{C})$ which is closed under the commutator and under $X \mapsto X^*$, and $\mathfrak{g} \cap i\mathfrak{g} = \{0\}$.*

Lemma 1.3 (Norms and the Adjoint Action)**.** *Let $X, Y \in M_N(\mathbb{C})$ and $U, V \in U(N)$.*

(i) $|\operatorname{tr} X| \le h(X) \le |X| \le \|X\| \le \sqrt{N}\, h(X)$. *Each of the three norms $\mathcal{N} \in \{h, |\cdot|, \|\cdot\|\}$ satisfies $\mathcal{N}(XY) \le \|X\|\,\mathcal{N}(Y)$, $\mathcal{N}(XY) \le \mathcal{N}(X)\|Y\|$, and $\mathcal{N}(X^*) = \mathcal{N}(X)$.*

(ii) $\mathcal{N}(UXV) = \mathcal{N}(X)$ *for each of the three norms, and*

$$1 - \operatorname{Re}\operatorname{tr} U = \frac{1}{2} h(U - \mathbf{1})^2 \,. \tag{1.1}$$

(iii) For $\mathbf{U} \in G^c$ the map $R(\mathbf{U})$ maps $\mathfrak{g}^c$ onto itself and commutes with π. For $U \in G$ it maps $\mathfrak{g}$ onto itself and preserves the three norms.

(iv) π is complex linear, $h(\pi X) \le h(X)$, $|\pi X| \le \sqrt{N}|X|$, and $\pi(X^*) = (\pi X)^*$.

Proof. (i) Let $s_1, \dots, s_N$ be the singular values of X. Then $h(X)^2 = N^{-1}\sum_i s_i^2$ and $\|X\| = \max_i s_i$, which gives $h(X) \le \|X\| \le \sqrt{N}\, h(X)$, and the bond norm is one of h and $\|\cdot\|$. Since $h(\mathbf{1}) = 1$, the Cauchy–Schwarz inequality gives $|\operatorname{tr} X| = |\langle \mathbf{1}, X\rangle| \le h(X)$. Next, $X^*X \le \|X\|^2\mathbf{1}$ as Hermitian matrices, so that $h(XY)^2 = \operatorname{tr} Y^*X^*XY \le \|X\|^2 \operatorname{tr} Y^*Y$, while the operator norm is submultiplicative; this is the first product inequality for h and for $\|\cdot\|$, hence for the bond norm. The matrices Z and Z^* have the same singular values, so that the three norms are invariant under the adjoint, and the second product inequality is the first applied to Y^*X^*.

(ii) The trace is cyclic and $U^*U = V^*V = \mathbf{1}$, so that $\operatorname{tr}(UXV)^*UXV = \operatorname{tr} X^*X$, and the operator norm is invariant because U and V are isometries of $\mathbb{C}^N$. Finally $h(U - \mathbf{1})^2 = \operatorname{tr}(2\mathbf{1} - U - U^*) = 2 - 2\operatorname{Re}\operatorname{tr} U$.

(iii) For $A \in \mathfrak{g}^c$ the map $R(e^{iA})$ is the exponential $\sum_n (n!)^{-1}(i \operatorname{ad} A)^n$ of the map $\operatorname{ad} A\colon X \mapsto [A, X]$, which preserves the finite-dimensional space $\mathfrak{g}^c$ by Lemma 1.2. Hence $R(e^{iA})\mathfrak{g}^c \subset \mathfrak{g}^c$, the same holds for products of such matrices and for their inverses, and so $R(\mathbf{U})\mathfrak{g}^c = \mathfrak{g}^c$ for all $\mathbf{U} \in G^c$. To see that $R(\mathbf{U})$ commutes with π, let $\mathfrak{n}$ be the orthogonal complement of $\mathfrak{g}^c$. Since $\mathfrak{g}^c$ is closed under the adjoint, a matrix Y lies in $\mathfrak{n}$ if and only if $\operatorname{tr} ZY = 0$ for all $Z \in \mathfrak{g}^c$. The bilinear form $(Z, Y) \mapsto \operatorname{tr} ZY$ is invariant under $R(\mathbf{U})$, and $R(\mathbf{U})^{-1} = R(\mathbf{U}^{-1})$ maps $\mathfrak{g}^c$ onto itself, so that $R(\mathbf{U})\mathfrak{n} \subset \mathfrak{n}$. A linear map which preserves

both $\mathfrak{g}^c$ and $\mathfrak{n}$ commutes with the projection onto the first along the second. For $U \in G$ and $A \in \mathfrak{g}$ the matrix UAU^{-1} is Hermitian and $e^{itUAU^{-1}} = Ue^{itA}U^{-1} \in G$, so that $R(U)\mathfrak{g} \subset \mathfrak{g}$, with equality because the same holds for U^{-1}; the norms are preserved by (ii).

(iv) An orthogonal projection onto a complex subspace is complex linear and does not increase the norm of the scalar product, which is h; with (i), $|\pi X| \le \sqrt{N}\, h(\pi X) \le \sqrt{N}\, h(X) \le \sqrt{N}|X|$. If $Y \in \mathfrak{n}$, then $\operatorname{tr} ZY^* = \overline{\operatorname{tr} YZ^*} = 0$ for all $Z \in \mathfrak{g}^c$, because $Z^* \in \mathfrak{g}^c$, so that $Y^* \in \mathfrak{n}$ by the criterion in the proof of (iii). Writing $X = \pi X + Y$ with $Y \in \mathfrak{n}$, we obtain $X^* = (\pi X)^* + Y^*$ with $(\pi X)^* \in \mathfrak{g}^c$ and $Y^* \in \mathfrak{n}$, which is the last assertion. *q.e.d.*

1.2 The Lattices of a Step

A renormalisation step relates fields on lattices whose spacings differ by powers of a fixed odd integer. We describe all of them inside one torus, in the units of [9, pp. 251–252], in which the coarsest lattice has the spacing one. The only difference from that description is the position of the sites, which we place at the points of the lattices $a\mathbb{Z}^4$ and not at the centres of their cells; since the spacings differ by odd factors, the lattices are nested in both descriptions, and a site of a coarse lattice is the centre of its block of finer sites.

Definition 1.4 (Lattices and Blocks)**.** Let $L \ge 3$ be an odd integer and let $k \ge 0$ and $m_{\mathbb{T}} \ge 1$ be integers. We put

$$\eta := L^{-k}\,, \qquad a_j := L^j \eta \quad (0 \le j \le k)\,, \qquad \ell_{\mathbb{T}} := 2L^{m_{\mathbb{T}}}\,. \tag{1.2}$$

(i) The *torus* is $\mathbb{T} := (\mathbb{R}/\ell_{\mathbb{T}}\mathbb{Z})^4$, with the coordinate unit vectors $e_0, \dots, e_3$. For $z \in \mathbb{T}$ we write $|z|_\infty$ and $|z|_1$ for the smallest value of the ℓ^∞-norm and of the ℓ^1-norm on the representatives of z in $\mathbb{R}^4$, and distances of points and of sets are taken with respect to $|\cdot|_\infty$ unless another norm is named.

(ii) The *lattice of the scale* j is $T^{(j)} := (a_j\mathbb{Z}/\ell_{\mathbb{T}}\mathbb{Z})^4 \subset \mathbb{T}$, for $0 \le j \le k$, and $T_\eta := T^{(0)}$ is the *fine lattice*. A length *in the units of the scale* j is a length divided by a_j.

(iii) For a number s of the form $L^r a_j$ with integers $0 \le j \le k$ and $r \ge 0$ and with $s \le L^{m_{\mathbb{T}}}$, the *grid of side* s is the family of the closed cubes $\{z \in \mathbb{T} \,|\, |z - c|_\infty \le s/2\}$ with centres $c \in (s\mathbb{Z}/\ell_{\mathbb{T}}\mathbb{Z})^4$. The cube of the grid of side a_j with the centre $y \in T^{(j)}$ is denoted by $\Delta_j(y)$.

(iv) For $y \in T^{(j)}$ and $0 \le n \le j$ the *block of order* n of y is $B^n(y) := \Delta_j(y) \cap T^{(j-n)}$, and $B(y) := B^1(y)$.

Since L is odd, $T^{(j)} \subset T^{(j-1)}$, every cube of the grid of side Ls is the union of L^4 cubes of the grid of side s, no point of a lattice $T^{(i)}$ lies on the boundary of a cube $\Delta_j(y)$ with $j \ge i$, and

$$B^n(y) = \{y + a_{j-n} \textstyle\sum_\mu n_\mu e_\mu \,|\, n_\mu \in \mathbb{Z},\ |n_\mu| \le \frac{1}{2}(L^n - 1)\}\,, \tag{1.3}$$

a set of L^{4n} points, which is the block of [9, (0.3), p. 252]. The blocks $B^n(y)$, $y \in T^{(j)}$, form a partition of $T^{(j-n)}$. The condition $m_{\mathbb{T}} \ge 1$ ensures that each lattice has at least six sites in every direction.

Definition 1.5 (Bonds, Paths, and Plaquettes)**.** Let $0 \le j \le k$ and $a := a_j$.

(i) A *bond* of $T^{(j)}$ is an ordered pair $b = (b_-, b_+)$ of sites with $b_+ = b_- + ae_\mu$ or $b_+ = b_- - ae_\mu$ for some μ; in the first case it is *positive*. The *reversed bond* is $b^{-1} := (b_+, b_-)$.

(ii) A *path* is a finite sequence $\gamma = (b_1, \dots, b_n)$ of bonds with $(b_i)_+ = (b_{i+1})_-$ for $1 \le i < n$; it leads from $(b_1)_-$ to $(b_n)_+$, its *length* is na, and its reverse is $\gamma^{-1} := (b_n^{-1}, \dots, b_1^{-1})$. A path from x to x' is *shortest* if its length is $|x - x'|_1$.

(iii) For $x \in T^{(j)}$ and $\mu \neq \nu$ the *plaquette* $p_{\mu\nu}(x)$ is the closed path

$$\partial p_{\mu\nu}(x) := ((x, x+ae_\mu),\ (x+ae_\mu, x+ae_\mu+ae_\nu),\ (x+ae_\mu+ae_\nu, x+ae_\nu),\ (x+ae_\nu, x)) ,$$

which is *positive* if $\mu < \nu$. The path $\partial p_{\nu\mu}(x)$ is the reverse of $\partial p_{\mu\nu}(x)$.

(iv) For a set $S \subset \mathbb{T}$ we write $S^{(j)} := S \cap T^{(j)}$. The *bonds of S of the scale j* are the bonds of $T^{(j)}$ with at least one end in S, and the *plaquettes of S of the scale j* are the plaquettes with at least one corner in S.

The convention (iv) is that of [3, p. 224] and [7, p. 277].

1.3 Configurations and the Action

The fields of the theory are functions on the bonds with values in the group. All later analyticity statements concern their extension to the complexified group, and so we define the objects for it at once.

Definition 1.6 (Configurations and Forms)**.** Let $0 \le j \le k$.

(i) A *configuration* on $T^{(j)}$ is a map $\mathbf{U}$ from the bonds of $T^{(j)}$ to G^c with $\mathbf{U}(b^{-1}) = \mathbf{U}(b)^{-1}$. It is a *gauge field* if its values lie in G; gauge fields are denoted by U, V. For a path $\gamma = (b_1, \dots, b_n)$ we put $\mathbf{U}(\gamma) := \mathbf{U}(b_1) \cdots \mathbf{U}(b_n)$.

(ii) A *gauge transformation* is a map $u \colon T^{(j)} \to G^c$. It acts on configurations by

$$\mathbf{U}^u(b) := u(b_-)\mathbf{U}(b)u(b_+)^{-1} , \tag{1.4}$$

so that $\mathbf{U}^u(\gamma) = u(x)\mathbf{U}(\gamma)u(x')^{-1}$ for a path γ from x to x'.

(iii) A 0-*form* is a map $\lambda \colon T^{(j)} \to M_N(\mathbb{C})$; a 1-*form* is a family $A = (A_\mu)_{0 \le \mu \le 3}$ of 0-forms, and $A(b) := A_\mu(x)$ for the positive bond $b = (x, x + a_j e_\mu)$; a 2-*form* is a family $F = (F_{\mu\nu})_{\mu \neq \nu}$ of 0-forms with $F_{\nu\mu} = -F_{\mu\nu}$. For a gauge transformation u and a form f we write $R(u)f$ for the form with the components $x \mapsto R(u(x))f(x)$.

(iv) For two forms of the same degree,

$$\langle \lambda, \lambda' \rangle := \sum_x a_j^4 \operatorname{tr} \lambda(x)\lambda'(x) , \qquad \langle A, A' \rangle := \sum_\mu \langle A_\mu, A'_\mu \rangle , \qquad \langle F, F' \rangle := \sum_{\mu<\nu} \langle F_{\mu\nu}, F'_{\mu\nu} \rangle , \tag{1.5}$$

and $\|f\|^2 := \sum_x a_j^4 \sum h(f(x))^2$, the inner sum over the components f_μ of a 1-form and $f_{\mu\nu}$, $\mu < \nu$, of a 2-form.

The pairing (1.5) is bilinear and not sesquilinear, as in [5, (3.11), p. 392]. On forms with Hermitian values it is the real scalar product whose norm is $\|\cdot\|$, and it is the form with respect to which adjoints are holomorphic in the configuration.

Definition 1.7 (The Action)**.** For a configuration $\mathbf{U}$ on $T^{(j)}$ and a plaquette p let

$$\operatorname{Re} \mathbf{U}(\partial p) := \frac{1}{2}(\mathbf{U}(\partial p) + \mathbf{U}(\partial p)^{-1}) , \qquad \operatorname{Im} \mathbf{U}(\partial p) := \frac{1}{2i}(\mathbf{U}(\partial p) - \mathbf{U}(\partial p)^{-1}) . \tag{1.6}$$

The *action* of $\mathbf{U}$ is

$$A(\mathbf{U}) \coloneqq \sum_p (1 - \operatorname{tr} \operatorname{Re} \mathbf{U}(\partial p)) \,, \tag{1.7}$$

the sum over the positive plaquettes of $T^{(j)}$.

This is the action [9, (0.2), p. 252] in dimension four, where it carries no power of the lattice spacing, with the extension to the complexified group of [5, p. 391].

Lemma 1.8 (The Action)**.** *(i) For a gauge field U, $A(U) = \sum_p (1 - \operatorname{Re} \operatorname{tr} U(\partial p)) = \frac{1}{2} \sum_p h(U(\partial p) - \mathbf{1})^2 \geq 0$.*

(ii) $A(\mathbf{U}^u) = A(\mathbf{U})$ for every configuration $\mathbf{U}$ and every gauge transformation u with values in G^c, and $\operatorname{Im} \mathbf{U}^u(\partial p_{\mu\nu}(x)) = R(u(x)) \operatorname{Im} \mathbf{U}(\partial p_{\mu\nu}(x))$.

(iii) The summand of (1.7) for p is a polynomial in the entries of the matrices $\mathbf{U}(b)$ and $\mathbf{U}(b)^{-1}$ for the four bonds b of ∂p.

Proof. For a unitary matrix W we have $W^{-1} = W^*$, so that $\operatorname{tr} \operatorname{Re} W = \operatorname{Re} \operatorname{tr} W$, and (i) follows from (1.1). By (1.4), $\mathbf{U}^u(\partial p_{\mu\nu}(x)) = u(x) \mathbf{U}(\partial p_{\mu\nu}(x)) u(x)^{-1}$, and the same holds for the inverse, which gives the second assertion of (ii); the first follows because the trace is invariant under conjugation. Finally (iii) is the definition, since $\mathbf{U}(\partial p)^{-1}$ is the product of the inverses in the reverse order. *q.e.d.*

1.4 Covariant Differences and the Current

The quadratic forms of the linear theory and the regularity conditions on the fields are written with difference operators which are covariant with respect to a configuration. We fix a scale j, write $a \coloneqq a_j$, and let $\mathbf{U}$ be a configuration on $T^{(j)}$.

Definition 1.9 (Covariant Differences)**.** For a 0-form λ and $0 \leq \mu \leq 3$,

$$(D_{\mathbf{U},\mu} \lambda)(x) \coloneqq a^{-1} (R(\mathbf{U}(x, x + a e_\mu)) \lambda(x + a e_\mu) - \lambda(x)) \,, \tag{1.8}$$

$$(D^*_{\mathbf{U},\mu} \lambda)(x) \coloneqq a^{-1} (R(\mathbf{U}(x, x - a e_\mu)) \lambda(x - a e_\mu) - \lambda(x)) \,. \tag{1.9}$$

With these we define, for a 0-form λ, a 1-form A, and a 2-form F:

(i) the 1-form $D_{\mathbf{U}} \lambda$ with the components $D_{\mathbf{U},\mu} \lambda$, and the 0-form $D^*_{\mathbf{U}} A \coloneqq \sum_\mu D^*_{\mathbf{U},\mu} A_\mu$;

(ii) the 2-form $D_{\mathbf{U}} A$ with the components $(D_{\mathbf{U}} A)_{\mu\nu} \coloneqq D_{\mathbf{U},\mu} A_\nu - D_{\mathbf{U},\nu} A_\mu$, and the 1-form $D^*_{\mathbf{U}} F$ with the components $(D^*_{\mathbf{U}} F)_\mu \coloneqq \sum_{\nu \neq \mu} D^*_{\mathbf{U},\nu} F_{\nu\mu}$;

(iii) the *covariant gradient* $\nabla_{\mathbf{U}} A$, the family of the sixteen 0-forms $(\nabla_{\mathbf{U}} A)_{\mu\nu} \coloneqq D_{\mathbf{U},\mu} A_\nu$, and likewise $\nabla_{\mathbf{U}} \lambda \coloneqq D_{\mathbf{U}} \lambda$;

(iv) the *covariant Laplacian* $\Delta_{\mathbf{U}} \coloneqq \sum_\mu D^*_{\mathbf{U},\mu} D_{\mathbf{U},\mu}$, which acts on 0-forms and, component by component, on 1-forms.

These are the operators of [5, (3.3), (3.4), (3.8), (3.9), pp. 390–392, and (3.23), p. 394]. In that paper a 1-form is extended to the negative bonds by $A(b^{-1}) = -A(b)$, and the difference of a 1-form on a plaquette $p_{\mu\nu}(x)$ is written as a sum over its four bonds [5, (3.4)]; with the transports to the corner x which that formula contains, the sum is the component $(D_{\mathbf{U}} A)_{\mu\nu}(x)$ above.

Lemma 1.10 (Adjoints and Covariance)**.** *Let* $\mathbf{U}$ *be a configuration on* $T^{(j)}$ *and* u *a gauge transformation, both with values in* G^c*.*

(i) For 0*-forms* λ*,* λ'*, a* 1*-form* A*, and a* 2*-form* F*,*

$$\langle D_{\mathbf{U},\mu}\lambda, \lambda'\rangle = \langle \lambda, D^*_{\mathbf{U},\mu}\lambda'\rangle \;, \qquad \langle D_{\mathbf{U}}\lambda, A\rangle = \langle \lambda, D^*_{\mathbf{U}}A\rangle \;, \qquad \langle D_{\mathbf{U}}A, F\rangle = \langle A, D^*_{\mathbf{U}}F\rangle \;.$$

(ii) $D_{\mathbf{U}^u,\mu}R(u)\lambda = R(u)D_{\mathbf{U},\mu}\lambda$ *and* $D^*_{\mathbf{U}^u,\mu}R(u)\lambda = R(u)D^*_{\mathbf{U},\mu}\lambda$*, and consequently each of the operators* $\mathcal{D}$ *of Definition 1.9 satisfies* $\mathcal{D}_{\mathbf{U}^u}R(u) = R(u)\mathcal{D}_{\mathbf{U}}$*.*

Proof. (i) Write $W \coloneqq \mathbf{U}(x, x+ae_\mu)$. By the cyclicity of the trace, $\mathrm{tr}(W\lambda(x+ae_\mu)W^{-1}\lambda'(x)) = \mathrm{tr}(\lambda(x+ae_\mu)R(W^{-1})\lambda'(x))$, and $W^{-1} = \mathbf{U}(x+ae_\mu, x)$. Summing over x and shifting the variable by ae_μ in this term, we obtain

$$\sum_x \mathrm{tr}((D_{\mathbf{U},\mu}\lambda)(x)\lambda'(x)) = a^{-1}\sum_x \mathrm{tr}\Big(\lambda(x)(R(\mathbf{U}(x, x-ae_\mu))\lambda'(x-ae_\mu) - \lambda'(x))\Big) \;,$$

which is the first identity. The second is its sum over μ with $\lambda' = A_\mu$. For the third, the first identity gives

$$\langle D_{\mathbf{U}}A, F\rangle = \sum_{\mu<\nu}(\langle A_\nu, D^*_{\mathbf{U},\mu}F_{\mu\nu}\rangle - \langle A_\mu, D^*_{\mathbf{U},\nu}F_{\mu\nu}\rangle) \;.$$

In the first sum we exchange the names of μ and ν, and in the second we write $F_{\mu\nu} = -F_{\nu\mu}$. The result is $\sum_\mu\langle A_\mu, \sum_{\nu>\mu} D^*_{\mathbf{U},\nu}F_{\nu\mu} + \sum_{\nu<\mu} D^*_{\mathbf{U},\nu}F_{\nu\mu}\rangle = \langle A, D^*_{\mathbf{U}}F\rangle$.

(ii) Since R is multiplicative, $R(\mathbf{U}^u(x, x+ae_\mu))R(u(x+ae_\mu)) = R(u(x))R(\mathbf{U}(x, x+ae_\mu))$ by (1.4), and therefore $(D_{\mathbf{U}^u,\mu}R(u)\lambda)(x) = a^{-1}R(u(x))(R(\mathbf{U}(x, x+ae_\mu))\lambda(x+ae_\mu) - \lambda(x))$. The same computation with $-e_\mu$ gives the identity for $D^*_{\mathbf{U},\mu}$, and the operators of Definition 1.9 are sums of products of these two. *q.e.d.*

The part of the action which is linear in a fluctuation of the field is the pairing of the fluctuation with a 1-form built from the plaquettes, and the regularity conditions of the later sections bound this 1-form separately. For this reason the terms of the effective actions are regarded as functions of a configuration and of an independent variable in the place of this 1-form [9, p. 261].

Definition 1.11 (Curvature, Current, and Pairs)**.** Let $\mathbf{U}$ be a configuration on $T^{(j)}$ and $a \coloneqq a_j$.

(i) The *curvature* of $\mathbf{U}$ is the 2-form $F(\mathbf{U})$ with the components $F(\mathbf{U})_{\mu\nu}(x) \coloneqq a^{-2}\pi\,\mathrm{Im}\,\mathbf{U}(\partial p_{\mu\nu}(x))$, and the *current* of $\mathbf{U}$ is the 1-form

$$J(\mathbf{U}) \coloneqq D^*_{\mathbf{U}}F(\mathbf{U}) \;. \tag{1.10}$$

(ii) A *pair* on a set of bonds is a pair $(\mathbf{U}, \mathbf{J})$ of a configuration $\mathbf{U}$ and a 1-form $\mathbf{J}$ with values in $\mathfrak{g}^c$, both restricted to the set. A gauge transformation acts on pairs by $(\mathbf{U}, \mathbf{J})^u \coloneqq (\mathbf{U}^u, R(u)\mathbf{J})$.

The family $F(\mathbf{U})$ is a 2-form because $\partial p_{\nu\mu}(x)$ is the reverse of $\partial p_{\mu\nu}(x)$, so that the two matrices $\mathrm{Im}\,\mathbf{U}(\partial p)$ differ by the sign. Its values lie in $\mathfrak{g}^c$, and in $\mathfrak{g}$ if $\mathbf{U}$ is a gauge field, by Lemma 1.3(iv). The current (1.10) is [9, (1.8), p. 261], and the action on pairs is [9, (1.10), p. 262]. By Lemma 1.8(ii), Lemma 1.3(iii), and Lemma 1.10(ii),

$$F(\mathbf{U}^u) = R(u)F(\mathbf{U}) \;, \qquad J(\mathbf{U}^u) = R(u)J(\mathbf{U}) \;, \tag{1.11}$$

for every gauge transformation u with values in G^c, so that $(\mathbf{U}, J(\mathbf{U}))^u = (\mathbf{U}^u, J(\mathbf{U}^u))$.

1.5 Sequences of Domains and the Weighted Distance

After several steps the field is an integration variable on lattices of different scales in different regions of the torus, the coarser lattices in the regions in which more steps could be performed. The regions are described by a decreasing sequence of domains, and all bounds on propagators decay in a distance which measures a length in each region in the units of its scale.

Definition 1.12 (Sequences of Domains)**.** Let M be a power of L with $M \le L^{m_{\mathbb{T}}}$ and let $R \ge 1$ be an integer. A *sequence of domains with the block size M and the separation R* is a sequence of subsets

$$\mathbb{T} = \Omega_0 \supset \Omega_1 \supset \cdots \supset \Omega_k \,, \qquad \Omega_{k+1} \coloneqq \emptyset \,, \tag{1.12}$$

of the torus such that, for $1 \le j \le k$, the set Ω_j is a union of cubes of the grid of side Ma_j, and

$$\operatorname{dist}(\Omega_j^c, \Omega_{j+1}) > RMa_j \tag{1.13}$$

for all $1 \le j < k$ for which neither Ω_j^c nor Ω_{j+1} is empty. Its *layers* are the sets

$$\Gamma_j \coloneqq \Omega_j^{(j)} \setminus \Omega_{j+1} \subset T^{(j)} \,, \qquad 0 \le j \le k \,, \tag{1.14}$$

and its *determining set* is the disjoint union $\mathbf{B} \coloneqq \bigsqcup_{j=0}^{k} \Gamma_j$. For $y \in \Gamma_j$ we call $j(y) \coloneqq j$ the *scale* of y, and we put

$$\Delta(y) \coloneqq B^j(y) \,, \qquad \tilde{\Delta}(y) \coloneqq \{x \in T_\eta \,|\, |x - y|_\infty < a_j\} \,. \tag{1.15}$$

The domains may be empty or equal to the torus, and the inclusions need not be strict. The sets $\Omega_j \setminus \Omega_{j+1}$ are pairwise disjoint and cover the torus. Both Ω_j and Ω_{j+1} are unions of cubes of the grid of side a_j, whose boundaries contain no point of the fine lattice, so that the fine points of $\Omega_j \setminus \Omega_{j+1}$ are those of the blocks $B^j(y)$ with $y \in \Gamma_j$. Hence the blocks $\Delta(y)$, $y \in \mathbf{B}$, form a partition of the fine lattice. For $x \in T_\eta$ we write $y(x)$ for the point of $\mathbf{B}$ with $x \in \Delta(y(x))$ and $j(x) \coloneqq j(y(x))$. The *bonds of the layer* Γ_j are the bonds of $T^{(j)}$ with an end in Ω_j and no end in Ω_{j+1}. Both ends of such a bond are points of $\mathbf{B}$ once $RM \ge L$: an end in Ω_j lies in Γ_j, and an end outside Ω_j is a point of $T^{(j)} \subset T^{(j-1)}$ in Ω_{j-1}, hence a point of Γ_{j-1}. This is the setting of [3, (2.1)–(2.4), p. 224] and [7, p. 277], with the sequence starting at $\Omega_0 = \mathbb{T}$, and with the names of [10, (2.2), p. 255], which writes Γ_j for the layers and $\mathbf{B}$ for the determining set. The condition (1.13) is imposed in the ℓ^∞-distance, which is the smaller one, so that a sequence in our sense satisfies the condition of the papers cited.

The distance in which the propagators decay is defined in [3, (2.46), p. 231] as the smallest length of a contour between two points, the part of the contour in a layer being composed of bonds of the scale of that layer and counted in its units. At the surface between two layers a bond of the finer scale does not end at a site of the coarser one, and so we define the distance by paths in the torus instead of lattice contours. The two properties which the later sections use, the cost of crossing a layer and the summability of the exponentials, are Lemma 1.14 and Lemma 1.15.

Definition 1.13 (The Weighted Distance)**.** Let a sequence of domains be given. For $z \in \mathbb{T}$ let $j(z)$ be the largest j with $z \in \Omega_j$. The *weighted length* of a Lipschitz path $\gamma\colon [0,1] \to \mathbb{T}$ is

$$\ell(\gamma) \coloneqq \int_0^1 \frac{|\dot\gamma(t)|_1}{a_{j(\gamma(t))}} \,\mathrm{d}t \,, \tag{1.16}$$

and the *weighted distance* $d(z, z')$ of two points of the torus is the infimum of $\ell(\gamma)$ over the Lipschitz paths from z to z'.

The function $z \mapsto j(z)$ is upper semicontinuous, because the domains are closed, so that the integrand of (1.16) is measurable. For a point x of the fine lattice, $j(x)$ is the scale of the block which contains x, in agreement with the notation introduced above.

Lemma 1.14 (The Weighted Distance)**.** *For every sequence of domains with the block size M and the separation R:*

(i) d is a metric on $\mathbb{T}$, and $|z - z'|_1 \le d(z, z') \le \eta^{-1}|z - z'|_1$;

(ii) if $z \in \Omega_j \setminus \Omega_{j+1}$ and $z' \in \Omega_{j'} \setminus \Omega_{j'+1}$, then $d(z, z') \ge RM \max\{|j - j'| - 1, 0\}$.

Proof. (i) The weights satisfy $\eta = a_0 \le a_{j(z)} \le a_k = 1$. Hence $\ell(\gamma)$ lies between the ℓ^1-length of γ and η^{-1} times that length, and the ℓ^1-length of a path from z to z' is at least $|z - z'|_1$, with equality for a suitable path. This gives the two inequalities, and with them $d(z, z') > 0$ for $z \ne z'$. The symmetry of d follows by reversing the paths, and the triangle inequality by concatenating them.

(ii) We may assume $j' \ge j + 2$. Let γ be a Lipschitz path from z to z' and let $j < i < j'$. Then $\gamma(0) = z \notin \Omega_i$, because $\Omega_i \subset \Omega_{j+1}$, and $\gamma(1) = z' \in \Omega_{i+1}$, because $\Omega_{j'} \subset \Omega_{i+1}$. Let t_1 be the smallest t with $\gamma(t) \in \Omega_{i+1}$, which exists since Ω_{i+1} is closed, and let t_0 be the supremum of the $t < t_1$ with $\gamma(t) \notin \Omega_i$. Then $\gamma(t_0)$ lies in the closure of Ω_i^c, and $\gamma(t) \in \Omega_i \setminus \Omega_{i+1}$ for $t_0 < t < t_1$, so that $j(\gamma(t)) = i$ on this interval. The ℓ^1-length of the arc $\gamma|_{[t_0,t_1]}$ is at least $|\gamma(t_1) - \gamma(t_0)|_\infty$, which is at least the distance of Ω_{i+1} from the closure of Ω_i^c, and this distance is greater than RMa_i by (1.13). The weighted length of the arc is therefore greater than RM. The intervals (t_0, t_1) which belong to different i are disjoint, since $j(\gamma(t))$ takes different values on them, and there are $j' - j - 1$ of them. Hence $\ell(\gamma) > RM(j' - j - 1)$. *q.e.d.*

The second property is the summability of the exponentials of the distance over the determining set, uniformly in the sequence and in the number of the scales. It is [3, Lemma 2.1, p. 234], stated here for an arbitrary rate δ, since the lemma is applied with the decay rates of several operators.

Lemma 1.15 (Summation over the Determining Set)**.** *For every $\delta > 0$ and every $0 < \alpha < 1$ there are numbers $r_1(\alpha, \delta) \ge 1$ and $c_1(\alpha, \delta) \ge 1$, which depend on α and δ only, with the following property. Let L, k, and $m_{\mathbb{T}}$ be as in Definition 1.4 and let a sequence of domains with the block size M and the separation R be given, with $RM \ge r_1(\alpha, \delta)$. Then*

$$\sup_{y \in \mathbf{B}} \sum_{y' \in \mathbf{B}} e^{-\alpha\delta d(y,y')} \le c_1(\alpha, \delta) \,, \tag{1.17}$$

and for every $n \ge 1$ and all $y_0, y_n \in \mathbf{B}$,

$$\sum_{y_1,\dots,y_{n-1} \in \mathbf{B}} \prod_{i=1}^{n} e^{-\delta d(y_{i-1},y_i)} \le c_1(\alpha, \delta)^{n-1} e^{-(1-\alpha)\delta d(y_0,y_n)} \,. \tag{1.18}$$

Proof of (1.18) *from* (1.17). By the triangle inequality of Lemma 1.14(i), $\sum_{i=1}^n d(y_{i-1}, y_i) \ge d(y_0, y_n)$, so that the product in (1.18) is at most $e^{-(1-\alpha)\delta d(y_0,y_n)} \prod_{i=1}^n e^{-\alpha\delta d(y_{i-1},y_i)}$. We bound the factor with $i = n$ by one and sum over $y_{n-1}, y_{n-2}, \dots, y_1$ in this order, each sum being at most $c_1(\alpha, \delta)$ by (1.17). *q.e.d.*

1.6 Norms of Fields

The bounds of the linear theory are stated in supremum norms, in Hölder norms, in norms which weight a field by a power of the spacing of the layer, and in L^2-norms. They are the norms of [2, (1.108), (1.109), p. 35] and [5, (3.39)–(3.41), p. 397].

Definition 1.16 (Norms)**.** Let a sequence of domains be given, let $\mathbf{U}$ be a configuration on T_η, and let f be a 0-form on T_η, or a family of 0-forms such as a 1-form or the gradient of one.

(i) The *supremum norm* $|f|$ is the supremum of $|f_\iota(x)|$ over the sites x and the components f_ι of f; for a subset S of the torus, $|f|_S$ is the same supremum over $x \in S^{(0)}$.

(ii) For $0 \le \beta \le 1$ and $0 \le j \le k$ the *Hölder norm at the scale* j is

$$\|f\|_\beta^{(j)} := \sup\Big(\frac{|x-x'|_1}{a_j}\Big)^{-\beta} |R(\mathbf{U}(\gamma))f_\iota(x') - f_\iota(x)| \,, \tag{1.19}$$

the supremum over the components, over the pairs of sites $x \neq x'$ of T_η with $|x - x'|_1 \le a_j$, and over the shortest paths γ from x to x'. We write $\|f\|_\beta := \|f\|_\beta^{(k)}$, and $\|A\|_{1,\beta} := \|\nabla_{\mathbf{U}} A\|_\beta$ for a 1-form A.

(iii) For a real number α the *weighted norm* is

$$|f|_{(\alpha)} := \sup_{x \in T_\eta} a_{j(x)}^{-\alpha} \max_\iota |f_\iota(x)| \,. \tag{1.20}$$

In the papers cited the Hölder norm $\|\cdot\|_\beta$ is taken "in the η-scale", which is our scale k, and a superscript $\xi = L^{-j}$ indicates that "functions and distances are on the ξ-lattice" [5, p. 397], which is (1.19). The norm (1.19) depends on the configuration through the transport along γ, and on scalar functions, for which we use it as well, it does not.

1.7 Localisation Domains

The terms of the effective actions are indexed by connected unions of large cubes, and they decay exponentially in the linear size of the union. We define these notions for the cubes of an arbitrary grid, together with a variant of the linear size in which a part of the domain is not counted, which is needed when a term has a prescribed support.

Definition 1.17 (Unions of Cubes and Their Linear Size)**.** Let s be the side of a grid in the sense of Definition 1.4(iii).

(i) Two cubes of the grid *touch* if they have a point in common, and they *share a wall* if their intersection is a three-dimensional cube. A union X of cubes of the grid is *connected* if any two of its cubes are joined by a chain of cubes of X in which consecutive cubes share a wall. The number of the cubes of X is denoted by $\#(X)$.

(ii) For a union X of cubes, $X^\sim$ is the union of X and of the cubes of the grid which touch a cube of X; $X^{\sim n}$ is the result of n such operations; and $X^{\sim -n} := ((X^c)^{\sim n})^c$, where X^c denotes the union of the cubes of the grid which are not in X.

(iii) For a connected union X of cubes and a union $D \subset X$ of cubes, possibly empty, the *linear size of X relative to D* is

$$d(X; D) := \frac{1}{s} \inf |G \setminus D| \,, \tag{1.21}$$

the infimum of the one-dimensional measure, with respect to the Euclidean length, over the finite unions G of segments and points with $G \subset X$ such that G meets every cube of X and $G \cup D$ is connected. For a union X of cubes with the connected components X^i we put $d(X;D) \coloneqq \sum_i d(X^i; D \cap X^i)$. The *linear size* of X is $d(X) \coloneqq d(X;\emptyset)$.

(iv) Let M be a power of L with $M \le L^{m_{\mathbb{T}}}$. For $0 \le j \le k$ the family π_j is the grid of side Ma_j, a *localisation domain* of the scale j is a connected union of cubes of π_j, the class of these domains is $\mathbf{D}_j$, and $d_j(X)$ and $d_j(X;D)$ denote the linear sizes (1.21) for the grid π_j.

The domains of $\mathbf{D}_j$ and the size $d_j(X)$ are those of [9, p. 257], where the size is the length of a shortest tree graph in X which meets all its cubes, divided by the side. In the papers the number of the cubes of a domain is bounded by a multiple of its linear size [11, (1.28), p. 8], which fails for a single cube, whose linear size is zero. The following count holds for every domain.

Lemma 1.18 (The Cubes of a Localisation Domain)**.** *Let X be a connected union of cubes of a grid of side s. Then*

$$\#(X) \le 2^8(2 + d(X)) \le 2^9(1 + d(X)) \,, \qquad \operatorname{diam} X \le s(2d(X) + 4) \,, \tag{1.22}$$

with the diameter in the Euclidean distance. Moreover, for every $\kappa \ge 2^9(6 \log 2 + 1)$ and every cube $\square$ of the grid,

$$\sum_{X \supset \square} e^{-\kappa d(X)} \le 2^{6 \cdot 2^{10}} + 1 \,, \tag{1.23}$$

the sum over the connected unions of cubes of the grid which contain $\square$.

Proof. Let G be a set which is admissible in (1.21) for $D = \emptyset$, that is, a connected finite union of segments in X which meets every cube of X, and let λ be the sum of the ℓ^∞-lengths of its segments, which is at most its Euclidean length. We first cover G by few boxes. Regard G as a connected graph whose arcs are segments, and double every arc. All degrees of the resulting graph are even, so that it has a closed walk which traverses every arc once, and parametrising this walk by the ℓ^∞-length we obtain a map $\gamma\colon [0, 2\lambda] \to G$ onto G with $|\gamma(t) - \gamma(t')|_\infty \le |t - t'|$. Let $n \coloneqq \lceil \lambda/s \rceil$ and $t_i \coloneqq \min(2is, 2\lambda)$ for $0 \le i \le n$. Every $t \in [0, 2\lambda]$ lies within s of some t_i, so that every point of G lies within the ℓ^∞-distance s of one of the $n + 1 \le \lambda/s + 2$ points $\gamma(t_i)$.

Next we count the cubes near one of these points. The cubes of the grid are the products of intervals $[c_\nu - s/2, c_\nu + s/2]$ with $c_\nu \in s\mathbb{Z}$, and such an interval meets $[q_\nu - s, q_\nu + s]$ only if c_ν lies in an interval of length $3s$, which contains at most four points of $s\mathbb{Z}$. A box $\{z \,|\, |z - q|_\infty \le s\}$ therefore meets at most $4^4 = 2^8$ cubes of the grid. Every cube of X contains a point of G and hence meets one of the $n + 1$ boxes around the points $\gamma(t_i)$, so that $\#(X) \le 2^8(2 + \lambda/s)$, and the infimum over G gives the first bound of (1.22). The second inequality there is $2 + d \le 2(1 + d)$.

For the diameter, let $z, z' \in X$. They lie in cubes of X which contain points q, q' of G, and the Euclidean diameter of a cube of side s in four dimensions is $2s$. Since G is connected, $|q - q'|_\infty \le \lambda$, and the Euclidean distance is at most twice the ℓ^∞-distance. Hence $|z - z'| \le 4s + 2\lambda$, and the infimum over G gives the bound.

For (1.23), a connected union of n cubes which contains $\square$ has a spanning tree with respect to the relation of sharing a wall, and a closed walk from $\square$ along this tree which

traverses every edge twice has $2(n-1)$ steps, each of which goes to one of the eight cubes which share a wall with the present one. The union is determined by the walk, so that there are at most $8^{2(n-1)} = 2^{6(n-1)}$ such unions. For $n \geq 2^{10}$ the first bound of (1.22) gives $d(X) \geq 2^{-8}n - 2 \geq 2^{-9}n$, and then $e^{-\kappa d(X)} \leq 2^{-6n}e^{-n}$ by the assumption on κ. The sum is therefore at most $\sum_{n=1}^{2^{10}-1} 2^{6(n-1)} + \sum_{n \geq 2^{10}} e^{-n}$, in which the first sum is less than $2^{6 \cdot 2^{10}}$ and the second less than one. *q.e.d.*

The cluster expansions of the later sections use five inequalities which relate the linear sizes of a domain, of the parts into which it is decomposed, and of the domain of the next scale which covers it. For $D = \emptyset$ the first is [11, (2.27), p. 18], with 3 in the place of 5, and the second is [11, (2.32), p. 18] with the constant 18 in the place of the printed 4. With 4 the inequality fails: if Z consists of one cube and of eight rods of $n \geq 2$ cubes each, one attached to each wall of the cube, and the rods are the sets Y_i, then $d_k(Y_i) = n - 2$, while a connected set which meets all the cubes of Z has length at least $(n-1)Ma_k$ beyond each of the eight walls, so that $d_k(Z) \geq \sum_i d_k(Y_i) + 8$. For a union X of cubes of π_k we write X' for the smallest union of cubes of the grid of side LMa_k which contains $X^\sim$, as in [11, p. 13], and we write d_k and d' for the linear sizes (1.21) of the grids of the sides Ma_k and LMa_k.

Lemma 1.19 (Five Geometric Inequalities)**.** *Let M be a power of L with $LM \leq L^{m_{\mathbb{T}}}$.*

(i) Let Y_0 be a connected union of cubes of π_k, $\mathbf{D}$ a finite family of connected unions $Y \subset Y_0$ of cubes, and $D \subset Y_0$ a union of cubes with $n_D \geq 0$ connected components, such that $Y_0 = \bigcup_{Y \in \mathbf{D}} Y \cup D$. Then $\sum_{Y \in \mathbf{D}} d_k(Y) \geq d_k(Y_0; D) - 3(|\mathbf{D}| + n_D - 1)$.

(ii) Let Z be a connected union of cubes of π_k, let $Y_1, \dots, Y_n \subset Z$ be connected unions of cubes no two of which have a cube in common or contain two cubes which share a wall, and let $D \subset \bigcup_i Y_i$ be a union of cubes. Then $\sum_i d_k(Y_i; D \cap Y_i) + 18\,\#(Z \setminus \bigcup_i Y_i) \geq d_k(Z; D)$.

(iii) Let Z be a connected union of cubes of π_k and let $D \subset Z$ be a nonempty union of cubes. Then $L\,d'(Z'; D') \leq (1 + 486/L)\,d_k(Z; D)$.

(iv) Let Z be a connected union of cubes of the grid of side LMa_k, let $Z_1, \dots, Z_n \subset Z$ be connected unions of cubes and $D_i \subset Z_i$ unions of cubes, of which $n' \geq 1$ are nonempty and $n_\emptyset$ are empty, and put $Z_0 := \bigcup_i Z_i$ and $D := \bigcup_i D_i$. Then $d'(Z; D) \leq \sum_i d'(Z_i; D_i) + 3\,\#(Z \setminus Z_0) + 3(n_\emptyset + n' - 1)$.

(v) Let X be a connected union of cubes of a grid and $D \subset X$ a nonempty union of cubes. Then $\#(X) \leq 81\,\#(D) + 324\,d(X; D)$.

2 The Averaging Operations

A renormalisation step replaces a gauge field on a lattice by its average on the lattice whose spacing is L times larger. Two operations are needed. The first is the average itself, a map between gauge fields which commutes with the gauge transformations and with the symmetries of the lattice. The second arises when the field is written as a fluctuation times a background: the average of the product is then the average of the background, multiplied by a function of the fluctuation whose linear part enters every quadratic form of the later sections. We define both, and we state for each what the later sections use: on which bonds it depends, how it transforms, where it is analytic, and how large it is.

2.1 The Chart of the Group

The averages are exponentials of means of logarithms, so that we first have to say which logarithm is meant. A Hermitian logarithm of an element of G need not lie in $\mathfrak{g}$, and for this reason the logarithm on the group is defined as the inverse of the exponential on a ball of the algebra, and not through the spectrum.

Definition 2.1 (The Chart)**.** Let $\varrho_G := \pi/\sqrt{N}$. The *chart domain* is $N_G := \{e^{iA} \,|\, A \in \mathfrak{g},\ h(A) < \varrho_G\}$, and the *chart* $\log\colon N_G \to i\mathfrak{g}$ is defined by $\log e^{iA} := iA$ for $A \in \mathfrak{g}$ with $h(A) < \varrho_G$. For a matrix W with $\|W - \mathbf{1}\| < 1$ we put $\operatorname{Log} W := \sum_{n\geq 1}(-1)^{n+1}n^{-1}(W-\mathbf{1})^n$.

Lemma 2.2 (The Chart)**.** *(i) The map $A \mapsto e^{iA}$ is injective on $\{A \in \mathfrak{g} \,|\, h(A) < \varrho_G\}$, so that the chart is well defined, and $\|A\| < \pi$ there. The chart domain is invariant under $W \mapsto W^{-1}$ and under conjugation by elements of G, with $\log W^{-1} = -\log W$ and $\log(uWu^{-1}) = R(u)\log W$.*

(ii) There is a number $r_G \in (0,1)$, which depends on G only, such that every $W \in G$ with $\|W - \mathbf{1}\| \leq r_G$ lies in N_G and satisfies $\log W = \operatorname{Log} W$.

(iii) For a Hermitian matrix X with $\|X\| \leq \pi$ and each of the three norms $\mathcal{N}$ of Lemma 1.3, $\mathcal{N}(e^{iX} - \mathbf{1}) \leq \mathcal{N}(X) \leq \frac{\pi}{2}\mathcal{N}(e^{iX} - \mathbf{1})$.

(iv) The function Log *is holomorphic on $\{\|W - \mathbf{1}\| < 1\}$, $e^{\operatorname{Log} W} = W$, $\operatorname{Log}(\mathbf{u}W\mathbf{u}^{-1}) = R(\mathbf{u})\operatorname{Log} W$ for every invertible $\mathbf{u}$ with $\|\mathbf{u}W\mathbf{u}^{-1} - \mathbf{1}\| < 1$, and $\operatorname{Log} W \in i\mathfrak{g}^c$ if $W = e^{iX}$ with $X \in \mathfrak{g}^c$ and $\|X\| < \log 2$.*

2.2 The Average of One Step

We fix a scale $1 \leq j \leq k$ and write $a := a_{j-1}$ for the spacing of the finer lattice $T^{(j-1)}$. The average of a bond c of $T^{(j)}$ is built from the fine paths which join the block of c_- to the block of c_+ [9, p. 252, (0.4), p. 253].

Definition 2.3 (The Paths of a Coarse Bond)**.** (i) For $y \in T^{(j)}$ and $x \in B(y)$, write $x - y = a\sum_\mu n_\mu e_\mu$ with $|n_\mu| \leq \frac{1}{2}(L-1)$. The family $\mathbf{G}(y,x)$ consists of the paths of $T^{(j-1)}$ from y to x which traverse the coordinates μ with $n_\mu \neq 0$ one after the other, $|n_\mu|$ bonds in the direction of the sign of n_μ for each, one path for every ordering of these coordinates. For $x = y$ it consists of the empty path.

(ii) For a bond c of $T^{(j)}$ and $x \in B(c_-)$ let $x' := x + (c_+ - c_-) \in B(c_+)$, and let $[x,x']$ be the straight path of L bonds of $T^{(j-1)}$ from x to x'; in particular $[c] := [c_-, c_+]$.

(iii) For $x \in B(c_-)$, $\Gamma \in \mathbf{G}(c_-,x)$, and $\Gamma' \in \mathbf{G}(c_+,x')$, the *loop* $\ell_c(x,\Gamma,\Gamma')$ is the closed path which consists of Γ, then $[x,x']$, then Γ'^{-1}, then $[c]^{-1}$. It begins and ends at c_-.

All these paths run through sites of $B(c_-) \cup B(c_+)$.

Definition 2.4 (The Average)**.** Let U be a gauge field on $T^{(j-1)}$ and c a bond of $T^{(j)}$. Let $\mathcal{O}_c$ be the set of the gauge fields for which $U(\ell_c(x,\Gamma,\Gamma')) \in N_G$ for all loops of c, and for $U \in \mathcal{O}_c$ let

$$\beta_c(U) := \frac{1}{L^4}\sum_{x\in B(c_-)} \frac{1}{|\mathbf{G}(c_-,x)||\mathbf{G}(c_+,x')|}\sum_{\Gamma,\Gamma'}\frac{1}{i}\log U(\ell_c(x,\Gamma,\Gamma'))\,. \tag{2.1}$$

The *average* of U is the gauge field $M(U)$ on $T^{(j)}$ with

$$M(U)(c) \coloneqq \begin{cases} e^{i\beta_c(U)}U([c]) & \text{if } U \in \mathcal{O}_c \,, \\ U([c]) & \text{otherwise} \,. \end{cases} \tag{2.2}$$

The *average of order* n is the n-fold iterate M^n, which maps gauge fields on $T^{(j)}$ to gauge fields on $T^{(j+n)}$.

On $\mathcal{O}_c$ this is [9, (0.4), p. 253]: the straight transport along c is multiplied by the exponential of a mean, with weights which are positive and sum to one, of the logarithms of the holonomies around the loops of c. The second line of (2.2) makes the average a map on all gauge fields, which the definition of the renormalisation transformation requires, and it plays no role in any estimate, since every estimate concerns fields in the set of the following lemma. The iterates are those of [4, (43), p. 24].

Lemma 2.5 (Properties of the Average)**.** *(i)* Consistency. *$\mathcal{O}_{c^{-1}} = \mathcal{O}_c$ and $M(U)(c^{-1}) = M(U)(c)^{-1}$, so that $M(U)$ is a gauge field, and $\beta_c(U) \in \mathfrak{g}$.*

(ii) Locality. *$M(U)(c)$ depends only on the values of U on the bonds of $T^{(j-1)}$ with both ends in $B(c_-) \cup B(c_+)$.*

(iii) Covariance. *For every gauge transformation $u\colon T^{(j-1)} \to G$, the set $\mathcal{O}_c$ is invariant and $M(U^u)(c) = u(c_-)M(U)(c)u(c_+)^{-1}$.*

(iv) Symmetry. *Let θ be a translation of the torus by a point of $T^{(j)}$, a reflection $x_\mu \mapsto -x_\mu$, or a permutation of the coordinates, and let $(\theta U)(b) \coloneqq U(\theta^{-1}b)$. Then $M(\theta U) = \theta M(U)$.*

(v) The domain. *There are numbers $\alpha_M > 0$ and C_M, which depend on L and G only, such that if $|U(\partial p) - \mathbf{1}| \le \alpha \le \alpha_M$ for all plaquettes p of $T^{(j-1)}$ whose corners lie in $B(c_-) \cup B(c_+)$, then $U \in \mathcal{O}_c$, every loop of c satisfies $\|U(\ell) - \mathbf{1}\| \le r_G$, and $|\beta_c(U)| \le C_M\alpha$.*

(vi) Analyticity. *There is a number $r_M > 0$, which depends on L and G only, with the following property. Let U be as in* (v)*, and let A be a 1-form with values in $\mathfrak{g}^c$ on the bonds of $B(c_-) \cup B(c_+)$ with $|aA| < r_M$. Then* (2.1) *and the first line of* (2.2)*, with* Log *in the place of* log*, define a holomorphic function $A \mapsto M(e^{iaA}U)(c)$ with values in G^c, which is the average of the gauge field $e^{iaA}U$ when A has values in $\mathfrak{g}$, and which has the properties* (i)–(iv)*, the covariance for gauge transformations with values in G^c which map the configuration into the domain.*

By (vi) the average, and with it every iterate, is defined for the configurations $e^{iaA}U$ of that statement, and we use the notation $M(\mathbf{U})$ for them.

The first quantitative property is that the average of a field with small plaquettes has small plaquettes, in the units of the coarser lattice, and that this is a local statement. It is [4, Propositions 1, 2, pp. 26], in the form which the test functions of the construction need, with an arbitrary bound in the place of a fixed threshold.

Proposition 2.6 (Averages of Regular Fields)**.** *There are numbers $C_0 \ge 1$ and $c_2 \in (0, \alpha_M]$, which depend on L and G only, with the following properties.*

(i) Let U be a gauge field on $T^{(j-1)}$ and p' a plaquette of $T^{(j)}$ with the first corner y_0, and suppose that $|U(\partial p) - \mathbf{1}| \le \alpha \le c_2 L^{-2}$ for all plaquettes p of $T^{(j-1)}$ whose corners lie in

the blocks of the four corners of p'. Then

$$|M(U)(\partial p') - \mathbf{1}| \le \frac{1}{L^4} \sum_{x \in B(y_0)} \sum_{p \subset p'_x} |U(\partial p) - \mathbf{1}| + C_0 L^4 \alpha^2 \le L^2 \alpha + C_0 L^4 \alpha^2 \,, \qquad (2.3)$$

where p'_x is the translate of p' by $x - y_0$ and the inner sum runs over the L^2 plaquettes of $T^{(j-1)}$ which it contains.

(ii) Let $0 \le i < j \le k$, let U be a gauge field on $T^{(i)}$ and p' a plaquette of $T^{(j)}$, and suppose that $|U(\partial p) - \mathbf{1}| \le \alpha (a_i/a_j)^2$ with $\alpha \le c_2$ for all plaquettes p of $T^{(i)}$ whose corners lie in the blocks of order $j - i$ of the four corners of p'. Then $|M^{j-i}(U)(\partial p') - \mathbf{1}| \le \alpha + 2C_0\alpha^2 \le 2\alpha$.

2.3 The Average Relative to a Background

Let the field be $e^{iaA}\mathbf{U}$, with a background $\mathbf{U}$ and a fluctuation A. We want to write its average as a function of the fluctuation times the average of the background. The quotient $M(e^{iaA}\mathbf{U})(c)M(\mathbf{U})(c)^{-1}$ does this, but it is not the function which the linear theory uses: under a gauge transformation of the fluctuation which is trivial at the coarse sites it changes, through the paths Γ and Γ' which join the coarse sites to the points of their blocks. The function used instead is obtained by first transforming the fluctuation, at each coarse site, to the gauge in which the mean of its transports to the points of the block is trivial [4, pp. 26–31], [5, p. 418]. The transports are taken along all the paths of $\mathbf{G}(y, x)$ at once, through a mean of group elements [9, (0.5)–(0.11), p. 253].

Definition 2.7 (The Mean of Group Elements and the Transports of a Block)**.** (i) Let $\mathbf{W}_1, \dots, \mathbf{W}_n \in G^c$ with $\|\mathbf{W}_i \mathbf{W}_1^{-1} - \mathbf{1}\| \le \frac{1}{4} r_G$ for all i. Their *mean* $\mathsf{m}(\{\mathbf{W}_i\})$ is the element $\mathbf{W} \in G^c$ with $\|\mathbf{W}\mathbf{W}_1^{-1} - \mathbf{1}\| \le \frac{1}{2} r_G$ and

$$\sum_{i=1}^{n} \mathrm{Log}(\mathbf{W}_i \mathbf{W}^{-1}) = 0 \,. \qquad (2.4)$$

(ii) For a configuration $\mathbf{U}$ on $T^{(j-1)}$, a site $y \in T^{(j)}$, and $x \in B(y)$, the *transport* from y to x is $\mathbf{U}(y, x) \coloneqq \mathsf{m}(\{\mathbf{U}(\Gamma)\}_{\Gamma \in \mathbf{G}(y,x)})$, whenever the mean is defined.

Lemma 2.8 (The Mean of Group Elements)**.** *Under the condition of Definition 2.7(i) the equation* (2.4) *has exactly one solution with $\|\mathbf{W}\mathbf{W}_1^{-1} - \mathbf{1}\| \le \frac{1}{2} r_G$. It does not depend on the order of the elements, it is a holomorphic function of them, and*

$$\mathsf{m}(\{\mathbf{u}\mathbf{W}_i\mathbf{v}\}) = \mathbf{u}\, \mathsf{m}(\{\mathbf{W}_i\})\, \mathbf{v} \,, \qquad \mathsf{m}(\{\mathbf{W}_i^{-1}\}) = \mathsf{m}(\{\mathbf{W}_i\})^{-1} \,,$$

for $\mathbf{u}, \mathbf{v} \in G^c$ for which both sides are defined. If all $\mathbf{W}_i$ lie in G, then so does the mean. If $\mathbf{W}_i = e^{iX_i}$ with $X_i \in \mathfrak{g}^c$ and $\|X_i\| \le \varrho$ for a sufficiently small ϱ, which depends on G only, then $\|\frac{1}{i} \mathrm{Log}\, \mathsf{m}(\{\mathbf{W}_i\}) - \frac{1}{n} \sum_i X_i\| \le C_{\mathsf{m}} \varrho^2$ with a constant C_{m} which depends on G only.

These are the properties [9, (0.5)–(0.9), p. 253] of the mean [9, (0.10)]. By the covariance, $\mathbf{U}^u(y, x) = u(y)\mathbf{U}(y, x)u(x)^{-1}$.

Definition 2.9 (The Relative Average)**.** Let $\mathbf{U}$ be a configuration on $T^{(j-1)}$ and A a 1-form on $T^{(j-1)}$ with values in $\mathfrak{g}^c$, and write $\mathbf{U}_A \coloneqq e^{iaA}\mathbf{U}$ for the configuration with $\mathbf{U}_A(b) = e^{iaA(b)}\mathbf{U}(b)$ on the positive bonds.

(i) The *site factor* of the fluctuation at $y \in T^{(j)}$ is

$$v_{\mathbf{U}}(A)(y) \coloneqq \exp\Big(\frac{1}{L^4} \sum_{x \in B(y)} \mathrm{Log}(\mathbf{U}_A(y,x)\mathbf{U}(y,x)^{-1})\Big) \,. \tag{2.5}$$

(ii) The *relative average* of the fluctuation on a bond c of $T^{(j)}$ is

$$\mathcal{M}_{\mathbf{U}}(A)(c) \coloneqq v_{\mathbf{U}}(A)(c_-)^{-1}\, M(\mathbf{U}_A)(c)\, v_{\mathbf{U}}(A)(c_+)\, M(\mathbf{U})(c)^{-1} \,, \tag{2.6}$$

so that

$$M(e^{iaA}\mathbf{U})(c) = v_{\mathbf{U}}(A)(c_-)\, \mathcal{M}_{\mathbf{U}}(A)(c)\, M(\mathbf{U})(c)\, v_{\mathbf{U}}(A)(c_+)^{-1} \,. \tag{2.7}$$

(iii) The *linear part* $Q(\mathbf{U})$ and the *remainder* $C(\mathbf{U};\cdot)$ of the relative average are defined by

$$\frac{1}{i}\,\mathrm{Log}\,\mathcal{M}_{\mathbf{U}}(A)(c) = La(Q(\mathbf{U})A)(c) + C(\mathbf{U};aA)(c) \,, \tag{2.8}$$

where $Q(\mathbf{U})$ is linear in A and $C(\mathbf{U};\cdot)$ vanishes to the second order at zero. The *linear part of the average* is the linear map $Q^{\mathrm{pl}}(\mathbf{U})$ with $\frac{1}{i}\,\mathrm{Log}(M(e^{iaA}\mathbf{U})(c)M(\mathbf{U})(c)^{-1}) = La(Q^{\mathrm{pl}}(\mathbf{U})A)(c) + O(A^2)$.

(iv) For a 0-form λ on $T^{(j-1)}$, the *block average* is the 0-form on $T^{(j)}$

$$(Q'(\mathbf{U})\lambda)(y) \coloneqq \frac{1}{L^4} \sum_{x \in B(y)} R(\mathbf{U}(y,x))\lambda(x) \,. \tag{2.9}$$

The identity (2.7) is [5, (3.113), p. 418] with the transports of Definition 2.7, and Q' is [5, (3.19), p. 393]. The factor La in (2.8) is the spacing of the coarse lattice, so that $Q(\mathbf{U})$ maps 1-forms to 1-forms without a change of their dimension.

Proposition 2.10 (The Relative Average of One Step)**.** *There are numbers $c_4 \in (0, \alpha_M]$, C_1, C_2, and C_3, which depend on L and G only, with the following properties. Let U be a gauge field on $T^{(j-1)}$ with $|U(\partial p) - \mathbf{1}| \le \alpha_0 \le c_4$ on the plaquettes with corners in $B(c_-) \cup B(c_+)$, let A' and A be 1-forms with values in $\mathfrak{g}^c$ on the bonds of $B(c_-) \cup B(c_+)$ with $|aA'| \le \alpha_1 \le c_4$ and $|aA| < c_4$, and let $\mathbf{U} \coloneqq e^{iaA'}U$.*

(i) Analyticity and locality. *The transports, the site factors at $c_\pm$, and $\mathcal{M}_{\mathbf{U}}(A)(c)$ are defined, they are holomorphic functions of (A', A), and they depend only on the values of U, A', and A on the bonds with both ends in $B(c_-) \cup B(c_+)$, the site factor at y only on those in $B(y)$. For $A' = 0$ and A with values in $\mathfrak{g}$ their values lie in G.*

(ii) Covariance. *For a gauge transformation u of $T^{(j-1)}$ with values in G,*

$$v_{\mathbf{U}^u}(R(u)A)(y) = u(y)v_{\mathbf{U}}(A)(y)u(y)^{-1} \,, \qquad \mathcal{M}_{\mathbf{U}^u}(R(u)A)(c) = u(c_-)\mathcal{M}_{\mathbf{U}}(A)(c)u(c_-)^{-1} \,,$$

so that $Q(\mathbf{U}^u)R(u) = R(u)Q(\mathbf{U})$ and $Q'(\mathbf{U}^u)R(u) = R(u)Q'(\mathbf{U})$, with $R(u)$ on the right sides taken at the sites of $T^{(j)}$. The same holds for u with values in G^c near G, as long as $\mathbf{U}^u$ is of the form assumed.

(iii) The two linear parts. *Let $\Lambda(\mathbf{U})A$ be the 0-form on $T^{(j)}$ which is the linear part in A of $\frac{1}{i}\,\mathrm{Log}\, v_{\mathbf{U}}(A)$. Then $Q^{\mathrm{pl}}(\mathbf{U}) = Q(\mathbf{U}) - D_{M(\mathbf{U})}\Lambda(\mathbf{U})$, with the covariant difference of the lattice $T^{(j)}$ at the configuration $M(\mathbf{U})$.*

(iv) Gauge directions. *For a* 0*-form* λ *on* $T^{(j-1)}$ *with values in* $\mathfrak{g}^c$*,*

$$Q(\mathbf{U})D_{\mathbf{U}}\lambda = D_{M(\mathbf{U})}Q'(\mathbf{U})\lambda\ , \qquad \Lambda(\mathbf{U})D_{\mathbf{U}}\lambda = Q'(\mathbf{U})\lambda - \lambda|_{T^{(j)}}\ ,$$
$$Q^{\mathrm{pl}}(\mathbf{U})D_{\mathbf{U}}\lambda = D_{M(\mathbf{U})}(\lambda|_{T^{(j)}})\ .$$

(v) The form of the linear part. *For a bond* b *of a line* $[x, x']$ *and* $\Gamma \in \mathbf{G}(c_-, x)$ *let* Γ_b *be the path which consists of* Γ *and of the part of the line from* x *to* b_-*, and let*

$$(Q_0(\mathbf{U})A)(c) \coloneqq \frac{1}{L^5}\sum_{x\in B(c_-)}\sum_{b\in[x,x']}\frac{1}{|\mathbf{G}(c_-,x)|}\sum_{\Gamma}R(\mathbf{U}(\Gamma_b))A(b)\ , \tag{2.10}$$

the mean of the values of A *on the lines of* c*, transported to* c_-*. Then* $Q(\mathbf{1}) = Q_0(\mathbf{1})$*, and* $|((Q(\mathbf{U}) - Q_0(\mathbf{U}))A)(c)| \le C_1(\alpha_0+\alpha_1)|A|$*, the supremum of* A *being taken over the bonds with both ends in* $B(c_-)\cup B(c_+)$*.*

(vi) The remainder. $|C(\mathbf{U}; aA)(c)| \le C_2|aA|^2$*, and the derivative of* $A \mapsto C(\mathbf{U}; aA)(c)$ *in a direction* δA *is at most* $C_3|aA||a\,\delta A|$*, the norms of* A *and* δA *being suprema over the bonds of* $B(c_-)\cup B(c_+)$*.*

(vii) Dependence on the background. *The maps* $Q(\mathbf{U})$*,* $Q'(\mathbf{U})$*, and* $C(\mathbf{U};\cdot)$ *are holomorphic in* A'*, and* $\|Q(e^{iaA'_1}U) - Q(e^{iaA'_2}U)\| \le C_1|a(A'_1 - A'_2)|$ *in the norms of* (v)*, and likewise for* Q'*.*

(viii) One function for all scales. *Identify the bonds with both ends in* $B(c_-)\cup B(c_+)$ *with those of a reference bond by the translation and the symmetry of Lemma 2.5*(iv) *which map* c *to it. Then* $\mathcal{M}_{\mathbf{U}}(A)(c)$*,* $v_{\mathbf{U}}(A)(c_\pm)$*, and* $M(\mathbf{U})(c)$ *are the values of functions of the matrices* $\mathbf{U}(b)$ *and* $aA(b)$ *on these bonds which do not depend on* c*, on the scale* j*, on* k*, or on the torus.*

The parts (v)–(vii) are [4, Propositions 3–7, (121)–(125)] for the relative average of this section, and (iii), (iv) are the relations [5, (3.114), (3.115), p. 418]; the last identity of (iv) is the derivative of the covariance of the average, Lemma 2.5(iii). At the trivial configuration the operator (2.10) is the average over the straight lines of [2, (1.18), p. 20], with centred blocks. It does not increase the ℓ^1-, the ℓ^2-, and the supremum norm of a 1-form, each taken with the weight of its lattice, because every bond of $T^{(j-1)}$ lies on exactly L of the lines $[x, x']$ of the bonds c of one direction.

The averages of higher order are compositions, and so are their relative versions: the background of the step $l+1$ is the average of order l of the background [4, (90), (91), p. 31], [5, (3.15), p. 393].

Definition 2.11 (Relative Averages of Higher Order)**.** Let $0 \le i < j \le k$, $n \coloneqq j - i$, let $\mathbf{U}$ be a configuration on $T^{(i)}$ and A a 1-form on $T^{(i)}$ with values in $\mathfrak{g}^c$. Put $\mathbf{U}^{(l)} \coloneqq M^l(\mathbf{U})$, $A^{(0)} \coloneqq A$, and, for $0 \le l < n$,

$$a_{i+l+1}A^{(l+1)} \coloneqq \frac{1}{i}\operatorname{Log}\mathcal{M}_{\mathbf{U}^{(l)}}(A^{(l)})\ .$$

The *linear part* and the *remainder of order* n are defined by $A^{(n)} = Q_n(\mathbf{U})A + a_j^{-1}C_n(\mathbf{U}; a_jA)$ with $Q_n(\mathbf{U})$ linear and $C_n(\mathbf{U};\cdot)$ vanishing to the second order, so that

$$Q_n(\mathbf{U}) = Q(\mathbf{U}^{(n-1)})\cdots Q(\mathbf{U}^{(1)})Q(\mathbf{U}^{(0)})\ , \tag{2.11}$$

and $Q'_n(\mathbf{U}) \coloneqq Q'(\mathbf{U}^{(n-1)})\cdots Q'(\mathbf{U}^{(0)})$.

Proposition 2.12 (Relative Averages of Higher Order)**.** *There are numbers $c_4' \in (0, c_4]$, C_2', and C_3', which depend on L and G only, with the following properties. Let $0 \le i < j \le k$ and $n = j - i$, let c be a bond of $T^{(j)}$, and let B_c be the set of the bonds of $T^{(i)}$ with both ends in $B^n(c_-) \cup B^n(c_+)$. Let U be a gauge field on $T^{(i)}$ with $|U(\partial p) - \mathbf{1}| \le \alpha_0 (a_i/a_j)^2$ on the plaquettes with corners in $B^n(c_-) \cup B^n(c_+)$, let A' and A be 1-forms with values in $\mathfrak{g}^c$ on B_c with $|a_j A'| \le \alpha_1$ and $|a_j A| < c_4'$, where $\alpha_0, \alpha_1 \le c_4'$, and let $\mathbf{U} \coloneqq e^{i a_i A'} U$.*

(i) The averages $M^l(e^{i a_i A}\mathbf{U})$, $l \le n$, on the bonds between the blocks concerned, and the forms $A^{(l)}$ are defined, they are holomorphic in (A', A), and they depend only on U, A', and A on B_c. There are gauge transformations $v_n(A)$ of $T^{(j)}$, holomorphic and local in the same sense, with $v_n(0) = \mathbf{1}$, such that

$$\begin{gathered} M^n(e^{i a_i A}\mathbf{U})(c) = v_n(A)(c_-)\, e^{i X_n(c)}\, M^n(\mathbf{U})(c)\, v_n(A)(c_+)^{-1} \,, \\ X_n(c) \coloneqq a_j (Q_n(\mathbf{U})A)(c) + C_n(\mathbf{U}; a_j A)(c) \,. \end{gathered} \tag{2.12}$$

(ii) $|(Q_n(\mathbf{U})A)(c)| \le (1 + C_2'(\alpha_0 + \alpha_1))|A|_{\mathsf{B}_c}$ *and* $|(Q_n'(\mathbf{U})\lambda)(y)| \le (1 + C_2'(\alpha_0 + \alpha_1))|\lambda|_{B^n(y)}$.

(iii) $|C_n(\mathbf{U}; a_j A)(c)| \le C_2' |a_j A|^2$, *and its derivative in a direction δA is at most* $C_3' |a_j A||a_j\, \delta A|$, *with the suprema over B_c.*

(iv) $|\frac{1}{i} \operatorname{Log} v_n(A)(y)| \le C_2' |a_j A|_{B^n(y)}$.

(v) The covariance of Proposition 2.10(ii) *and the relation $Q_n(\mathbf{U}) D_{\mathbf{U}} = D_{M^n(\mathbf{U})} Q_n'(\mathbf{U})$ hold, and Q_n, Q_n', and C_n are Lipschitz continuous in $a_j A'$ with a constant C_3' in the norms of* (ii) *and* (iii).

(vi) $|M^n(e^{i a_i A}\mathbf{U})(c) M^n(\mathbf{U})(c)^{-1} - \mathbf{1}| \le C_3' |a_j A|_{\mathsf{B}_c}$.

The constants do not depend on n.

This is [4, Propositions 4–7, pp. 38–43] with [4, (88), (92), (97), (99), pp. 31–32] for the factors v_n. The point of the statement is the uniformity in the order n. The plaquettes of the averages $M^l(U)$ and the fields $A^{(l)}$ are measured in the units of their scales, in which the hypotheses reproduce themselves from one step to the next by Proposition 2.6 and by (iii), and the complex part of the background is carried along as a second fluctuation: by (2.12) for the pair (U, A'), each $M^l(\mathbf{U})$ is a gauge transform of $e^{i a_{i+l} A_l'} M^l(U)$ with $|a_{i+l} A_l'|$ of the order $\alpha_1 a_{i+l}/a_j$. No bound on the differences of A' is assumed.

The gauge transformation which takes a field to the axial gauge of a background [4, (106), p. 33] and the averages of gauge transformations [4, Sect. F] are built from the site factors v_n. They are stated in Section 5 together with the gauge conditions, since their statements refer to a sequence of domains.

3 The Trivial Background

At the trivial configuration and for the sequence of domains with a single layer, $\Omega_j = \mathbb{T}$ for all $j \le k$, the operators of the linear theory commute with the translations of the coarse lattice and are computed in momentum space [2, Sects. B–D]. The main paper uses these formulae for the coefficients of the coupling flow at zero coupling. In this section $\mathbf{U} = \mathbf{1}$, the covariant differences are the ordinary ones, written ∂_μ, ∂, and ∂^*, and the operators act on each component of a form with respect to a basis of $\mathfrak{g}$ separately, so that we may regard the forms as scalar.

Definition 3.1 (Momenta and Symbols)**.** (i) The *dual tori* are $\tilde{T}_\eta := \{p \in (2\pi/\ell_{\mathbb{T}})\mathbb{Z}^4 \mid -\pi/\eta \le p_\mu < \pi/\eta\}$ and $\tilde{T}^{(k)} := \{p' \in (2\pi/\ell_{\mathbb{T}})\mathbb{Z}^4 \mid -\pi \le p'_\mu < \pi\}$. For a function f on T_η, $\tilde{f}(p) := \sum_x \eta^4 e^{-ip\cdot x} f(x)$ for $p \in \tilde{T}_\eta$, and likewise with the weight one for functions on $T^{(k)}$ and $p' \in \tilde{T}^{(k)}$. Every $p \in \tilde{T}_\eta$ is uniquely $p' + l$ with $p' \in \tilde{T}^{(k)}$ and $l \in 2\pi\mathbb{Z}^4$, $l_\mu = 2\pi m_\mu$ with $|m_\mu| \le \frac{1}{2}(L^k - 1)$. We write $\int \mathrm{d}p'$ for $\ell_{\mathbb{T}}^{-4}(2\pi)^4 \sum_{p' \in \tilde{T}^{(k)}}$.

(ii) The *symbols* are

$$\partial_\mu(p) := \frac{e^{i\eta p_\mu} - 1}{\eta}\,, \qquad \Delta(p) := \sum_\mu |\partial_\mu(p)|^2\,,$$
$$\partial^1_\mu(p') := e^{ip'_\mu} - 1\,, \qquad \Delta_0(p') := \sum_\mu |\partial^1_\mu(p')|^2\,,$$
$$u_k(p) := \prod_\mu \frac{\sin(\frac{1}{2}p_\mu)}{L^k \sin(\frac{1}{2}\eta p_\mu)}\,, \qquad v_\mu(p) := \frac{e^{ip_\mu} - 1}{L^k(e^{i\eta p_\mu} - 1)}\,,$$

where a quotient with vanishing denominator has the value one.

Proposition 3.2 (The Averages and the Minimal Form in Momentum Space)**.** *Let $Q_k := Q_k(\mathbf{1})$ and $Q'_k := Q'_k(\mathbf{1})$ be the averages of order k of Definition 2.11 at the trivial configuration, from forms on T_η to forms on $T^{(k)}$.*

(i) $(Q'_k\lambda)(y)$ is the mean of λ over $B^k(y)$, $(Q_kA)_\mu(y)$ is the mean over $x \in B^k(y)$ of the mean of A_μ over the L^k bonds of the straight path from x to $x + e_\mu$, and

$$\widetilde{Q'_k\lambda}(p') = \sum_l u_k(p'+l)\tilde{\lambda}(p'+l)\,, \qquad \widetilde{(Q_kA)}_\mu(p') = \sum_l u_k(p'+l)v_\mu(p'+l)\tilde{A}_\mu(p'+l)\,. \tag{3.1}$$

Moreover $Q_k\partial\lambda = \partial^1 Q'_k\lambda$, with the difference ∂^1 of the lattice $T^{(k)}$.

(ii) $|u_k(p'+l)| \le \prod_\mu 2(1+|m_\mu|)^{-1}$, $|v_\mu(p)| \le 1$, $\Delta(p) \ge 4\pi^{-2}|p|^2$, and $4\pi^{-2}|p'|^2 \le \Delta_0(p') \le |p'|^2$.

(iii) For $p' \ne 0$ let

$$\phi_\mu(p') := \sum_l \frac{|u_k(p'+l)|^2 |v_\mu(p'+l)|^2}{\Delta(p'+l)}\,. \tag{3.2}$$

There are absolute constants $0 < \gamma_0 \le \gamma_1$ with $\gamma_0 \le \Delta_0(p')\phi_\mu(p') \le \gamma_1$ for all k, L, $\ell_{\mathbb{T}}$, μ, and $p' \ne 0$.

(iv) For a 1-form B on $T^{(k)}$ let $\langle B, \Delta_k B\rangle$ be the minimum of $\langle \partial A, \partial A\rangle$ over the 1-forms A on T_η with $Q_kA = B$. Then

$$\langle B, \Delta_k B\rangle = \frac{1}{2}\sum_{\mu,\nu} \frac{1}{(2\pi)^4} \int \sigma_{\mu\nu}(p') |\widetilde{(\partial^1 B)}_{\mu\nu}(p')|^2 \,\mathrm{d}p'\,,$$
$$\sigma_{\mu\nu} := \Big(\Delta_0\phi_\mu\, \Delta_0\phi_\nu \sum_\lambda \frac{|\partial^1_\lambda|^2}{\Delta_0^2\phi_\lambda}\Big)^{-1}\,, \tag{3.3}$$

the term $p' = 0$ being absent, and $\gamma'_0\langle \partial^1 B, \partial^1 B\rangle \le \langle B, \Delta_k B\rangle \le \gamma'_1\langle \partial^1 B, \partial^1 B\rangle$ with absolute constants $0 < \gamma'_0 \le \gamma'_1$.

These are [2, (1.18), (1.20), (1.31), (1.36), (1.61), (1.62), (1.65)–(1.67)] for centred blocks, for which the symbol u_k is real. The form Δ_k is defined here by the minimum, which does not refer to a gauge; by Theorem 4.13(iv) at the trivial configuration the minimum is attained at HB.

4 Propagators in a Background Field

The fluctuation integral of a renormalisation step is Gaussian to leading order. Its quadratic form is the second variation of the action at the background, to which the constraints on the averages and a gauge-fixing term are added, and everything which the later sections compute is expressed through the inverses of a small number of operators built in this way. This section defines the operators, on a sequence of domains and at a pair of a complex configuration and a current, and states their properties: their analyticity and covariance, the decay and the regularity of their kernels, and their expansions into sums over walks of local operators. The expansions are part of the statements and not only of the proofs, because the main paper compares them term by term on two lattices.

4.1 The Forms and the Operators

Throughout this section a sequence of domains with the block size M and the separation R is given, with its layers Γ_j and its determining set $\mathbf{B}$, and $\mathbf{U}$ is a configuration on the fine lattice T_η for which the averages $\mathbf{U}^{(j)} \coloneqq M^j(\mathbf{U})$ and the operators $Q_j(\mathbf{U})$, $Q'_j(\mathbf{U})$ of Definition 2.11 are defined on the bonds and sites concerned. Forms are forms on T_η with values in $\mathfrak{g}^c$, and D, D^*, ∇, and the covariant Laplacian carry the index $\mathbf{U}$ unless another configuration is named. In addition a *Dirichlet domain* $\Omega_{\mathrm{D}} \subset \mathbb{T}$ is given, which is either the torus or a union of cubes of the grid of side $M\eta$ which contains Ω_1; forms are then required to vanish at the sites and bonds outside Ω_{D}, and an operator $\mathcal{A}$ is replaced by its compression $\chi\mathcal{A}\chi$ with the indicator χ of Ω_{D}, which we do not write. The case of a proper subset is needed for the local operators of Section 4.3.

Definition 4.1 (The Second Variation of the Action)**.** The operator $\Delta(\mathbf{U})$ on 1-forms is the operator which is symmetric with respect to the pairing (1.5) and satisfies

$$A(e^{i\eta A}\mathbf{U}) = A(\mathbf{U}) + \langle A, J(\mathbf{U})\rangle + \frac{1}{2}\langle A, \Delta(\mathbf{U})A\rangle + O(A^3) \tag{4.1}$$

for 1-forms A with values in $\mathfrak{g}^c$, where $e^{i\eta A}\mathbf{U}$ is the configuration with the values $e^{i\eta A(b)}\mathbf{U}(b)$ on the positive bonds.

The expansion (4.1) is [5, (3.7), (3.12), pp. 391–392]. Its linear term is the pairing with the current (1.10): the first variation of a plaquette term pairs $D_{\mathbf{U}}A$ with $\eta^{-2}\operatorname{Im}\mathbf{U}(\partial p)$, and since A has values in $\mathfrak{g}^c$ only the projection π of the latter contributes.

Definition 4.2 (The Operators of a Sequence)**.** Let $\mathsf{a} > 0$, and let $\mathsf{a}_0 \coloneqq \mathsf{a}_1 \coloneqq \mathsf{a}$ and $\mathsf{a}_{j+1} \coloneqq \mathsf{a}\mathsf{a}_j/(\mathsf{a}L^{-2} + \mathsf{a}_j)$.

(i) The *averages of the sequence* are the maps Q, from 1-forms on T_η to functions on the bonds of the layers, and Q', from 0-forms on T_η to functions on $\mathbf{B}$, with

$$(QA)(c) \coloneqq (Q_j(\mathbf{U})A)(c) \text{ for a bond } c \text{ of } \Gamma_j\ , \qquad (Q'\lambda)(y) \coloneqq (Q'_j(\mathbf{U})\lambda)(y) \text{ for } y \in \Gamma_j\ ,$$

where Q_0 and Q'_0 are the identity. The *constraint forms* are

$$\begin{aligned} \langle A, Q^*\mathsf{a}QA\rangle &\coloneqq \mathsf{a}\sum_{j=0}^{k}\sum_{c} a_j^2 \operatorname{tr}((QA)(c))^2\ , \\ \langle \lambda, Q'^*\mathsf{a}Q'\lambda\rangle &\coloneqq \sum_{j=0}^{k}\mathsf{a}_j\sum_{y\in\Gamma_j} a_j^2 \operatorname{tr}((Q'\lambda)(y))^2\ , \end{aligned} \tag{4.2}$$

the inner sum of the first over the positive bonds of Γ_j, and $Q^*\mathsf{a}Q$, $Q'^*\mathsf{a}Q'$ are the symmetric operators of these forms.

(ii) $\Delta'_{\mathsf{a}} \coloneqq \Delta_{\mathbf{U}} + Q'^*\mathsf{a}Q'$ on 0-forms, and $G' \coloneqq (\Delta'_{\mathsf{a}})^{-1}$.

(iii) $R \coloneqq I - G'Q'^*(Q'G'^2Q'^*)^{-1}Q'G'$ and $P \coloneqq I - R$, operators on 0-forms, where Q'^* is the transpose of Q' with respect to the pairing (1.5) and the pairing $\sum_{y\in\mathbf{B}} a_{j(y)}^4 \operatorname{tr}\mu(y)\mu'(y)$ of functions on $\mathbf{B}$.

(iv) $\Delta_{\mathsf{a}} \coloneqq \Delta(\mathbf{U}) + DRD^* + Q^*\mathsf{a}Q$ on 1-forms, and $G \coloneqq (\Delta_{\mathsf{a}})^{-1}$.

The inverses are taken on the forms which vanish outside the Dirichlet domain, and each operator is defined whenever the inverses in its definition exist.

These are the operators [5, (3.16), (3.18)–(3.27), pp. 393–395]. For a gauge field the pairings are real scalar products, R is the orthogonal projection onto the image of the kernel of Q' under the covariant Laplacian [5, (3.21), (3.25)], and DRD^* is the quadratic form of the gauge-fixing term. For a complex configuration the formulae are the definitions.

Definition 4.3 (Regular Pairs)**.** (i) A *cube of the class* of the sequence, of the index j, is a cube $\square$ which is a union of cubes of the grid of side Ma_j, has a side $s_\square$ with $10Ma_j \le s_\square \le 20Ma_j$, is contained in $\Omega_j \setminus \Omega_{j+2}$, and meets $\Omega_j \setminus \Omega_{j+1}$.

(ii) A gauge field U on T_η is *regular with the constant* α_0 if for every cube $\square$ of the class, of the index j, there is a gauge transformation u on $\square$ with values in G such that $U^u = e^{i\eta A}$ on the bonds of $\square$ with a 1-form A with values in $\mathfrak{g}$ and

$$|A| \le \frac{s_\square}{a_j}\,\alpha_0 a_j^{-1}\,, \qquad |\nabla_{\mathbf{1}} A| \le \frac{s_\square}{a_j}\,\alpha_0 a_j^{-2} \tag{4.3}$$

on $\square$. It is *regular to the second order* if in addition $|D^*_{\mathbf{1}} D_{\mathbf{1}} A| \le (s_\square/a_j)\alpha_0 a_j^{-3}$ on $\square$.

(iii) A configuration $\mathbf{U}$ on T_η is *regular with the constants* (α_0, α_1) if $\mathbf{U} = e^{i\eta A'}U$ with a gauge field U which is regular with the constant α_0 and a 1-form A' with values in $\mathfrak{g}^c$ such that

$$|A'(b)| \le \alpha_1 a_j^{-1}\,, \qquad |(\nabla_U A')(x)| \le \alpha_1 a_j^{-2} \tag{4.4}$$

for the bonds b and the sites x of Ω_j, $0 \le j \le k$. It is *regular to the second order* if U is and $|D^*_U D_U A'| \le \alpha_1 a_j^{-3}$ on Ω_j.

(iv) A pair $(\mathbf{U}, \mathbf{J})$ is *regular with the constants* $(\alpha_0, \alpha_1, \gamma)$ if $\mathbf{U}$ is regular with (α_0, α_1) and $|\mathbf{J}(b)| \le \gamma a_j^{-3}$ for the bonds b of Ω_j, $0 \le j \le k$.

These are the conditions [5, (3.35)–(3.38), p. 396]; the operators of this subsection do not involve the current, which enters with the operators of the Landau gauge.

4.2 Decay and Regularity

Theorem 4.4 (The Operators G', $(Q'G'^2Q'^*)^{-1}$, P, and G)**.** *Let* $\mathsf{a} = 1$. *There are numbers* M_1, R_1, $\delta_0 > 0$, $a_0 > 0$, *and* B_0, *and numbers* $B_0(\beta)$ *for* $0 \le \beta < 1$ *and* $B'_0(\epsilon, \beta)$ *for* $0 < \epsilon \le 1$ *and* $0 \le \beta < 1$, *all depending on* L *and* G *only, with the following properties. Let a sequence of domains with* $M \ge M_1$ *and* $R \ge R_1$ *be given, with a Dirichlet domain, and let* $\mathbf{U} = e^{i\eta A'}U$ *be regular with constants* (α_0, α_1) *such that* $M\alpha_0 + \alpha_1 \le a_0$.

(i) Existence. *The operators Δ'_{a}, $Q'G'^2Q'^*$, and Δ_{a} are invertible, so that G', R, P, and G are defined. For a gauge field, Δ'_{a}, G', $Q'G'^2Q'^*$, Δ_{a}, and G are symmetric and positive definite with respect to the scalar products of the forms with values in $\mathfrak{g}$.*

(ii) Analyticity. *For a fixed regular gauge field U the four operators are holomorphic functions of A' on the set* (4.4).

(iii) Covariance. *For a gauge transformation u of T_η with values in G, $G'(\mathbf{U}^u) = R(u)G'(\mathbf{U})R(u)^{-1}$, and the same holds for R, P, Δ_{a}, and G; the kernel of $(Q'G'^2Q'^*)^{-1}$ is conjugated by the values of u at the points of $\mathbf{B}$. All the bounds below are invariant under these transformations.*

(iv) Pointwise bounds. *Let $y \in \Gamma_j$, $y' \in \Gamma_{j'}$, $x \in \Delta(y)$, and let λ be a 0-form supported in $\Delta(y')$. Then*

$$|(G'\lambda)(x)| \le B_0 a_j^2 e^{-\delta_0 d(y,y')}|\lambda| \,, \qquad |(\nabla G'\lambda)(x)|,\ |(G'\nabla^*\lambda)(x)| \le B_0 a_j e^{-\delta_0 d(y,y')}|\lambda| \,, \tag{4.5}$$

and $|(\Delta_{\mathbf{U}} G'\lambda)(x)| \le B_0 e^{-\delta_0 d(y,y')}|\lambda|$.

(v) Hölder bounds. *Let $0 \le \beta \le \beta_0 < 1$ and $0 < \epsilon \le 1$, let ζ be a real function supported in $\tilde\Delta(y)$, and let λ be supported in $\tilde\Delta(y')$. Then*

$$\|\zeta\nabla G'\lambda\|_\beta,\ \|\zeta G'\nabla^*\lambda\|_\beta \le B_0(\beta_0) a_j^{1-\beta}(\|\zeta\|_\beta^{(j)} + |\zeta|)e^{-\delta_0 d(y,y')}|\lambda| \,, \tag{4.6}$$

$$|(\nabla G'\nabla^*\lambda)(x)| \le B'_0(\epsilon,0)e^{-\delta_0 d(y,y')}(\|\lambda\|_\epsilon^{(j')} + |\lambda|) \,, \tag{4.7}$$

$$\|\zeta\nabla G'\nabla^*\lambda\|_\beta \le B'_0(\epsilon,\beta_0) a_j^{-\beta}(\|\zeta\|_\beta^{(j)} + |\zeta|)e^{-\delta_0 d(y,y')}(\|\lambda\|_{\beta+\epsilon}^{(j')} + |\lambda|) \,, \tag{4.8}$$

the last for $\beta + \epsilon \le 1$.

(vi) Bounds in L^2. *For a function h supported in $\Delta(y)$ and λ supported in $\Delta(y')$, the L^2-norms of $hG'\lambda$, of $h\nabla G'\lambda$ and $hG'\nabla^*\lambda$, and of $h\nabla\nabla G'\lambda$, $h\nabla G'\nabla^*\lambda$, and $hG'\nabla^*\nabla^*\lambda$ are at most $B_0|h|e^{-\delta_0 d(y,y')}\|\lambda\|$ times a_j^2, a_j, and 1, respectively.*

(vii) The block inverse and the projection. *For $y \in \Gamma_j$ and $y' \in \Gamma_{j'}$ the kernel of $(Q'G'^2Q'^*)^{-1}$ with respect to the pairing of Definition 4.2*(iii) *satisfies*

$$|(Q'G'^2Q'^*)^{-1}(y,y')| \le B_0 a_j^{-4} a_{j'}^{-4} e^{-\delta_0 d(y,y')} \,, \tag{4.9}$$

and for $x \in \Delta(y)$, $x' \in \Delta(y')$ the kernels of P, ∇P, $P\nabla^*$, and $\nabla P\nabla^*$ with respect to (1.5) *are at most $B_0 a_{j'}^{-4} e^{-\frac{1}{2}\delta_0 d(y,y')}$ times 1, a_j^{-1}, a_j^{-1}, and a_j^{-2}, respectively.*

(viii) The operator G. *The bounds* (iv)–(vi) *hold for G in the place of G' and 1-forms J with values in $\mathfrak{g}^c$ in the place of λ, the gradients being those of Definition 1.9*(iii).

In (iv)–(viii) *the norm of a kernel or of a value is the bond norm of the matrix, or the norm of the linear map of $\mathfrak{g}^c$ which it defines, and d is the weighted distance of the sequence.*

This combines [5, Theorems 3.1–3.4, Corollaries 3.5, 3.6, pp. 397–408] with [5, (3.48), (3.49)]; at the trivial configuration it contains [3, Propositions 2.2, 2.3, 2.6]. The constants do not depend on the sequence, on the Dirichlet domain, on k, or on the torus, so that the theorem applies to the local sequences of the cubes in Section 4.3 and to sequences with a block size other than that of the localisation domains. A power of the spacing in a bound may be divided between the two points: by Lemma 1.14(ii), $a_j^\sigma e^{-\delta d(y,y')}$ is at most a constant times $a_{j'}^\sigma e^{-\delta' d(y,y')}$ for every $\delta' < \delta$ once RM is large, depending on σ, $\delta - \delta'$, and L.

4.3 Walks

The bounds of Theorem 4.4 are obtained from the same bounds for operators on single cubes, where the background can be made small by a gauge transformation, and the passage from the cubes to the lattice is an expansion of each inverse into a sum over walks through overlapping cubes [5, Sect. C]. We describe the cubes, the operators attached to them, and the expansions. In this subsection the Dirichlet domain of the given sequence is the torus.

Definition 4.5 (The Cubes of the Walks and Their Local Sequences)**.** Let R_0 and M_0 be powers of L, and let $M = KR_0M_0$ with a positive integer K which is a power of L.

(i) For $0 \le j \le k$ the family $\mathscr{D}_j$ consists of the closed cubes of side $2Ma_j$ whose centre is a corner of a cube of the grid of side Ma_j, which are contained in Ω_j, and which contain a cube of that grid which is not contained in Ω_{j+1}. We put $\mathscr{D} := \bigcup_j \mathscr{D}_j$, and for $\square \in \mathscr{D}_j$ and $n \ge 1$ we write $\tilde{\square}^n$ for the cube of side $(2+2n)Ma_j$ with the centre of $\square$.

(ii) The *local sequence* of $\square \in \mathscr{D}_j$ is the sequence of domains $\Omega_0(\square) \supset \cdots \supset \Omega_{j+1}(\square)$, and $\Omega_n(\square) := \emptyset$ for $n > j+1$, with

$$\Omega_{j+1}(\square) := \tilde{\square}^3 \cap \Omega_{j+1} \ , \qquad \Omega_j(\square) := \tilde{\square}^4 \ ,$$

and, for $0 \le n < j$, with $\Omega_n(\square)$ the cube with the centre of $\square$ for which

$$\operatorname{dist}(\Omega_n(\square)^c, \Omega_{n+1}(\square)) = 2R_0M_0a_n \ . \tag{4.10}$$

It is a sequence of domains with the block size M_0 and the separation R_0, and its Dirichlet domain is $\Omega_0(\square)$, which is contained in $\tilde{\square}^5$.

(iii) The *local operators* $G'_\square$, $C_\square := (Q'G'^2_\square Q'^*)^{-1}$, $P_\square := G'_\square Q'^* C_\square Q' G'_\square$, and $G_\square$ are the operators of Definition 4.2 for the local sequence of $\square$, at the same configuration $\mathbf{U}$.

By (4.10), $\operatorname{dist}(\Omega_n(\square)^c, \tilde{\square}^4) = 2R_0M_0a_n(L^{j-n}-1)/(L-1) < 2R_0M_0a_j \le 2Ma_j$, which gives $\Omega_0(\square) \subset \tilde{\square}^5$, and the boundaries of the cubes $\Omega_n(\square)$ lie on the boundaries of cubes of the grid of side M_0a_n, because M/M_0 and L are odd. The cubes of $\mathscr{D}$ cover the torus: a point of $\Omega_j \setminus \Omega_{j+1}$ lies in a cube of side $2Ma_j$ centred at a corner of the grid at the distance at least Ma_j from Ω_j^c, which is contained in Ω_j. A cube of $\mathscr{D}_j$ may reach into Ω_{j+1}, by one cube of the grid, and it does not meet Ω_{j+2} once $R \ge 2$. On such a cube the layers of the local sequence are therefore those of the given one, $\square \cap \Omega_{j+1}(\square) = \square \cap \Omega_{j+1}$ and $\square \subset \Omega_j(\square)$, so that the averages of the two sequences coincide on the forms supported in $\square$. The local sequences are those of [5, p. 408], and the local operators are controlled by Theorem 4.4 for these sequences.

Lemma 4.6 (A Partition of Unity)**.** *For every sequence of domains there are functions $h_\square \colon \mathbb{T} \to [0,1]$, $\square \in \mathscr{D}$, with the following properties.*

(i) $\sum_{\square \in \mathscr{D}} h_\square^2 = 1$*, and $h_\square$ vanishes outside the cube with the centre of $\square$ and the side $\frac{4}{3}Ma_j$, for $\square \in \mathscr{D}_j$.*

(ii) $h_\square$ is constant on each cube of the grid of side a_0, and $|h_\square(x) - h_\square(x')| \le C_h|x - x'|_\infty (Ma_j)^{-1}$ with an absolute constant C_h, with the corresponding bounds $C_h(Ma_j)^{-2}$ for its second differences.

(iii) $h_\square$ *is determined by the domains of the sequence inside* $\tilde\square^2$*, and the family is covariant under the translations, reflections, and permutations of the coordinates which map every grid to itself: if* θ *maps the sequence* $\{\Omega_j\}$ *to* $\{\theta\Omega_j\}$*, then the functions of the latter are* $h_{\theta\square} = h_\square \circ \theta^{-1}$*.*

Definition 4.7 (Commutators with the Partition of Unity)**.** For a function h on T_η, regarded as a multiplication operator on forms, and on functions on $\mathbf{B}$ through its values at the points of $\mathbf{B}$,

$$K'(h) \coloneqq [h, \Delta'_{\mathsf{a}}] \,, \qquad K(h) \coloneqq [h, \Delta(\mathbf{U}) + DD^* + Q^*\mathsf{a}Q] \,,$$
$$P_1(h) \coloneqq [DPD^*, h] \,, \qquad P_{\square,1}(h) \coloneqq [DP_\square D^*, h] \,.$$

So $\Delta'_{\mathsf{a}} h = h\Delta'_{\mathsf{a}} - K'(h)$ and $\Delta_{\mathsf{a}} h = h\Delta_{\mathsf{a}} - K(h) - P_1(h)$, which are [5, (3.88), (3.104)]. The operators $K'(h_\square)$ and $K(h_\square)$ are sums of difference operators of the first order with coefficients which are differences of $h_\square$ and of operators which couple the points of a block.

Theorem 4.8 (The Walk Expansions)**.** *There are numbers* R_0*,* M_0*,* K_1*, and* C_{w}*, which depend on* L *and* G *only, such that for* $M = KR_0M_0$ *with* $K \geq K_1$*, for every sequence of domains with the block size* M*, the separation* $R \geq R_1$*, and the torus as its Dirichlet domain, and for every configuration* $\mathbf{U}$ *as in Theorem 4.4, the following holds, with the constants* δ_0 *and* B_0 *of that theorem.*

(i) The first approximations. *Let* $G'_0 \coloneqq \sum_\square h_\square G'_\square h_\square$*,* $C_0 \coloneqq \sum_\square h_\square C_\square h_\square$*, and* $G_0 \coloneqq \sum_\square h_\square G_\square h_\square$*, the sums over* $\square \in \mathscr{D}$*. Then*

$$\Delta'_{\mathsf{a}} G'_0 = I - R' \,, \qquad R' \coloneqq \sum_\square K'(h_\square) G'_\square h_\square \,, \tag{4.11}$$

$$Q'G'^2Q'^*C_0 = I - R_C \,, \quad R_C \coloneqq -\sum_\square \Big(Q'(G'^2 - G'^2_\square)Q'^* h_\square + [Q'G'^2_\square Q'^*, h_\square] \Big) C_\square h_\square \,, \tag{4.12}$$

$$\Delta_{\mathsf{a}} G_0 = I - R_G \,, \quad R_G \coloneqq \sum_\square \Big(K(h_\square) + (DPD^* - DP_\square D^*)h_\square + P_{\square,1}(h_\square) \Big) G_\square h_\square \,. \tag{4.13}$$

(ii) The series. $G' = G'_0 \sum_{n\geq 0} R'^n$*,* $(Q'G'^2Q'^*)^{-1} = C_0 \sum_{n\geq 0} R_C^n$*, and* $G = G_0 \sum_{n\geq 0} R_G^n$*, each series converging in the norms of the bounds of Theorem 4.4 for its operator.*

(iii) The walks of G'*. For a sequence* $\omega = (\square_0, \dots, \square_n)$ *of cubes of* $\mathscr{D}$ *with* $\square_{i-1} \cap \square_i \neq \emptyset$*, of the length* $|\omega| \coloneqq n$*, let*

$$G'_\omega \coloneqq h_{\square_0} G'_{\square_0} h_{\square_0} \prod_{i=1}^{n} K'(h_{\square_i}) G'_{\square_i} h_{\square_i} \,. \tag{4.14}$$

Then $G' = \sum_\omega G'_\omega$*. Each* G'_ω *is a holomorphic function of the configuration which depends only on its values on the bonds of* $\tilde\square_0^5 \cup \cdots \cup \tilde\square_n^5$*, and it is covariant under gauge transformations as* G' *is.*

(iv) The bound of a factor and of a walk. *For* $\square \in \mathscr{D}_j$*,* $y, y' \in \square \cap \mathbf{B}$*,* $x \in \Delta(y)$*, and* λ *supported in* $\Delta(y')$*,*

$$|(K'(h_\square) G'_\square h_\square \lambda)(x)| \leq C_{\mathrm{w}} M^{-1} e^{-\delta_0 |y-y'|_1 / a_j} |\lambda| \,, \tag{4.15}$$

and for a walk ω, $y \in \square_0 \cap \Gamma_j$, $y' \in \square_n \cap \mathbf{B}$, $x \in \Delta(y)$, *and* λ *supported in* $\Delta(y')$,

$$|(G'_\omega \lambda)(x)| \le B_0 a_j^2 (C_{\mathrm{w}} M^{-1})^{|\omega|} e^{-\frac{1}{2}\delta_0 d(\omega,y,y')} |\lambda| \,, \tag{4.16}$$

where $d(\omega, y, y')$ *is the infimum of* $d(y, y_1) + d(y_1, y_2) + \cdots + d(y_n, y')$ *over the points* $y_i \in \square_i \cap \mathbf{B}$. *The corresponding bounds hold in the other norms of Theorem 4.4*(iv)–(vi).

(v) The remainders. *The kernels of* R' *and of* R_G *are bounded by* $C_{\mathrm{w}} M^{-1} e^{-\frac{1}{2}\delta_0 d(y,y')}$ *in the sense of* (4.15), *and that of* R_C *by* $C_{\mathrm{w}} M^{-1} a_{j'}^{-4} e^{-\frac{1}{2}\delta_0 d(y,y')}$ *for* $y \in \Gamma_j$, $y' \in \Gamma_{j'}$.

The parts (i)–(iv) are [5, (3.87)–(3.96), (3.104)–(3.106), Theorem 3.7, Corollary 3.8]. The remainders R_C and R_G are written here in a form in which the cancellation between the given operators and the local ones is visible; they equal those of [5, (3.95), (3.105)], which insert the indicator of $\tilde{\square}^1$ and the function $\zeta_\square$ in order to separate the terms with a small factor from an overall decay from those with a small commutator. The identities (4.11)–(4.13) use that the given operators and the local ones coincide on forms supported in $\square$, apart from the projections: for example $h_\square \Delta_{\mathsf{a}} G_\square h_\square = h_\square^2 - h_\square (DPD^* - DP_\square D^*) G_\square h_\square$, since Δ_{a} and the local operator differ by $D(P_\square - P)D^*$ there.

The remainders R_C and R_G still contain the operators G' and P of the whole lattice, and the expansions of the block inverse and of G into walks whose terms are products of local operators only [5, Theorems 3.9, 3.10] are obtained by inserting the walk expansion of G' into them. For these expansions the papers give a bound of a whole term and not of its factors [5, p. 413]. The main paper, however, compares the terms of two lattices by replacing one factor at a time, and for this it needs a term to be a product whose bound follows from bounds of the factors. We therefore state the expansions in this form. The bounds of the factors are bounds between three kinds of objects: sources, fields, and data on the determining set.

Definition 4.9 (Block Norms and Block Bounds)**.** Fix $0 < \beta < 1$. For $y \in \Gamma_j$ we define the following *block norms*:

(i) for a form f on T_η, regarded as a *source*, $|f|_y$ is the supremum of the components of f over $\Delta(y)$;

(ii) for a form u on T_η, regarded as a *field*, $N_y(u)$ is the supremum over $\Delta(y)$ of $a_j^{-1}|u| + |\nabla u|$, plus a_j^β times the supremum of $(|x - x'|_1 / a_j)^{-\beta} |R(\mathbf{U}(\gamma))(\nabla u)(x') - (\nabla u)(x)|$ over the pairs of sites $x \ne x'$ of $\tilde{\Delta}(y)$ and the shortest paths γ between them;

(iii) for a function μ on $\mathbf{B}$, or on the positive bonds of the layers, regarded as *data*, the block norm is $|\mu(y)|$, or the largest of the norms $|\mu(c)|$ over the bonds c with $c_- = y$.

Let $\mathcal{T}$ be a linear operator which maps objects of one of the three kinds to objects of one of the three kinds, let $X \subset \mathbb{T}$, and let $b \ge 0$, $p \in \mathbb{Z}$, and $\delta > 0$. We say that $\mathcal{T}$ has the *block bound* (b, p, δ) *in* X if $\mathcal{T}f$ depends only on the restriction of f to X, if $[\mathcal{T}f]_y = 0$ for every $y \in \mathbf{B} \setminus X$, and if for every f and every $y \in \mathbf{B} \cap X$

$$[\mathcal{T}f]_y \le b\, a_{j(y)}^p \sum_{y' \in \mathbf{B} \cap X} e^{-\delta d(y,y')} [f]_{y'} \,, \tag{4.17}$$

where $[\cdot]_y$ denotes the block norm of the kind of the object.

The exponent p counts the powers of the spacing which the operator carries: a propagator on sources has $p = 1$ as a map into fields, since the norm N_y measures a field through its first differences, and a difference operator of the first order from fields to sources has $p = 0$.

Lemma 4.10 (Composition of Block Bounds). *Let $\delta > 0$, $0 < \alpha < 1$, and $S \geq 0$, and let the sequence of domains satisfy $RM \geq r_1(\frac{1}{2}\alpha, \delta)$ and $\alpha\delta RM \geq 2S \log L$, with the number r_1 of Lemma 1.15. Let $\mathcal{T}_1, \dots, \mathcal{T}_n$ be operators such that $\mathcal{T}_i$ has the block bound (b_i, p_i, δ) in X_i and the product $\mathcal{T}_1 \cdots \mathcal{T}_n$ is defined, and suppose that $|p_i + p_{i+1} + \cdots + p_n| \leq S$ for $2 \leq i \leq n$. Then for every f and every $y \in \mathbf{B} \cap X_1$,*

$$[\mathcal{T}_1 \cdots \mathcal{T}_n f]_y \leq (c_1 L^S)^{n-1} \Big(\prod_{i=1}^{n} b_i \Big) a_{j(y)}^{p_1 + \cdots + p_n} \sum_{y' \in \mathbf{B} \cap X_n} e^{-(1-\alpha)\delta d_{\mathbf{X}}(y,y')} [f]_{y'} \ , \tag{4.18}$$

where $c_1 \coloneqq c_1(\frac{1}{2}\alpha, \delta)$ and $d_{\mathbf{X}}(y, y')$ is the infimum of $d(y, y_1) + d(y_1, y_2) + \cdots + d(y_{n-1}, y')$ over the points $y_i \in \mathbf{B} \cap X_i \cap X_{i+1}$.

Proof. Applying (4.17) to $\mathcal{T}_1$, then to $\mathcal{T}_2$, and so on, and writing $y_0 \coloneqq y$, we obtain

$$[\mathcal{T}_1 \cdots \mathcal{T}_n f]_{y_0} \leq \Big(\prod_i b_i \Big) \sum_{y_1, \dots, y_n} \Big(\prod_{i=1}^{n} a_{j(y_{i-1})}^{p_i} e^{-\delta d(y_{i-1}, y_i)} \Big) [f]_{y_n} \ ,$$

where y_i runs over $\mathbf{B} \cap X_i \cap X_{i+1}$ for $i < n$, because the bound of $\mathcal{T}_i$ contains the block norms of its argument at the points of X_i only and the block norms of $\mathcal{T}_{i+1}(\cdot)$ vanish outside X_{i+1}, and y_n runs over $\mathbf{B} \cap X_n$. We first collect the powers of the spacings at the point y_0. With $s_i \coloneqq p_i + \cdots + p_n$,

$$\prod_{i=1}^{n} a_{j(y_{i-1})}^{p_i} = a_{j(y_0)}^{s_1} \prod_{i=2}^{n} \Big(\frac{a_{j(y_{i-1})}}{a_{j(y_{i-2})}} \Big)^{s_i} \ ,$$

as one sees by collecting the exponent of each $a_{j(y_l)}$ on the right side, which is $s_{l+1} - s_{l+2} = p_{l+1}$. The quotient of the spacings of two points is $L^{j(y_{i-1}) - j(y_{i-2})}$, and by Lemma 1.14(ii) the difference of the scales of two points is at most $1 + d/(RM)$, with d their weighted distance. Since $|s_i| \leq S$ and $S \log L/(RM) \leq \frac{1}{2}\alpha\delta$, each quotient is at most $L^S e^{\frac{1}{2}\alpha\delta d(y_{i-2}, y_{i-1})}$. The summand is therefore at most $L^{S(n-1)} a_{j(y_0)}^{s_1} \prod_i e^{-(1-\frac{1}{2}\alpha)\delta d(y_{i-1}, y_i)} [f]_{y_n}$. By the definition of $d_{\mathbf{X}}$, $\sum_i d(y_{i-1}, y_i) \geq d_{\mathbf{X}}(y_0, y_n)$, so that the product of the exponentials is at most $e^{-(1-\alpha)\delta d_{\mathbf{X}}(y_0, y_n)} \prod_i e^{-\frac{1}{2}\alpha\delta d(y_{i-1}, y_i)}$. In the last product we bound the factor with $i = n$ by one and sum over $y_{n-1}, \dots, y_1$ in this order, each sum being at most $c_1(\frac{1}{2}\alpha, \delta)$ by (1.17). *q.e.d.*

Theorem 4.11 (Walks as Products of Local Factors). *Under the hypotheses of Theorem 4.8, and with a constant C_1 which depends on L, G, and β only:*

(i) The factors of the walks of G'. *For $\square \in \mathscr{D}_j$, the operator $h_\square G'_\square h_\square$ maps sources to fields with the block bound $(C_1, 1, \delta_0)$ in $\square$, and $K'(h_\square) G'_\square h_\square$ maps sources to sources with the block bound $(C_1 M^{-1}, 0, \delta_0)$ in $\square$. If $G'_\square$ is replaced in them by an operator $\mathcal{O}$ with $N_y(\mathcal{O}f) \leq \theta a_j e^{-\delta d(y,y')} |f|_{y'}$ for all $y, y' \in \mathbf{B} \cap \square$ and all sources f supported in $\Delta(y')$, the two operators have the block bounds $(C_1\theta, 1, \delta)$ and $(C_1 M^{-1}\theta, 0, \delta)$ in $\square$.*

(ii) The block inverse and the operator on 1-forms. *There are sets $\mathscr{W}_C$ and $\mathscr{W}_G$ and, for each of their elements ω, a number $|\omega| \geq 0$, a sequence of cubes $\square_0, \dots, \square_m$ of $\mathscr{D}$ in which consecutive cubes intersect, and an operator W_ω, such that*

$$(Q' G'^2 Q'^*)^{-1} = \sum_{\omega \in \mathscr{W}_C} W_\omega \ , \qquad G = \sum_{\omega \in \mathscr{W}_G} W_\omega \ , \tag{4.19}$$

and such that each W_ω is a product $\mathcal{T}_1 \cdots \mathcal{T}_n$ of $n \leq C_1(|\omega| + 1)$ operators with the following properties. Each $\mathcal{T}_i$ has the form $E_i \mathcal{O}_i E'_i$, where $\mathcal{O}_i$ is a local operator G'_X, C_X,

or G_X of the local sequence of a cube X which is a cube of the walk or its enlargement $\tilde{\square}_l^2$, or is absent, and where E_i and E_i' are products of at most C_1 operators of the following kinds: the multiplication by a function which is determined by the sequence of domains; the commutator of such a multiplication with Δ_{a}' or with $\Delta(\mathbf{U})+DD^*+Q^*\mathsf{a}Q$; one of the operators D, D^*, Q, Q', Q^*, Q'^*; and the difference $\Delta_{\mathsf{a}}'-\Delta_{\mathsf{a},X}'$ of the scalar operators of the given sequence and of the local sequence of X. Each $\mathcal{T}_i$ has a block bound $(b_i, p_i, \frac{1}{2}\delta_0)$ in a cube of the walk, between objects of the three kinds, the sums $p_i + \cdots + p_n$ are at most 4 in absolute value, and

$$\prod_{i=1}^{n} b_i \le C_1 (C_{\mathrm{w}} M^{-1})^{|\omega|} , \qquad \sum_{i=1}^{n} p_i = -4 \text{ for } \mathscr{W}_C \text{ and } 1 \text{ for } \mathscr{W}_G . \tag{4.20}$$

Moreover, if in one $\mathcal{T}_i$ the local operator $\mathcal{O}_i$ is replaced by an operator which satisfies, between the blocks of X, θ times the bounds which Theorem 4.4 gives for $\mathcal{O}_i$, with a rate $\delta \ge \frac{1}{2}\delta_0$, then $\mathcal{T}_i$ has the block bound $(C_1\theta b_i, p_i, \frac{1}{2}\delta_0)$.

(iii) Naturality. *The sets $\mathscr{W}_C$ and $\mathscr{W}_G$, the numbers $|\omega|$, the cubes of a walk, and for each factor its kind, its cube, and its function are determined by the sequence of domains as a sequence of subsets of the torus and by M, R_0, and M_0. They do not depend on the configuration or on the spacing η of the fine lattice, and they are mapped to those of the image sequence by the translations, reflections, and permutations of the coordinates which map every grid to itself.*

(iv) Analyticity and locality. *Each W_ω is a holomorphic function of the configuration on the set of Theorem 4.4, it depends only on its values on the bonds of $\tilde{\square}_0^5 \cup \cdots \cup \tilde{\square}_m^5$, and it is covariant under gauge transformations as the operator which it expands. The block bounds of the factors hold on that whole set.*

The term (4.14) of a walk of G' is the product of $h_{\square_0} G'_{\square_0} h_{\square_0}$ and of the operators $K'(h_{\square_i}) G'_{\square_i} h_{\square_i}$, so that (i) and Lemma 4.10 with $S = 1$ give (4.16) in the supremum norm and in the norm N_y, which contains the Hölder norm of the first differences. In the same way (ii) and the lemma with $S = 4$ give the bounds of the terms of the other two expansions [5, (3.99), (3.108)], in the distance through the cubes of the walk, the constant $c_1 L^4$ of each factor being absorbed into C_{w}; and they give the same bound, multiplied by $C_1\theta$, for the product in which one local operator is replaced by an operator θ times as large. This last statement is the one on which the comparison of two lattices in the main paper rests.

4.4 The Operators of the Landau Gauge

The background field of a step is the minimiser of the action with prescribed averages, and to second order the problem is linear: one minimises $\frac{1}{2}\langle A, \Delta(\mathbf{U})A\rangle$ among the 1-forms with $QA = B$. The functional and the constraint are invariant under the gauge transformations $A \mapsto A - D\lambda$ with $Q'\lambda = 0$, by Proposition 2.12(v), but only up to terms which contain the current, and the gauge is fixed by the condition $RD^*A = 0$. The solution operator of this problem and of a variant of it, in which the constraint is corrected to the second order, are the operators through which the background field and the covariance of the fluctuation are written in the later sections [5, Sect. D]. In the formulae the current of the configuration appears explicitly, and following [9, p. 261] we define the operators at a pair, with the variable $\mathbf{J}$ in its place.

Definition 4.12 (The Operators at a Pair)**.** Let $(\mathbf{U}, \mathbf{J})$ be a pair on T_η for which the operators of Definition 4.2 exist. For a 1-form A let $\lambda_A \coloneqq G'RD^*A$ and $\mathcal{P}A \coloneqq A - D\lambda_A$.

(i) $\Delta'_\pi = \Delta'_\pi(\mathbf{U}, \mathbf{J})$ is the symmetric operator on 1-forms with $\langle A, \Delta'_\pi A\rangle = \langle X(A), \mathbf{J}\rangle$, where $X(A)$ is the 1-form with

$$X(A)(b) := i[\lambda_A(b_-), A(b)] - i[A(b), R(\mathbf{U}(b))\lambda_A(b_+)] - i[\lambda_A(b_-), (D\lambda_A)(b)] \tag{4.21}$$

on the positive bonds b.

(ii) $G_\pi := (\Delta_{\mathsf{a}} - \Delta'_\pi)^{-1}$ and $H := G_\pi Q^*(QG_\pi Q^*)^{-1}$, where Q^* is the transpose of Q with respect to (1.5) and the pairing $\sum_c a^4_{j(c)} \operatorname{tr} \mu(c)\mu'(c)$ of functions on the positive bonds c of the layers.

(iii) For a 1-form A let $\mathsf{C}^{(2)}(A)$ be the function on the bonds of the layers which on a bond c of Γ_j equals a_j^{-1} times the part of the second order in A of the remainder $C_j(\mathbf{U}; a_j A)(c)$ of Definition 2.11. Then $\Delta^{(2)} = \Delta^{(2)}(\mathbf{U}, \mathbf{J})$ is the symmetric operator with $\langle A, \Delta^{(2)} A\rangle = 2\langle H\mathsf{C}^{(2)}(A), \mathbf{J}\rangle$, and $\Delta^{(2)}_\pi := \mathcal{P}^T \Delta^{(2)} \mathcal{P}$, with the transpose $\mathcal{P}^T = I - DRG'D^*$.

(iv) $G_1 := (\Delta_{\mathsf{a}} - \Delta'_\pi - \Delta^{(2)}_\pi)^{-1}$ and $H_1 := G_1 Q^*(QG_1Q^*)^{-1}$.

These are [5, (3.119)–(3.129), (3.134), (3.135)] with $\mathbf{J}$ in the place of the current. The map $\mathcal{P}$ takes a 1-form to the gauge $RD^*A = 0$ by a transformation with $Q'\lambda = 0$: indeed $Q'G'R = 0$ and $R^2 = R$ by the definition of R, so that $Q'\lambda_A = 0$, $\Delta_{\mathbf{U}}\lambda_A = RD^*A$, and $RD^*\mathcal{P}A = 0$.

Theorem 4.13 (The Operators of the Landau Gauge)**.** *Let* $\mathsf{a} = 1$. *There are numbers* $a_2 > 0$, $\delta_1 > 0$, B_1, *and* $B_1(\beta)$, $B'_1(\epsilon, \beta)$ *as in Theorem 4.4, which depend on* L *and* G *only, such that for every sequence of domains with* $M \geq M_1$ *and* $R \geq R_1$ *and every regular pair* $(\mathbf{U}, \mathbf{J})$ *with constants* $M\alpha_0 + \alpha_1 + \gamma \leq a_2$ *the following holds.*

(i) Existence and analyticity. *The operators of Definition 4.12 exist, they are holomorphic in* $(A', \mathbf{J})$ *for a fixed regular gauge field* U, *and*

$$G_\pi = \sum_{n\geq 0} G(\Delta'_\pi G)^n\,, \qquad G_1 = \sum_{n\geq 0} G((\Delta'_\pi + \Delta^{(2)}_\pi)G)^n\,, \tag{4.22}$$

with the operator G *of Definition 4.2, the series converging in the norms of* (iii).

(ii) The vertices. *For* 1*-forms* A_1 *supported in* $\Delta(y)$, $y \in \Gamma_j$, *and* A_2 *supported in* $\Delta(y')$, $y' \in \Gamma_{j'}$,

$$|\langle A_1, \Delta'_\pi A_2\rangle| \leq B_1\gamma(\|D^*A_1\|_{L^1} + a_j^{-1}\|A_1\|_{L^1})e^{-\frac{1}{2}\delta_0 d(y,y')}(|D^*A_2| + a_{j'}^{-1}|A_2|)\,, \tag{4.23}$$

where $\|f\|_{L^1} := \sum_x \eta^4 |f(x)|$, *and for a bond* b *of* $\Delta(y)$, $|(\Delta^{(2)}A)(b)| \leq B_1\gamma a_j^{-2}|A|$ *with the supremum of* A *over the blocks* $\Delta(y'')$ *with* $d(y, y'') \leq 2$.

(iii) Bounds. *The operators* G_π *and* G_1 *satisfy the bounds* (iv)–(vi) *of Theorem 4.4 for* 1*-forms, with the constants* B_1, $B_1(\beta)$, $B'_1(\epsilon, \beta)$, *and* δ_1, *except the bound of the covariant Laplacian. For bonds* c *of* Γ_j *and* c' *of* $\Gamma_{j'}$,

$$|(QG_\pi Q^*)^{-1}(c, c')| \leq B_1 a_j^{-2} a_{j'}^{-4} e^{-\delta_1 d(c_-, c'_-)}\,, \tag{4.24}$$

and the same holds with G_1. *For* $x \in \Delta(y)$, $y \in \Gamma_j$, *a bond* c' *of* $\Gamma_{j'}$, *a real function* ζ *supported in* $\tilde{\Delta}(y)$, *and* $0 \leq \beta \leq \beta_0 < 1$, *the kernels of* H *and of* H_1 *satisfy*

$$\begin{aligned} |H(x, c')| + a_j|\nabla H(x, c')| &\leq B_1 a_{j'}^{-4} e^{-\frac{1}{2}\delta_1 d(y, c'_-)}\,, \\ \|\zeta \nabla H(\cdot, c')\|_\beta &\leq B_1(\beta_0) a_j^{-1-\beta}(\|\zeta\|^{(j)}_\beta + |\zeta|) a_{j'}^{-4} e^{-\frac{1}{2}\delta_1 d(y, c'_-)}\,. \end{aligned} \tag{4.25}$$

(iv) The current of the configuration. *If* $\mathbf{J} = J(\mathbf{U})$*, then* $\Delta(\mathbf{U}) - \Delta'_\pi = \mathcal{P}^T \Delta(\mathbf{U}) \mathcal{P}$*, the form* $\langle A, (\Delta(\mathbf{U}) - \Delta'_\pi) A \rangle$ *is invariant under* $A \mapsto A - D\lambda$ *for* $Q'\lambda = 0$*, and*

$$RD^* G_\pi Q^* = 0 \,, \qquad QG_\pi DR = 0 \,, \tag{4.26}$$

so that $RD^*H = 0$*, while* $QH = I$ *at every pair by the definition of* H*; the same identities hold for* G_1 *and* H_1*. If moreover* $\mathbf{U}$ *is a gauge field, then* G_π*,* G_1*,* $QG_\pi Q^*$*, and* QG_1Q^* *are symmetric and positive definite, and* HB *is the unique minimiser of* $\frac{1}{2}\langle A, \Delta(\mathbf{U}) A \rangle$ *among the* 1*-forms with values in* $\mathfrak{g}$ *which satisfy* $QA = B$ *and* $RD^*A = 0$*.*

This is [5, Theorem 3.12] with [5, (3.124), (3.126), (3.130)–(3.138)]. The identities (4.26) rest on the gauge invariance of the action and do not hold at a pair whose second entry is not the current of the first. The bound (4.24) is [5, (3.132)], which is printed there without a proof.

4.5 The Propagator of the Fluctuation

After the background field has been split off, the fluctuation is a 1-form with vanishing averages in the gauge $RD^*A = 0$, and its Gaussian measure has the quadratic form $\Delta(\mathbf{U}) - \Delta'_\pi - \Delta^{(2)}_\pi$ on this subspace. Its covariance is expressed through G_1 [5, (3.139)–(3.153)].

Definition 4.14 (The Projection and the Covariance of the Constrained Fluctuation)**.** At a pair $(\mathbf{U}, \mathbf{J})$ for which the operators of Definition 4.12 exist,

$$\mathfrak{P} \coloneqq I - G_1 Q^* (QG_1Q^*)^{-1} Q - G_1 DRD^* \,, \qquad \mathfrak{G} \coloneqq G_1 - G_1 DRD^* G_1 - G_1 Q^* (QG_1Q^*)^{-1} QG_1 \,. \tag{4.27}$$

Theorem 4.15 (The Propagator of the Fluctuation)**.** *Under the hypotheses of Theorem 4.13:*

(i) $\mathfrak{P}$ *and* $\mathfrak{G}$ *are holomorphic in* $(A', \mathbf{J})$*, and* $\mathfrak{G}$ *satisfies the bounds* (iv)–(vi) *of Theorem 4.4 for* 1*-forms, except that of the covariant Laplacian, with the constants of Theorem 4.13.*

(ii) If $\mathbf{J} = J(\mathbf{U})$*, then*

$$RD^*G_1 = RG'D^* \,, \qquad G_1 DR = DG'R \,, \qquad RD^*G_1DR = R \,, \tag{4.28}$$

and consequently $Q\mathfrak{P} = 0$*,* $RD^*\mathfrak{P} = 0$*,* $\mathfrak{P}^2 = \mathfrak{P}$*,* $\mathfrak{P}A = A$ *for every* A *with* $QA = 0$ *and* $RD^*A = 0$*, and* $\mathfrak{G} = \mathfrak{P}G_1 = G_1\mathfrak{P}^T$*.*

(iii) If moreover $\mathbf{U}$ *is a gauge field, then* $\mathfrak{G}$ *is symmetric and positive semidefinite, its range is the space* $\mathcal{S}$ *of the* 1*-forms with values in* $\mathfrak{g}$ *which satisfy* $QA = 0$ *and* $RD^*A = 0$*, and for every* 1*-form* J *with values in* $\mathfrak{g}$*,*

$$\int_{\mathcal{S}} e^{\langle A, J \rangle} \, \mathrm{d}\mu(A) = e^{\frac{1}{2}\langle J, \mathfrak{G} J \rangle} \,, \tag{4.29}$$

where μ *is the Gaussian probability measure on* $\mathcal{S}$ *with the density proportional to* $\exp(-\frac{1}{2}\langle A, (\Delta(U) - \Delta'_\pi - \Delta^{(2)}_\pi) A \rangle)$ *with respect to the Lebesgue measure of* $\mathcal{S}$*.*

This is [5, Theorem 3.13] with [5, (3.147)–(3.153)]. In (iii) the quadratic form is positive definite on $\mathcal{S}$ because on $\mathcal{S}$ it equals $\langle A, G_1^{-1} A \rangle$.

4.6 The Dependence on the Sequence

A term of the density is computed with the operators of its own sequence of domains, and different terms have different sequences. Where two sequences agree, their operators agree up to an error which decays in the distance to the region in which the sequences differ [5, Theorem 3.14, pp. 426–427]. We state this for two sequences which agree on a region of any shape, which is the form in which the localisation of the later sections uses it.

Theorem 4.16 (Change of the Sequence)**.** *Let $\{\Omega_j\}$ and $\{\Omega'_j\}$ be two sequences of domains with the same block size and separation, as in Theorem 4.13, and let $W \subset \mathbb{T}$ be a union of cubes of the grid of side $M\eta$ such that $\Omega_j \cap W = \Omega'_j \cap W$ for all j. Let $(\mathbf{U}, \mathbf{J})$ be a pair which is regular with the constants of Theorem 4.13 for both sequences. Let $\mathcal{A}$ and $\mathcal{A}'$ be the operators of one of the kinds G', $(Q'G'^2Q'^*)^{-1}$, P, G, G_π, G_1, $(QG_\pi Q^*)^{-1}$, $(QG_1Q^*)^{-1}$, H, H_1, $\mathfrak{G}$ for the two sequences. Then, for points y, y' of the determining sets which lie in W, the difference $\mathcal{A} - \mathcal{A}'$ satisfies every bound which Theorems 4.4, 4.13, and 4.15 state for $\mathcal{A}$ at the localisations y, y', with a constant which depends on L and G only in the place of the constant of the bound, with half the decay rate, and with the additional factor*

$$\exp\Big(-\frac{1}{2}\delta_1(d(y, W^c) + d(y', W^c))\Big) \,, \tag{4.30}$$

where $d(y, W^c)$ is the infimum of the weighted distance of y from the points of the complement of W, which is the same for both sequences.

For $W = \Omega_k \cap \Omega'_k$ and points of the lattice $T^{(k)}$ this is [5, Theorem 3.14], whose factor [5, (3.154)] is the exponential of the smallest length of a path from y to y' through the complement of W. A path from a point of W which leaves W and stays in W until then has the same weighted length for both sequences, so that $d(y, W^c)$ is well defined.

5 Regular Fields and Gauge Conditions

The fields over which a step integrates have prescribed averages on the determining set, and two such fields which differ by a gauge transformation which is trivial on the determining set are equivalent. To compute with them one chooses a representative in each class. Two choices are used. The *axial gauge* is algebraic: the representative is found by an explicit formula, block by block, and this is the gauge in which the variables of the construction are defined. It gives, however, no control of the size of a field in terms of its plaquettes beyond a bound which does not improve with the scale. The *Landau gauge* relative to a background is the gauge of the operators of Section 4, in which a regular field differs from the background by a fluctuation of the natural size, and the passage from the first gauge to the second is the main result of [6]. Throughout this section a sequence of domains with the block size M and the separation R is given.

5.1 Fields with Given Averages

Definition 5.1 (Regular Fields and Data)**.** (i) For $\alpha_0 > 0$ the set $\mathfrak{U}(\alpha_0)$ consists of the gauge fields U on T_η with

$$|U(\partial p) - \mathbf{1}| \le \alpha_0 \Big(\frac{\eta}{a_j}\Big)^2 \,, \qquad |J(U)(b)| \le \alpha_0 a_j^{-3} \,, \tag{5.1}$$

for the plaquettes p and the bonds b of Ω_j of the scale 0, for $0 \le j \le k$.

(ii) *Data* on the determining set are a gauge field V on the bonds of the layers, that is, a map which assigns an element of G to every bond of every layer, compatibly with the reversal of bonds. The set $\mathfrak{V}(V)$ consists of the gauge fields U on T_η with $M^j(U)(c) = V(c)$ for the bonds c of Γ_j, $0 \le j \le k$.

(iii) $\mathcal{G}_{\mathbf{B}}$ is the group of the gauge transformations $u\colon T_\eta \to G$ with $u(y) = \mathbf{1}$ for $y \in \mathbf{B}$.

These are the spaces [6, (1.7)–(1.14), pp. 77–78], with the current (1.10) in the second condition. By Lemma 2.5(iii) and Lemma 1.8(ii), $\mathfrak{U}(\alpha_0)$ is invariant under all gauge transformations with values in G, and $\mathfrak{V}(V)$ under those of $\mathcal{G}_{\mathbf{B}}$.

Inside a block the points are reached from the centre along paths, and the gauge conditions are conditions on the transports along them. The construction uses two kinds of transports: along one fixed path, which is simpler but distinguishes an order of the coordinates, and the mean over the paths of all orders [9, p. 253].

Definition 5.2 (Transports in a Block)**.** Let $1 \le n \le k$, let $\mathbf{U}$ be a configuration on $T^{(n-1)}$, $y \in T^{(n)}$, and $x \in B(y)$. The *axial transport* $\mathbf{U}_{\mathrm{ax}}(y,x)$ is $\mathbf{U}(\Gamma)$ for the path Γ of $\mathbf{G}(y,x)$ which traverses the coordinates in increasing order, and the *mean transport* $\mathbf{U}_{\mathrm{m}}(y,x)$ is the transport $\mathbf{U}(y,x)$ of Definition 2.7(ii). A *transport rule* is one of the symbols $\mathbf{ax}$ and $\mathbf{m}$, fixed throughout an argument and written $\mathfrak{t}$. For both rules $\mathbf{U}^u_{\mathfrak{t}}(y,x) = u(y)\mathbf{U}_{\mathfrak{t}}(y,x)u(x)^{-1}$.

5.2 The Axial Gauge

For $y \in \Gamma_j$ and a point x_0 of the block $B^j(y)$ there is a unique chain $x_0, x_1, \dots, x_j = y$ with $x_n \in T^{(n)}$ and $x_n \in B(x_{n+1})$, the *chain* of x_0.

Definition 5.3 (Axial Gauges)**.** Let U_0 and U be gauge fields on T_η for which the averages and the transports below are defined. The field U is *in the axial gauge of* U_0, with respect to the rule $\mathfrak{t}$, if

$$M^n(U)_{\mathfrak{t}}(x_{n+1}, x_n) = M^n(U_0)_{\mathfrak{t}}(x_{n+1}, x_n) \tag{5.2}$$

for every $1 \le j \le k$, every $y \in \Gamma_j$, every chain in $B^j(y)$, and every $0 \le n < j$. It is *in the axial gauge* if this holds with the right side replaced by $\mathbf{1}$.

With the axial transports these are the conditions [6, (1.15), p. 78] and [6, (1.19), (1.20), p. 79]; the latter are written there as a product along the path of the relative variables $M^n(U)M^n(U_0)^{-1}$ transported to the centre, which telescopes to (5.2).

Lemma 5.4 (The Representative in the Axial Gauge)**.** *Let U_0 and U be as in Definition 5.3. There is exactly one $u \in \mathcal{G}_{\mathbf{B}}$ for which U^u is in the axial gauge of U_0. On the chain of a point $x_0 \in B^j(y)$, $y \in \Gamma_j$, it is given by $u(x_j) = \mathbf{1}$ and*

$$u(x_n) = M^n(U_0)_{\mathfrak{t}}(x_{n+1}, x_n)^{-1}\, u(x_{n+1})\, M^n(U)_{\mathfrak{t}}(x_{n+1}, x_n)\,, \qquad n = j-1, \dots, 0\,. \tag{5.3}$$

In particular u on $B^j(y)$ depends only on U and U_0 on the bonds with both ends in $B^j(y)$, it depends on them analytically, and if w is a gauge transformation which is constant on each of the blocks $\Delta(y)$, $y \in \mathbf{B}$, then the representative of U^w in the axial gauge of U_0^w is $(U^u)^w$.

Proof. By Lemma 2.5(iii), $M^n(U^u)(c) = u(c_-)M^n(U)(c)u(c_+)^{-1}$ for the bonds c of $T^{(n)}$, so that $M^n(U^u)$ is the gauge transform of $M^n(U)$ by the restriction of u to $T^{(n)}$, and by Definition 5.2 its transport from x_{n+1} to x_n is $u(x_{n+1})M^n(U)_{\mathfrak{t}}(x_{n+1}, x_n)u(x_n)^{-1}$. The condition (5.2) for U^u is therefore equivalent to (5.3), which determines u at the points of $T^{(n)}$ in the

block from its values at the points of $T^{(n+1)}$, beginning with $u(y) = \mathbf{1}$. A point of $T^{(n+1)}$ in the block is the centre of its own block of order one, where the transport is trivial, so that the values assigned to it at the different levels agree. The transformation so defined equals $\mathbf{1}$ at the points of $\mathbf{B}$. The locality and the analyticity are those of the averages and of the transports. If w is constant on the blocks, then $M^n(U^w)$ and $M^n(U_0^w)$ are the transforms of $M^n(U)$ and $M^n(U_0)$ by the same constant in each block, both sides of (5.2) are conjugated by it, and the condition is preserved. *q.e.d.*

The next statement is the only quantitative property of the axial gauge. It is [6, Lemma 1, p. 79], for one level and for either transport rule.

Lemma 5.5 (Smallness in the Axial Gauge)**.** *There are numbers C_{ax} and $c_{\mathrm{ax}} > 0$, which depend on L and G only, with the following property. Let $1 \le n \le k$, let c be a bond of $T^{(n)}$, and let V_0 and V be gauge fields on $T^{(n-1)}$ such that*

(i) $|V_0(\partial p) - \mathbf{1}| \le \alpha_0 L^{-2}$ *and* $|V(\partial p) - \mathbf{1}| \le \alpha_0 L^{-2}$ *for the plaquettes p of $T^{(n-1)}$ with corners in $B(c_-) \cup B(c_+)$;*

(ii) $V_{\mathrm{t}}(y, x) = V_{0,\mathrm{t}}(y, x)$ *for $y \in \{c_-, c_+\}$ and $x \in B(y)$;*

(iii) $|M(V)(c)M(V_0)(c)^{-1} - \mathbf{1}| \le \alpha_1$*;*

with $\alpha_0 + \alpha_1 \le c_{\mathrm{ax}}$. Then $|V(b)V_0(b)^{-1} - \mathbf{1}| \le C_{\mathrm{ax}}\alpha_0 + \alpha_1$ for every bond b of $T^{(n-1)}$ with both ends in $B(c_-) \cup B(c_+)$.

5.3 The Landau Gauge

Definition 5.6 (The Landau Gauge and Averaged Gauge Transformations)**.** Let $\mathbf{U}_0$ be a configuration on T_η for which the operator $R = R(\mathbf{U}_0)$ of Definition 4.2 exists.

(i) A configuration $e^{i\eta A}\mathbf{U}_0$, with a 1-form A with values in $\mathfrak{g}^c$, is *in the Landau gauge of* $\mathbf{U}_0$ if $RD^*_{\mathbf{U}_0}A = 0$.

(ii) For a gauge transformation v of $T^{(n-1)}$, a configuration $\mathbf{V}_0$ on $T^{(n-1)}$, and $y \in T^{(n)}$, let

$$\overline{v}(y) \coloneqq v(y) \exp\Big(\frac{1}{L^4} \sum_{x \in B(y)} \mathrm{Log}(v(y)^{-1} R(\mathbf{V}_{0,\mathrm{t}}(y, x))v(x)) \Big) . \tag{5.4}$$

The *average of order j* of a gauge transformation u of T_η relative to $\mathbf{U}_0$ is the gauge transformation $N^j_{\mathbf{U}_0}(u)$ of $T^{(j)}$ obtained by applying (5.4) j times, at the step n with $\mathbf{V}_0 = M^{n-1}(\mathbf{U}_0)$. The transformation u is *admissible* if $N^j_{\mathbf{U}_0}(u)(y) = \mathbf{1}$ for all $y \in \Gamma_j$, $0 \le j \le k$.

The condition (i) is [6, (1.27), p. 80], and the averages of (ii) are [4, (78)–(80), p. 30]; the linear part of $\frac{1}{i}\mathrm{Log}\, N^j_{\mathbf{U}_0}(e^{i\lambda})$ in λ is $Q'_j(\mathbf{U}_0)\lambda$. The admissible transformations take the place of the group $\mathcal{G}_{\mathbf{B}}$: they are not a group, and for this reason the Landau representative is defined by starting from the representative in the axial gauge [6, pp. 80–81].

An admissible transformation is not trivial on the determining set, so that it changes the averages of a field on the layers. What it preserves is the average relative to the background, up to an explicit correction on the bonds which join two layers, and this is the form in which the constraint of a step is carried to the Landau gauge [6, (1.30), (1.31), pp. 81–82].

Definition 5.7 (The Relative Average of a Sequence and the Constraint Field)**.** Let $RM \geq 2L$.

(i) Let $\mathbf{U}$ be a configuration on T_η for which the operators of Definition 4.2(i) are defined, and let A be a 1-form with values in $\mathfrak{g}^c$. For a bond c of Γ_j put $\mathsf{C}(A)(c) \coloneqq a_j^{-1} C_j(\mathbf{U}; a_j A)(c)$, with $C_0 \coloneqq 0$, and

$$\mathsf{X}(A) \coloneqq QA + \mathsf{C}(A) \,, \tag{5.5}$$

a function on the bonds of the layers, so that (2.12) reads $M^j(e^{i\eta A}\mathbf{U})(c) = v_j(A)(c_-)\, e^{ia_j \mathsf{X}(A)(c)}\, M^j(\mathbf{U})(c)\, v_j(A)(c_+)^{-1}$.

(ii) Let V_0 and V be maps which assign an element of G^c to every bond of every layer, compatibly with the reversal of bonds. For $1 \leq j \leq k$ and an end y of a bond of Γ_j put $m_j(y) \coloneqq \mathbf{1}$ if $y \in \Omega_j$, and otherwise

$$m_j(y) \coloneqq \exp\Big(\frac{1}{L^4} \sum_{x \in B(y)} \mathrm{Log}(V_{\mathrm{t}}(y,x) V_{0,\mathrm{t}}(y,x)^{-1})\Big) \,, \tag{5.6}$$

where $B(y) \subset T^{(j-1)}$ is the block of y and the transports are those of the restrictions of V and V_0 to the bonds with both ends in $B(y)$, which are bonds of Γ_{j-1}; put $m_0 \coloneqq \mathbf{1}$. The *constraint field* of V relative to V_0 is the function $\mathsf{b}(V, V_0)$ on the positive bonds of the layers with

$$\mathsf{b}(V, V_0)(c) \coloneqq \frac{1}{ia_j} \mathrm{Log}(m_j(c_-)^{-1}\, V(c)\, m_j(c_+)\, V_0(c)^{-1}) \tag{5.7}$$

for a bond c of Γ_j, defined when the arguments of the logarithms lie in the domain of the chart.

An end y of a bond of Γ_j which is not in Ω_j is a point of $T^{(j)}$ within the ℓ^∞-distance a_j of Ω_j, and its block consists of points of $T^{(j-1)}$ within the distance $\frac{3}{2}a_j = \frac{3}{2}La_{j-1}$ of Ω_j which do not lie in Ω_j, because Ω_j is a union of cubes of the grid of side a_j; by (1.13) and $RM \geq 2L$ they lie in Ω_{j-1}, so that the bonds between them are bonds of Γ_{j-1}. On a bond with both ends in Γ_j the constraint field is $(ia_j)^{-1}\mathrm{Log}(V(c)V_0(c)^{-1})$.

Theorem 5.8 (From the Axial to the Landau Gauge)**.** *There are numbers $c_5 > 0$, B_2, and $B_2(\beta_0)$ for $0 \leq \beta_0 < 1$, which depend on L and G only, with the following property. Let the sequence of domains be as in Theorem 4.13, let $U_0 \in \mathfrak{U}(\alpha_0)$ be regular with the constant α_0 in the sense of Definition 4.3, and let $U \in \mathfrak{U}(\alpha_0)$ be in the axial gauge of U_0 and satisfy $|M^j(U)(c)M^j(U_0)(c)^{-1} - \mathbf{1}| \leq \alpha_1$ for the bonds c of Γ_j, $0 \leq j \leq k$, with $\alpha_0 + \alpha_1 \leq c_5$. Write $U = U'U_0$. Then there is exactly one admissible gauge transformation u with values in G such that the configuration U_1 with*

$$U_1(b) \coloneqq u(b_-)^{-1} U'(b) R(U_0(b)) u(b_+)$$

on the positive bonds is of the form $e^{i\eta A}$ with a 1-form A with values in $\mathfrak{g}$ which satisfies $RD^*_{U_0}A = 0$ and, on Ω_j for $0 \leq j \leq k$ and with $\epsilon \coloneqq \alpha_0 + \alpha_1$,

$$\begin{gathered} |A| \leq B_2 \epsilon a_j^{-1} \,, \qquad |\nabla_{U_0} A| \leq B_2 \epsilon a_j^{-2} \,, \qquad \|\nabla_{U_0} A\|_\beta \leq B_2(\beta_0) \epsilon a_j^{-2-\beta} \quad (\beta \leq \beta_0) \,, \\ |D^*_{U_0} D_{U_0} A| \leq B_2 \epsilon a_j^{-3} \,, \qquad |\Delta_{U_0} A| \leq B_2 \epsilon a_j^{-3} \,. \end{gathered} \tag{5.8}$$

Moreover U_1U_0 is the gauge transform of U by u^{-1}, and the relative average of A is determined by the averages of U and of U_0 on the layers alone: if V and V_0 denote the data with $V(c) = M^j(U)(c)$ and $V_0(c) = M^j(U_0)(c)$ for the bonds c of Γ_j, then

$$\mathsf{X}(A) = \mathsf{b}(V, V_0) \,, \qquad |a_j \mathsf{b}(V, V_0)(c)| \leq 8L\alpha_1 \text{ for the bonds } c \text{ of } \Gamma_j \,, \tag{5.9}$$

with the relative average (5.5) *at* $\mathbf{U} = U_0$.

This is [6, Theorem 2, p. 83] with the bounds [6, (1.36)–(1.39)]; the statement on the averages is [6, (1.30), (1.31), (1.37)], where the factors (5.6) are written for the transports along the fixed paths. The transformation u is expressed through U_1 by the site factors of Proposition 2.12 [4, (106), p. 33].

6 The Variational Problem and the Background Field

The effective action of a step is expanded about a configuration on the fine lattice which is determined by the field of the step: the configuration of least action among the regular fields whose averages on the layers are that field. This section states that such a configuration exists, that it is unique up to the gauge transformations which are trivial on the determining set, that it is as regular as its data, and that it depends analytically on the data [7]. The proof is a perturbation argument about a configuration which is already close to the minimiser, and its objects are needed afterwards in their own right, because the terms of the effective action are functions of them: the nonlinear part of the action in the fluctuation, the change of variables which makes the constraint linear, the potential which remains once the quadratic form has been split off, and the two fixed-point equations for the critical point. We therefore define these objects at a pair, as in Section 4.4, state their bounds, and express the minimiser through them.

6.1 The Problem

The data of the problem are a gauge field on the bonds of the layers, and their regularity is measured by their plaquettes. At the surface between two layers a plaquette of the coarser lattice has bonds which are not bonds of its layer, and on these the data are replaced by their averages [7, p. 278].

Definition 6.1 (Regular Data)**.** Let $RM \geq 2L$ and let V be data on the determining set (Definition 5.1). For $0 \leq j \leq k$ a *plaquette of the data* of the layer Γ_j is a plaquette p' of $T^{(j)}$ with a corner in Ω_j and no corner in Ω_{j+1}. For a bond c in the boundary of p' which has no end in Ω_j we put $V(c) \coloneqq M(V)(c)$, the average of Definition 2.4 of the restriction of V to the bonds of $T^{(j-1)}$ with both ends in $B(c_-) \cup B(c_+)$. The data are *regular with the constant* ε_1 if

$$|V(\partial p') - \mathbf{1}| \leq \varepsilon_1 \tag{6.1}$$

for every plaquette of the data.

The definition makes sense for the following reason. A bond of p' with an end in Ω_j has no end in Ω_{j+1}, because no corner of p' lies there, so that it is a bond of Γ_j. A bond c of p' with no end in Ω_j occurs only for $j \geq 1$, its ends are points of $T^{(j)}$ within the ℓ^∞-distance a_j of a corner in Ω_j, and as in the remark after Definition 5.7 the bonds of $T^{(j-1)}$ with both ends in $B(c_-) \cup B(c_+)$ are bonds of Γ_{j-1}, on which V is given. This is the condition [7, (7), p. 278].

Definition 6.2 (Critical and Minimal Configurations)**.** Let V be data, let $0 < \varepsilon_0 \le L^{-2}c_4'$, and let $U \in \mathfrak{U}(\varepsilon_0) \cap \mathfrak{V}(V)$, so that the operator $Q = Q(U)$ of Definition 4.2 is defined by Proposition 2.12.

(i) U is *critical* if $\langle \delta A, J(U) \rangle = 0$ for every 1-form δA with values in $\mathfrak{g}$ and $Q\,\delta A = 0$.

(ii) U is *minimal in* $\mathfrak{U}(\varepsilon_0) \cap \mathfrak{V}(V)$ if $A(U) \le A(U')$ for every U' in this set.

The linearisation of the constraint $M^j(U) = V$ at U is not Q but the operator which Proposition 2.10 calls Q^{pl}, and the two differ by the gauge transformation of the site factors. Since the current is orthogonal to the gauge directions, the two operators give the same notion of a critical configuration, which is the content of the first part of the next lemma.

Lemma 6.3 (Critical Configurations)**.** *Let V, ε_0, and U be as in Definition 6.2.*

(i) U is critical if and only if $\frac{\mathrm{d}}{\mathrm{d}t} A(U_t)|_{t=0} = 0$ for every continuously differentiable curve $t \mapsto U_t$ in $\mathfrak{V}(V)$ with $U_0 = U$.

(ii) If U is minimal in $\mathfrak{U}(\varepsilon_0) \cap \mathfrak{V}(V)$ and $U \in \mathfrak{U}(\varepsilon_0')$ for some $\varepsilon_0' < \varepsilon_0$, then U is critical.

Proof of (ii) from (i). The functions $U \mapsto U(\partial p)$ and $U \mapsto J(U)(b)$ are continuous, so that a curve as in (i) lies in $\mathfrak{U}(\varepsilon_0)$ for small $|t|$. On this part of the curve $A(U_t) \ge A(U)$ by the minimality, and a differentiable function with a minimum at $t = 0$ has a vanishing derivative there. *q.e.d.*

Lemma 6.4 (Gauge Transformations of the Problem)**.** *Let V and ε_0 be as in Definition 6.2, let u be a gauge transformation of T_η with values in G, and let V^u be the data with $V^u(c) \coloneqq u(c_-)V(c)u(c_+)^{-1}$. Then $U \mapsto U^u$ is a bijection of $\mathfrak{U}(\varepsilon_0) \cap \mathfrak{V}(V)$ onto $\mathfrak{U}(\varepsilon_0) \cap \mathfrak{V}(V^u)$ which preserves the action and maps critical configurations to critical ones and minimal configurations to minimal ones, and V^u is regular with the same constant as V. In particular the critical and the minimal configurations in $\mathfrak{U}(\varepsilon_0) \cap \mathfrak{V}(V)$ form unions of orbits of $\mathcal{G}_{\mathbf{B}}$.*

Proof. The action and the set $\mathfrak{U}(\varepsilon_0)$ are invariant under u by Lemma 1.8(ii) and (1.11), the norm being invariant under conjugation by elements of G (Lemma 1.3). By Lemma 2.5(iii), $M^j(U^u)(c) = u(c_-)M^j(U)(c)u(c_+)^{-1}$ for every bond c of $T^{(j)}$, so that $U \in \mathfrak{V}(V)$ if and only if $U^u \in \mathfrak{V}(V^u)$; the inverse map is the transformation by u^{-1}. The same identity for one step, applied to the data, shows that the plaquettes of V^u, with the convention of Definition 6.1, are the conjugates of those of V by the values of u at a corner. Minimality is preserved because the action is. For criticality, $J(U^u) = R(u)J(U)$ by (1.11) and $Q(U^u)R(u) = R(u)Q(U)$ by Proposition 2.12(v), where $R(u)$ acts on a 1-form at a bond b and on a function on the bonds of the layers at a bond c by the adjoint action of $u(b_-)$ and of $u(c_-)$. Hence $\delta A \mapsto R(u)\delta A$ maps the kernel of $Q(U)$ onto that of $Q(U^u)$, and $\langle R(u)\delta A, J(U^u)\rangle = \langle \delta A, J(U)\rangle$ by the invariance of the trace. The last statement is the case $u \in \mathcal{G}_{\mathbf{B}}$, for which $V^u = V$. *q.e.d.*

For u constant on the blocks $\Delta(y)$ this contains the covariance [7, (181), p. 307] of the minimisers in the axial gauge, by the last statement of Lemma 5.4.

Theorem 6.5 (The Minimal Orbit)**.** *Let M_1 be the number of Theorem 4.4, increased if necessary to a power of L. There are numbers $R_2 \ge R_1$, $a_0^{\mathrm{v}} > 0$, $a_1^{\mathrm{v}} > 0$, $B_3 \ge 1$ with $B_3 a_1^{\mathrm{v}} \le a_0^{\mathrm{v}} \le L^{-2}c_4'$, and $B_4(\beta_0)$ for $0 \le \beta_0 < 1$, which depend on L and G only, with the following properties. Let a sequence of domains with the block size M_1 and a separation $R \ge R_2$ be given, and let V be data which are regular with a constant $\varepsilon_1 \le a_1^{\mathrm{v}}$.*

(i) Existence. *There is a gauge field* $U \in \mathfrak{U}(B_3\varepsilon_1) \cap \mathfrak{V}(V)$ *which is minimal in* $\mathfrak{U}(a_0^{\mathrm{v}}) \cap \mathfrak{V}(V)$.

(ii) Uniqueness. *The critical configurations in* $\mathfrak{U}(a_0^{\mathrm{v}}) \cap \mathfrak{V}(V)$ *form exactly one orbit of* $\mathcal{G}_{\mathbf{B}}$, *the orbit of* U.

(iii) Regularity. *Let* $0 \le j \le k$, *and let* $\square$ *be a cube which is a union of cubes of the grid of side* $M_1 a_j$, *of a side* $s_\square$ *with*

$$2R_2 M_1 a_j \le s_\square \,, \qquad s_\square \varepsilon_1 \le 2 a_1^{\mathrm{v}} R_2 M_1 a_j \,, \tag{6.2}$$

which meets Ω_j *and for which the set of the points at an* ℓ^∞*-distance of at most* $2R_2M_1a_j$ *from* $\square$ *is contained in* $\Omega_{j-1} \setminus \Omega_{j+1}$, *where* $\Omega_{-1} \coloneqq \Omega_0$. *Then there is a gauge transformation* u *on* $\square$ *with values in* G *such that* $U^u = e^{i\eta A}$ *on the bonds of* $\square$, *with a* 1*-form* A *with values in* $\mathfrak{g}$ *which satisfies on* $\square$

$$\begin{aligned} |A| &\le \frac{s_\square}{a_j} \frac{B_3\varepsilon_1}{2} a_j^{-1} \,, \qquad |\nabla_{\mathbf{1}} A| \le \frac{s_\square}{a_j} \frac{B_3\varepsilon_1}{2} a_j^{-2} \,, \\ |D_{\mathbf{1}}^* D_{\mathbf{1}} A| &\le \frac{s_\square}{a_j} \frac{B_3\varepsilon_1}{2} a_j^{-3} \,, \qquad |\Delta_{\mathbf{1}} A| \le \frac{s_\square}{a_j} \frac{B_3\varepsilon_1}{2} a_j^{-3} \,, \end{aligned} \tag{6.3}$$

and, for $0 \le \beta_0 < 1$ *and all sites* x, x' *of* $\square$ *with* $|x - x'|_1 \le a_j$,

$$|(\nabla_{\mathbf{1}} A)(x') - (\nabla_{\mathbf{1}} A)(x)| \le \frac{s_\square}{a_j} B_4(\beta_0) \varepsilon_1 a_j^{-2} \Big(\frac{|x - x'|_1}{a_j} \Big)^{\beta_0} \,. \tag{6.4}$$

In (iii) *a cube is a cube of* $\mathbb{R}^4$, *and the grids, the domains, and the field are replaced by their preimages and lifts under the covering map* $\mathbb{R}^4 \to \mathbb{T}$.

This is [7, Theorem 1, p. 279], with the class of cubes of its proof [7, p. 300]; the number $s_\square/(2a_j)$ is the size M of the cube there, and the second condition (6.2) is $M \le M(\varepsilon_1)$. Four points of the statement are chosen for the use which the later sections make of it. First, a sequence of domains with a block size $M \ge M_1$ and a separation R is also a sequence with the block size M_1 and the separation RM/M_1, so that the theorem applies to it when $RM \ge R_2 M_1$; the operators of Section 4 depend on the domains and not on the block size by which they are described. Secondly, minimality is asserted on the whole of $\mathfrak{U}(a_0^{\mathrm{v}}) \cap \mathfrak{V}(V)$, and not only near U. Thirdly, the Hölder bound is stated for the exponents less than one: the Laplacian of A has jumps across the faces of the blocks, where the constraints act, and the second differences of A are not bounded uniformly in η. Lastly, the cubes are taken in the covering space, so that the statement has a content for the cubes which are larger than the torus.

In the form in which the propagators of Section 4 require it, the regularity reads as follows.

Corollary 6.6 (The Minimiser Is Regular)**.** *Let a sequence of domains with a block size* $M \ge LR_2M_1$ *and a separation* $R \ge 27$ *be given, let* V *be data which are regular with a constant* $\varepsilon_1 \le a_1^{\mathrm{v}}$ *such that* $40M\varepsilon_1 \le a_1^{\mathrm{v}} R_2 M_1$, *and let* U *be the configuration of Theorem 6.5. Then for every cube* $\square$ *of the class of the sequence, of an index* j *(Definition 4.3(i), in the covering space), there is a gauge transformation* u *on* $\square$ *with values in* G *such that* $U^u = e^{i\eta A}$ *on the bonds of* $\square$ *and*

$$|A| \le \frac{s_\square}{a_j} B_3 \varepsilon_1 a_j^{-1} \,, \qquad |\nabla_{\mathbf{1}} A| \le \frac{s_\square}{a_j} B_3 \varepsilon_1 a_j^{-2} \,, \qquad |D_{\mathbf{1}}^* D_{\mathbf{1}} A| \le \frac{s_\square}{a_j} B_3 \varepsilon_1 a_j^{-3} \tag{6.5}$$

on $\square$. *In particular* U *is regular to the second order with the constant* $B_3\varepsilon_1$.

Proof. The sequence has the separation $RM/M_1 \geq R_2$ with respect to the block size M_1, so that Theorem 6.5 applies. Let $\square$ be a cube of the class of the index j: a union of cubes of the grid of side Ma_j with $10Ma_j \leq s_\square \leq 20Ma_j$, contained in $\Omega_j \setminus \Omega_{j+2}$, which contains a point $x \notin \Omega_{j+1}$. We write $m \coloneqq 2R_2M_1$, so that $m \leq 2M/L$, and $\tilde{\square}$ for the set of the points within the ℓ^∞-distance ma_j of $\square$. Since M_1 divides M, the cube $\square$ is a union of cubes of the grid of side M_1a_j.

Suppose first that $\tilde{\square}$ does not meet Ω_{j+1}. The points of $\tilde{\square}$ are within the distance $ma_j \leq 2Ma_{j-1}$ of $\square \subset \Omega_j$, which is less than RMa_{j-1}, so that $\tilde{\square} \subset \Omega_{j-1}$ by (1.13). The conditions (6.2) hold because $s_\square \geq 10Ma_j \geq ma_j$ and $s_\square\varepsilon_1 \leq 20M\varepsilon_1 a_j \leq 2a_1^{\mathrm{v}}R_2M_1a_j$. Hence Theorem 6.5(iii) applies to $\square$ with the index j and gives (6.5) with half the constant.

Suppose next that $\tilde{\square}$ contains a point $z \in \Omega_{j+1}$, so that $j < k$. Let $\square'$ be a cube which is a union of cubes of the grid of side M_1a_{j+1}, contains $\square$ and z, and has the smallest side s' for which this is possible. The set $\square \cup \{z\}$ lies in a box with sides of at most $s_\square + ma_j$, and a box with sides of at most ℓ lies in a cube of the grid of side g whose side is the smallest multiple of g not less than $\ell + g$; hence

$$s_\square \leq s' \leq s_\square + ma_j + 2M_1a_{j+1} \leq s_\square + \frac{2M}{L}a_j + 2M_1La_j \leq s_\square + 4Ma_j \leq \frac{7}{5}s_\square \,,$$

where we used $M_1L \leq M$ and $s_\square \geq 10Ma_j$. We verify the hypotheses of Theorem 6.5(iii) for $\square'$ with the index $j+1$. The cube meets Ω_{j+1} at z. The first condition (6.2) holds since $s' \geq 10Ma_j \geq 10LR_2M_1a_j = 5ma_{j+1}$, and the second since $s' \leq 24Ma_j = 24ML^{-1}a_{j+1}$ and $12M\varepsilon_1 \leq a_1^{\mathrm{v}}R_2M_1$. Every point of the set $\tilde{\square}'$ of the points within the distance $ma_{j+1} \leq 2Ma_j$ of $\square'$ is within the distance $s'+2Ma_j \leq 26Ma_j$ of z and of x. Since $z \in \Omega_{j+1}$ and $26M < RM$, the set $\tilde{\square}'$ lies in Ω_j by (1.13); and since $x \notin \Omega_{j+1}$ and $26Ma_j < RMa_{j+1}$, it does not meet Ω_{j+2}. The theorem gives a gauge transformation on $\square'$ and a 1-form with the bounds (6.3) in which $s_\square/a_j$ is replaced by s'/a_{j+1} and a_j^{-p} by a_{j+1}^{-p}, $p = 1, 2, 3$. Their ratio to the bounds (6.5) is $\frac{1}{2}(s'/s_\square)L^{-1-p} \leq \frac{7}{10}L^{-2} \leq 1$, and we restrict the transformation to $\square$.

In both cases the form has values in $\mathfrak{g}$, which is the condition of Definition 4.3(ii) with the constant $B_3\varepsilon_1$, to the second order. *q.e.d.*

6.2 The Expansion about a Pair

Throughout this subsection and the next a sequence of domains as in Theorem 4.13 is given, and $(\mathbf{U}, \mathbf{J})$ is a regular pair with constants $(\alpha_0, \alpha_1, \gamma)$ which satisfy the hypothesis of that theorem, so that the operators of Sections 4.1–4.5 exist at the pair. The arguments of the functionals below are 1-forms on T_η with values in $\mathfrak{g}^c$, which we write A, A', and $\mathfrak{A}$ as in [7]; the complex part of $\mathbf{U}$ in Definition 4.3 does not appear explicitly in this section. A function of such a form and of the pair is called *holomorphic* if it is holomorphic in the form and, for a fixed regular gauge field, in the complex part of $\mathbf{U}$ and in $\mathbf{J}$, as in Theorem 4.13(i).

Definition 6.7 (Scaled Norms)**.** For a 1-form A, a point $y \in \Gamma_j$, and a function μ on the bonds of the layers,

$$\begin{gathered} |A|_* \coloneqq \max\{|A|_{(-1)}, |\nabla_{\mathbf{U}}A|_{(-2)}\} \,, \qquad \|A\|_y \coloneqq a_j|A|_{\Delta(y)} + a_j^2|\nabla_{\mathbf{U}}A|_{\Delta(y)} \,, \\ |\mu|_{(\alpha)} \coloneqq \sup_c a_{j(c)}^{-\alpha}|\mu(c)| \,, \end{gathered} \tag{6.6}$$

with the weighted norms of Definition 1.16(iii), and with $j(c) \coloneqq j$ for a bond c of Γ_j.

The first object is the change of variables which makes the constraint linear. The relative average $\mathsf{X}(A) = QA + \mathsf{C}(A)$ of (5.5) has a remainder of the second order, and the remainder

is removed by subtracting from the field the minimiser of the quadratic form whose average is the remainder [7, Sect. C].

Proposition 6.8 (The Linearising Transformation)**.** *There are numbers $a_3 > 0$, $\delta_2 \in (0, \delta_1]$, and C_5, which depend on L and G only, such that for every 1-form A' with values in $\mathfrak{g}^c$ and $|A'|_{(-1)} \le a_3$ the following holds.*

(i) There is exactly one function $\mathsf{D} = \mathsf{D}(A')$ on the bonds of the layers with $|\mathsf{D}|_{(-1)} \le a_3$ and

$$\mathsf{D}(A') = \mathsf{C}(A' - H\mathsf{D}(A')) \; . \tag{6.7}$$

It is holomorphic, it vanishes on the bonds of Γ_0, and $|\mathsf{D}(A')|_{(-1)} \le C_5 |A'|^2_{(-1)}$.

(ii) The linearising transformation *$T(A') \coloneqq A' - H\mathsf{D}(A')$ satisfies $\mathsf{X}(T(A')) = QA'$ and $|T(A') - A'|_* \le C_5 |A'|^2_{(-1)}$. It is injective on the set of the A' with $|A'|_{(-1)} \le a_3$, and for every A with $|A|_{(-1)} \le \frac{1}{4} a_3$ there is an A' in this set with $T(A') = A$, $|A'|_{(-1)} \le 2|A|_{(-1)}$, and $|A'|_* \le 2|A|_*$.*

(iii) Let $\mathfrak{D}(A')$ be the derivative of D at A', a linear map from 1-forms to functions on the bonds of the layers. For a 1-form δA supported in $\Delta(y)$, $y \in \mathbf{B}$, and a bond c of the layers,

$$a_{j(c)} |(\mathfrak{D}(A')\delta A)(c)| \le C_5 |A'|_{(-1)} e^{-\delta_2 d(c_-, y)} a_{j(y)} |\delta A| \; , \tag{6.8}$$

and the derivative of $\mathsf{D}_3 \coloneqq \mathsf{D} - \mathsf{C}^{(2)}$ satisfies the same bound with $|A'|^2_{(-1)}$ in the place of $|A'|_{(-1)}$.

This is [7, Proposition 3, p. 289] with [7, (49)–(62), (73)], in the normalisation of Definition 4.12, in which $QH = I$; the function D and the operator H of [7] are $a_{j(\cdot)}\mathsf{D}$ and $H a^{-1}_{j(\cdot)}$. The identity $\mathsf{X}(T(A')) = QA'$ is (6.7): $QT(A') = QA' - \mathsf{D}(A')$ and $\mathsf{C}(T(A')) = \mathsf{D}(A')$.

Next we split the action. The second variation $\Delta(\mathbf{U})$ and the form $\Delta_\pi \coloneqq \Delta(\mathbf{U}) - \Delta'_\pi$ agree on the forms in the Landau gauge when the second entry of the pair is the current, and [7] writes V for the potential formed with either. Their gradients differ, however, and the two fixed-point equations of Section 6.3 need one each, so that we keep both.

Definition 6.9 (The Potentials)**.** (i) $\mathsf{V}_0(A) \coloneqq A(e^{i\eta A}\mathbf{U}) - A(\mathbf{U}) - \langle A, J(\mathbf{U})\rangle - \frac{1}{2}\langle A, \Delta(\mathbf{U})A\rangle$, an entire function of the 1-form A which vanishes to the third order at $A = 0$ by Definition 4.1.

(ii) For $|A'|_{(-1)} \le a_3$,

$$\mathsf{V}(A') \coloneqq -\langle H\mathsf{D}_3(A'), \mathbf{J}\rangle - \langle A', \Delta(\mathbf{U})H\mathsf{D}(A')\rangle + \frac{1}{2}\langle H\mathsf{D}(A'), \Delta(\mathbf{U})H\mathsf{D}(A')\rangle + \mathsf{V}_0(T(A')) \; , \tag{6.9}$$

and $\mathsf{V}_\pi(A')$ is defined by the same formula with Δ_π in the place of $\Delta(\mathbf{U})$.

(iii) The *gradients* $\mathsf{N}(A')$ and $\mathsf{N}_\pi(A')$ are the 1-forms with $\langle \delta A, \mathsf{N}(A')\rangle = \frac{\mathrm{d}}{\mathrm{d}t}\mathsf{V}(A' + t\,\delta A)|_{t=0}$ for all 1-forms δA, and likewise for V_π.

The potential V_π is [7, (80), p. 290], and V is the functional which appears, under the same letter, in [7, (127), (128), p. 297] and in the equations derived from them.

Lemma 6.10 (The Action in the New Variable)**.** *Let $|A'|_{(-1)} \le a_3$ and $A \coloneqq T(A')$. Then*

$$A(e^{i\eta A}\mathbf{U}) + \langle A, \mathbf{J} - J(\mathbf{U})\rangle = A(\mathbf{U}) + \langle A', \mathbf{J}\rangle + \frac{1}{2}\langle A', (\Delta(\mathbf{U}) - \Delta^{(2)})A'\rangle + \mathsf{V}(A') \; , \tag{6.10}$$

and

$$\mathsf{V}(A') - \mathsf{V}_\pi(A') = \langle A', \Delta'_\pi H\mathsf{D}(A')\rangle - \frac{1}{2}\langle H\mathsf{D}(A'), \Delta'_\pi H\mathsf{D}(A')\rangle \ . \tag{6.11}$$

If $\mathbf{J} = J(\mathbf{U})$ *and* $RD^*A' = 0$*, then* $RD^*A = 0$*, the right side of* (6.11) *vanishes, and* $\langle A', \Delta^{(2)}A'\rangle = \langle A', \Delta^{(2)}_\pi A'\rangle$*.*

Proof. By the definition of V_0 the left side of (6.10) equals $A(\mathbf{U})+\langle A, \mathbf{J}\rangle+\frac{1}{2}\langle A, \Delta(\mathbf{U})A\rangle+\mathsf{V}_0(A)$. We insert $A = A' - H\mathsf{D}(A')$ and $\mathsf{D} = \mathsf{C}^{(2)} + \mathsf{D}_3$. The linear term becomes $\langle A', \mathbf{J}\rangle - \langle H\mathsf{C}^{(2)}(A'), \mathbf{J}\rangle - \langle H\mathsf{D}_3(A'), \mathbf{J}\rangle$, and $\langle H\mathsf{C}^{(2)}(A'), \mathbf{J}\rangle = \frac{1}{2}\langle A', \Delta^{(2)}A'\rangle$ by Definition 4.12(iii). The quadratic term becomes $\frac{1}{2}\langle A', \Delta(\mathbf{U})A'\rangle - \langle A', \Delta(\mathbf{U})H\mathsf{D}(A')\rangle + \frac{1}{2}\langle H\mathsf{D}(A'), \Delta(\mathbf{U})H\mathsf{D}(A')\rangle$, because $\Delta(\mathbf{U})$ is symmetric. The sum is the right side of (6.10), by (6.9). The difference (6.11) is read off from the two definitions, since $\Delta(\mathbf{U}) - \Delta_\pi = \Delta'_\pi$.

Let now $\mathbf{J} = J(\mathbf{U})$ and $RD^*A' = 0$. Then $RD^*H = 0$ by Theorem 4.13(iv), so that $RD^*A = 0$, and the gauge functions $\lambda_{A'} = G'RD^*A'$ and $\lambda_{H\mu} = G'RD^*H\mu$ of Definition 4.12 vanish. Every term of the form (4.21) contains a factor λ, and $\lambda_{A_1+A_2} = \lambda_{A_1} + \lambda_{A_2}$, so that the bilinear form of Δ'_π, which is obtained from $\langle X(A), \mathbf{J}\rangle$ by polarisation, vanishes on two forms whose gauge functions vanish. This gives the second statement. For the third, $\mathcal{P}A' = A' - D\lambda_{A'} = A'$, hence $\langle A', \Delta^{(2)}_\pi A'\rangle = \langle \mathcal{P}A', \Delta^{(2)}\mathcal{P}A'\rangle = \langle A', \Delta^{(2)}A'\rangle$. *q.e.d.*

At $\mathbf{J} = J(\mathbf{U})$ the left side of (6.10) is the action of $e^{i\eta A}\mathbf{U}$, and (6.10) is its expansion in the variable in which the constraint $\mathsf{X}(A) = \mathfrak{b}$ is the linear condition $QA' = \mathfrak{b}$. This is [7, (74)–(81), pp. 289–290].

Proposition 6.11 (The Potentials)**.** *The numbers* a_3 *and* δ_2 *of Proposition 6.8 can be chosen, and there are numbers* C_4 *and* C_N *which depend on* L *and* G *only, such that for every* 1*-form* A' *with values in* $\mathfrak{g}^c$ *and* $|A'|_* \le a_3$*:*

(i) V*,* V_π*,* N*, and* N_π *are holomorphic, and* V *and* V_π *vanish to the third order at* $A' = 0$*.*

(ii) $|\mathsf{N}(A')|_{(-3)} \le C_4|A'|^2_*$*.*

(iii) For a bond b *of* $\Delta(y)$*,* $y \in \Gamma_j$*,*

$$a_j^3|\mathsf{N}(A')(b)| \le C_N|A'|_* \sum_{y''\in\mathbf{B}} e^{-\delta_2 d(y,y'')}\|A'\|_{y''} \ . \tag{6.12}$$

(iv) For a 1*-form* $\mathfrak{A}$ *supported in* $\tilde{\Delta}(y')$*,* $y' \in \mathbf{B}$*, and a bond* b *of* $\Delta(y)$*,* $y \in \Gamma_j$*, the derivative of* N *at* A' *in the direction* $\mathfrak{A}$ *satisfies*

$$a_j^3\Big|\Big(\frac{\mathrm{d}}{\mathrm{d}t}\mathsf{N}(A' + t\mathfrak{A})\Big|_{t=0}\Big)(b)\Big| \le C_N|A'|_* e^{-\delta_2 d(y,y')}|\mathfrak{A}|_* \ . \tag{6.13}$$

The bounds (ii)–(iv) *hold for* N_π *as well.*

The bound (ii) is [7, Proposition 4, p. 292], and (iv) is [7, (189), p. 308], which is printed there without its proof. The form (iii), in which one factor of the field is kept at its place, is not in [7]; it is what the localisation of the minimiser needs.

6.3 The Equations of the Critical Point

A critical point of the action under the constraint and the gauge condition is a critical point of the functional (6.10) on an affine subspace, and the equation for it becomes a fixed-point equation once the quadratic form is inverted on that subspace. This can be done in two ways. In the first the subspace is that of the forms with given averages in the Landau gauge, the quadratic form is that of G_1, and the inverse is the propagator $\mathfrak{G}$ of the fluctuation [7, (99)–(111)]. In the second the gauge condition is dropped from the subspace, which is legitimate because the action is gauge invariant, and the inverse is formed with the operator G of Definition 4.2, corrected by the form $\Delta^{(2)}$ [7, (127)–(143), (179), (180)]. The second equation is the one with the better regularity, because G is bounded together with its covariant Laplacian, and it is the one through which the later sections write the background field.

Definition 6.12 (The Operators of the Second Equation)**.** With the operator G of Definition 4.2 and the form $\Delta^{(2)}$ of Definition 4.12,

$$H_0 \coloneqq GQ^*(QGQ^*)^{-1} , \qquad G_{\mathrm{c}} \coloneqq (\Delta_{\mathsf{a}} - \Delta^{(2)})^{-1} , \qquad \tilde{G} \coloneqq G_{\mathrm{c}} - G_{\mathrm{c}}Q^*(QG_{\mathrm{c}}Q^*)^{-1}QG_{\mathrm{c}} . \tag{6.14}$$

By their definitions $QH_0 = I$, $Q\tilde{G} = 0$, and $\tilde{G}Q^* = 0$. In [7, (129), p. 297] the operator H_0 is formed with G, and in [7, (180), p. 306] the operator $\tilde{G}$ is formed with the corrected operator, which is there again called G.

Proposition 6.13 (The Operators of the Second Equation)**.** *Under the hypotheses of Theorem 4.13, with smaller numbers a_2 and δ_1 and larger constants B_1, $B_1(\beta)$, $B_1'(\epsilon,\beta)$ if necessary:*

(i) The operators (6.14) *exist and are holomorphic in the pair, and $G_{\mathrm{c}} = \sum_{n\geq 0} G(\Delta^{(2)}G)^n$.*

(ii) G_{c} and $\tilde{G}$ satisfy the bounds (iv)–(vi) *of Theorem 4.4 for* 1*-forms, the bound of the covariant Laplacian included, and $(QGQ^*)^{-1}$ and $(QG_{\mathrm{c}}Q^*)^{-1}$ satisfy* (4.24)*.*

(iii) The kernel of H_0 satisfies (4.25) *and, for $x \in \Delta(y)$, $y \in \Gamma_j$, and a bond c' of $\Gamma_{j'}$,*

$$a_j^2 |(\Delta_{\mathbf{U}} H_0)(x,c')| \leq B_1 a_{j'}^{-4} e^{-\frac{1}{2}\delta_1 d(y,c'_-)} . \tag{6.15}$$

(iv) If $\mathbf{J} = J(\mathbf{U})$ and $\mathbf{U}$ is a gauge field in $\mathfrak{U}(\alpha_0)$, then for every function μ on the bonds of the layers

$$|D^*DH\mu|_{(-3)} + |\Delta_{\mathbf{U}} H\mu|_{(-3)} \leq B_1 |\mu|_{(-1)} , \tag{6.16}$$

and G_{c}, $\tilde{G}$, and $QG_{\mathrm{c}}Q^*$ are symmetric.

These are [7, (129), (130), (137)–(140), pp. 297–299] and the remark [7, p. 306] that the corrected operator "has exactly the same properties" as G. The bound (6.16) is derived there from the identity $RD^*H = 0$, and we do not state it at a pair whose second entry is not the current of the first.

Proposition 6.14 (The Two Fixed Points)**.** *There are numbers $a_4 \in (0, \frac{1}{2}a_3]$ and $C_6 \geq 1$, which depend on L and G only, with the following property. Let $\gamma \leq a_4/C_6$, and let $\mathfrak{A}$ be a* 1*-form with values in $\mathfrak{g}^c$ and $|\mathfrak{A}|_* \leq a_4$. Then each of the maps*

$$\Phi_1(A) \coloneqq -\mathfrak{G}\mathbf{J} - \mathfrak{G}\mathsf{N}_\pi(A+\mathfrak{A}) , \qquad \Phi_0(A) \coloneqq -\tilde{G}\mathbf{J} + \tilde{G}\Delta^{(2)}\mathfrak{A} - \tilde{G}\mathsf{N}(A+\mathfrak{A}) \tag{6.17}$$

maps the set of the 1*-forms* A *with values in* $\mathfrak{g}^c$ *and* $|A|_* \le a_4$ *into itself and satisfies* $|\Phi(A) - \Phi(A'')|_* \le \frac{1}{2}|A - A''|_*$ *on it. Consequently each has exactly one fixed point in this set,* $\mathcal{A}_1(\mathfrak{A})$ *and* $\mathcal{A}_0(\mathfrak{A})$*, respectively. The fixed points are holomorphic in* $\mathfrak{A}$ *and in the pair, and*

$$|\mathcal{A}_1(\mathfrak{A})|_* \le C_6(\gamma + |\mathfrak{A}|_*^2) \,, \qquad |\mathcal{A}_0(\mathfrak{A})|_* + |\Delta_{\mathbf{U}}\mathcal{A}_0(\mathfrak{A})|_{(-3)} \le C_6(\gamma + |\mathfrak{A}|_*^2) \,. \tag{6.18}$$

Moreover $Q\mathcal{A}_0(\mathfrak{A}) = 0$*, and if* $\mathbf{J} = J(\mathbf{U})$*, then* $Q\mathcal{A}_1(\mathfrak{A}) = 0$ *and* $RD^*\mathcal{A}_1(\mathfrak{A}) = 0$*.*

This is [7, Proposition 6, p. 295] with [7, (116)–(121)] for the first map, and the statement of [7, p. 300] that the second equation "has all the properties" of the first. The terms $\mathfrak{G}\mathbf{J}$ and $\tilde{G}\mathbf{J}$ are kept in the equations, since the functions defined by them are used at pairs at which they do not vanish.

Definition 6.15 (The Two Solutions)**.** Let B_H be a number such that $|H_0\mu|_* \le B_H|\mu|_{(-1)}$ and $|H_1\mu|_* \le B_H|\mu|_{(-1)}$ for all functions μ on the bonds of the layers, which exists and depends on L and G only by (4.25), Lemma 1.15, and the remark after Theorem 4.4 on the powers of the spacing. For $\gamma \le a_4/C_6$ and a function $\mathfrak{b}$ on the bonds of the layers with values in $\mathfrak{g}^c$ and $B_H|\mathfrak{b}|_{(-1)} \le a_4$ we put

$$\mathcal{H}_1(\mathfrak{b}) \coloneqq T(\mathcal{A}_1(H_1\mathfrak{b}) + H_1\mathfrak{b}) \,, \qquad \mathcal{H}(\mathfrak{b}) \coloneqq T(\mathcal{A}_0(H_0\mathfrak{b}) + H_0\mathfrak{b}) \,, \tag{6.19}$$

with the linearising transformation of Proposition 6.8. Both are functions of the pair as well, and we write $\mathcal{H}(\mathbf{U}, \mathbf{J}; \mathfrak{b})$ where the pair has to be named.

The arguments of T have the norm $|\cdot|_{(-1)}$ at most $2a_4 \le a_3$, so that both functions are defined, and they are holomorphic in $\mathfrak{b}$ and in the pair. By Proposition 6.8(ii) and $QH_0 = QH_1 = I$ they satisfy $\mathsf{X}(\mathcal{H}(\mathfrak{b})) = \mathfrak{b}$ at every pair, and $\mathsf{X}(\mathcal{H}_1(\mathfrak{b})) = \mathfrak{b}$ when $\mathbf{J} = J(\mathbf{U})$. The representations (6.19) are [7, (174), (175), p. 305] and [7, (179), (180), p. 306].

Theorem 6.16 (The Critical Configurations near a Regular Field)**.** *There are numbers* $a_5 > 0$ *and* B_5*, which depend on* L *and* G *only, with the following properties. Let* $U_0 \in \mathfrak{U}(\alpha_0)$ *be a gauge field which is regular to the second order with the constant* α_0*, where* $(M+1)\alpha_0 \le a_5$*, and let the operators and the functions of this section be taken at the pair* $(U_0, J(U_0))$*.*

(i) For every function $\mathfrak{b}$ *on the bonds of the layers with values in* $\mathfrak{g}$ *and* $|\mathfrak{b}|_{(-1)} \le a_5$*, the forms* $\mathcal{H}(\mathfrak{b})$ *and* $\mathcal{H}_1(\mathfrak{b})$ *are equal, have values in* $\mathfrak{g}$*, and satisfy* $RD^*\mathcal{H}(\mathfrak{b}) = 0$ *and* $\mathsf{X}(\mathcal{H}(\mathfrak{b})) = \mathfrak{b}$*. The gauge field* $e^{i\eta\mathcal{H}(\mathfrak{b})}U_0$ *is critical, and*

$$|\mathcal{H}(\mathfrak{b})|_* + |D^*D\mathcal{H}(\mathfrak{b})|_{(-3)} + |\Delta_{U_0}\mathcal{H}(\mathfrak{b})|_{(-3)} \le B_5(\alpha_0 + |\mathfrak{b}|_{(-1)}) \,. \tag{6.20}$$

(ii) Conversely, if A *is a* 1*-form with values in* $\mathfrak{g}$ *such that* $RD^*A = 0$*,* $|A|_* \le a_5$*, and* $e^{i\eta A}U_0$ *is critical, then* $A = \mathcal{H}(\mathsf{X}(A))$*.*

(iii) If U_0 *is critical, then* $\mathfrak{G}J(U_0) = 0$ *and* $\tilde{G}J(U_0) = 0$*, and* $\mathcal{H}(0) = 0$*.*

This collects [7, Propositions 2, 5, and 7], in a form in which the configurations in the Landau gauge are not required to have prescribed averages: a field $e^{i\eta A}U_0$ is critical for the data which are its own averages, in the sense of Definition 6.2, and by Lemma 6.4 this property does not depend on the gauge. The statement (iii) is [7, (170), (171), (178)], and it follows from the definitions: if U_0 is critical, then $J(U_0)$ is orthogonal to the kernel of Q, which contains the ranges of $\mathfrak{P}$ and of the projection $I - G_{\mathrm{c}}Q^*(QG_{\mathrm{c}}Q^*)^{-1}Q$, while $\mathfrak{G} = G_1\mathfrak{P}^T$ by Theorem 4.15(ii) and $\tilde{G}$ is G_{c} times the transpose of that projection; and then $A = 0$ is the fixed point of Φ_0 at $\mathfrak{A} = 0$, because $\mathsf{N}(0) = 0$.

6.4 The Background Field as a Function of the Data

We can now say how the minimiser moves when the data are changed. The minimisers of two nearby sets of data differ by a fluctuation which, in the Landau gauge of the first, is the function $\mathcal{H}$ at the constraint field of the two sets of data, and this function extends to complex data [7, Sect. G].

Theorem 6.17 (The Background Field)**.** *There are numbers $C_{\mathrm{d}} \geq 1$, $C'_{\mathrm{d}} \geq 1 + 4C_{\mathrm{d}}$, B_6, and $B_6(\beta_0)$ for $0 \leq \beta_0 < 1$, which depend on L and G only, such that the following holds, with smaller numbers a_1^{v} and δ_2 in Theorem 6.5 and Proposition 6.8 if necessary. Let the sequence of domains, with its block size M and its separation R, be as in Corollary 6.6, let V_0 be data which are regular with a constant ε_1 such that $C'_{\mathrm{d}}\varepsilon_1$ satisfies the two conditions which that corollary imposes on ε_1, let U_0 be a minimal configuration for V_0 as in Theorem 6.5, and let the functions of Section 6.3 be taken at the pair $(U_0, J(U_0))$. For a function B on the positive bonds of the layers with values in $\mathfrak{g}^c$ and $|B(c)| \leq C_{\mathrm{d}}\varepsilon_1$ for all c, let V_B be the map with $V_B(c) \coloneqq e^{iB(c)}V_0(c)$, and $\mathfrak{b}_B \coloneqq \mathfrak{b}(V_B, V_0)$ its constraint field* (5.7).

(i) The constraint field. *$\mathfrak{b}_B$ is defined and holomorphic in B, it vanishes at $B = 0$, and $a_j|\mathfrak{b}_B(c)| \leq 8L \sup|B|$ for a bond c of Γ_j. The value $\mathfrak{b}_B(c)$ depends on $B(c)$ and, if an end y of c does not lie in Ω_j, on B on the bonds with both ends in $B(y)$.*

(ii) The fluctuation. *$\mathcal{H}(\mathfrak{b}_B)$ is defined and holomorphic in B, it equals $\mathcal{H}_1(\mathfrak{b}_B)$, it vanishes at $B = 0$, and*

$$|\mathcal{H}(\mathfrak{b}_B)|_* + |D^*D\mathcal{H}(\mathfrak{b}_B)|_{(-3)} + |\Delta_{U_0}\mathcal{H}(\mathfrak{b}_B)|_{(-3)} \leq B_6 \sup|B| \,, \qquad RD^*\mathcal{H}(\mathfrak{b}_B) = 0 \,. \tag{6.21}$$

(iii) The minimiser. *There is a gauge transformation w_B of T_η with values in G^c, holomorphic in B, with $w_0 = \mathbf{1}$ and with values in G for B with values in $\mathfrak{g}$, such that the configuration*

$$U(B) \coloneqq (e^{i\eta\mathcal{H}(\mathfrak{b}_B)}U_0)^{w_B} \tag{6.22}$$

is in the axial gauge of U_0 and satisfies $M^j(U(B))(c) = V_B(c)$ for the bonds c of Γ_j, $0 \leq j \leq k$. For B with values in $\mathfrak{g}$ the transformation w_B is admissible (Definition 5.6), the data V_B are regular with the constant $C'_{\mathrm{d}}\varepsilon_1$, and $U(B)$ is the configuration of Theorem 6.5 for V_B which is in the axial gauge of U_0.

(iv) The derivative. *Let c' be a bond of $\Gamma_{j'}$, and let K be the derivative of $\mathcal{H}(\mathfrak{b}_B)$ with respect to B in a direction δB which is supported on c'. Then, for $x \in \Delta(y)$, $y \in \Gamma_j$, a real function ζ supported in $\tilde{\Delta}(y)$, and $0 \leq \beta \leq \beta_0 < 1$,*

$$\begin{aligned} a_j|K(x)| + a_j^2|(\nabla_{U_0}K)(x)| &\leq B_6 e^{-\delta_2 d(y,c'_-)}|\delta B| \,, \\ a_j^3|(D^*DK)(x)| + a_j^3|(\Delta_{U_0}K)(x)| &\leq B_6 e^{-\delta_2 d(y,c'_-)}|\delta B| \,, \\ a_j^{2+\beta}\|\zeta\nabla_{U_0}K\|_\beta &\leq B_6(\beta_0)(\|\zeta\|_\beta^{(j)} + |\zeta|)e^{-\delta_2 d(y,c'_-)}|\delta B| \,. \end{aligned} \tag{6.23}$$

This is [7, Proposition 9, p. 309]: the smallness of the data is [7, (172)], the bound (6.21) is [7, (173)] with [7, (19)–(21), p. 281], in the form which is linear in B and which follows from the bound by a constant and the vanishing at $B = 0$, and (6.23) is [7, (190), p. 308]. For real B the action of the minimiser for V_B is $A(e^{i\eta\mathcal{H}(\mathfrak{b}_B)}U_0)$, by (iii) and the invariance of the action, so that the least action is a real analytic function of the data. The expansion of $\mathcal{H}(\mathfrak{b})$ in powers of $\mathfrak{b}$ is obtained by iterating the fixed-point equations; by Theorem 6.16(iii) and the vanishing of N to the second order it begins with $H_1\mathfrak{b}$ [7, (176), (177), p. 306].

6.5 The Problem on the Support of a Determining Set

A field with given averages is given itself on the bonds of the layer Γ_0, that is, on every bond which has no end in Ω_1. The variational problem is therefore a problem for the field near Ω_1 only, and its hypotheses should concern the data near Ω_1 only. This is how the problem is used: the test functions of a step and the spaces of Section 7 are defined through the minimisers of determining sets which are built inside a cube or a localisation domain, and outside it no field is given. Following [10, p. 255] we call a neighbourhood of Ω_1 a support of the determining set.

Definition 6.18 (Supports)**.** Let a sequence of domains with the block size M_1 be given, where $M_1 \geq L^2$. A *support* is a set $W \subset \mathbb{T}$ which is a union of cubes of the grid of side $M_1\eta$ and contains every point at an ℓ^∞-distance of at most $M_1\eta$ from Ω_1. The bonds and the plaquettes *of* W are those with all ends and all corners in W. With these in the place of the bonds and the plaquettes of T_η, and with the bonds of the layers with both ends in W:

(i) $\mathfrak{U}_W(\varepsilon_0)$ is the set of the gauge fields U on the bonds of W which satisfy (5.1) for the plaquettes of W and for the bonds of W all of whose plaquettes are plaquettes of W;

(ii) *data on* W, their plaquettes, and their regularity are defined as in Definitions 5.1 and 6.1, and $\mathfrak{V}_W(V)$ is the set of the gauge fields U on the bonds of W with $M^j(U)(c) = V(c)$ for the bonds c of Γ_j with both ends in W;

(iii) $A_W(U)$ is the sum (1.7) over the plaquettes of W, and $U \in \mathfrak{U}_W(\varepsilon_0) \cap \mathfrak{V}_W(V)$ is *critical*, or *minimal*, on W if it satisfies the conditions of Definition 6.2 with A_W and with the 1-forms δA on the bonds of W.

A bond of a layer Γ_j with $j \geq 1$ has an end in $\Omega_j \subset \Omega_1$, its other end is within the distance a_j of Ω_j, and the blocks B^j of its ends are within $\frac{3}{2}a_j$ of Ω_j; for $j = 1$ this is less than $M_1\eta$, and for $j \geq 2$ these blocks lie in $\Omega_{j-1} \subset \Omega_1$ by (1.13). So the bonds of the layers Γ_j, $j \geq 1$, are bonds of W, and their averages are determined by the field on W (Lemma 2.5). A 1-form δA with $Q\,\delta A = 0$ vanishes on the bonds of Γ_0, so that it is supported on the bonds with an end in Ω_1; the plaquettes of these bonds have their corners within the distance 2η of Ω_1 and are plaquettes of W. The pairing $\langle \delta A, J(U)\rangle$ and the operator Q on such forms are therefore defined by the field on W.

Lemma 6.19 (Restriction to a Support)**.** *Let W be a support, let V and ε_0 be as in Definition 6.2, and let $U \in \mathfrak{U}(\varepsilon_0) \cap \mathfrak{V}(V)$. Then the restriction $U|_W$ lies in $\mathfrak{U}_W(\varepsilon_0) \cap \mathfrak{V}_W(V|_W)$, the map $U' \mapsto U'|_W$ is a bijection of $\mathfrak{U}(\varepsilon_0) \cap \mathfrak{V}(V)$ onto $\mathfrak{U}_W(\varepsilon_0) \cap \mathfrak{V}_W(V|_W)$ along which $A(U') - A_W(U'|_W)$ is constant, and U is critical, or minimal, if and only if $U|_W$ is critical, or minimal, on W.*

Proof. The conditions which define $\mathfrak{U}_W(\varepsilon_0)$ and $\mathfrak{V}_W(V|_W)$ are among those which define $\mathfrak{U}(\varepsilon_0)$ and $\mathfrak{V}(V)$, by the remark before the lemma; this gives the first statement. Every $U' \in \mathfrak{V}(V)$ equals V on the bonds with no end in Ω_1, since M^0 is the identity, and a bond which is not a bond of W has no end in Ω_1. Hence U' is determined by $U'|_W$, and the restriction is injective. Conversely let U'' lie in $\mathfrak{U}_W(\varepsilon_0) \cap \mathfrak{V}_W(V|_W)$, and let U' equal U'' on the bonds of W and V on the others, which are bonds of Γ_0. Then $U' \in \mathfrak{V}(V)$, because every bond of a layer is a bond of W or one of these. A plaquette which is not a plaquette of W has a corner outside W, hence no corner in Ω_1 and no bond with an end in Ω_1, so that on its bonds $U' = V = U$; the same holds for the plaquettes of a bond which has a plaquette that is not a plaquette of W. The conditions (5.1) for these plaquettes and bonds therefore hold for U'

because they hold for U, and for the others they hold because $U'' \in \mathfrak{U}_W(\varepsilon_0)$. So $U' \in \mathfrak{U}(\varepsilon_0)$, and the restriction is onto. The difference $A(U') - A_W(U'|_W)$ is the sum over the plaquettes which are not plaquettes of W, on which $U' = V$. Minimality is thus the same condition on both sides, and so is criticality, by the remark before the lemma. *q.e.d.*

Theorem 6.20 (The Variational Problem on a Support)**.** *Lemma 6.4, Theorem 6.5, Corollary 6.6, and Theorems 6.16 and 6.17 hold, with the same constants, for a sequence of domains with a support W, when the fields and the data are given on W, the sets $\mathfrak{U}(\varepsilon_0)$ and $\mathfrak{V}(V)$ and the action are replaced by $\mathfrak{U}_W(\varepsilon_0)$, $\mathfrak{V}_W(V)$, and A_W, the gauge transformations are those of the sites of W, the operators of Section 4 are taken with the Dirichlet domain W, and the cubes of Theorem 6.5*(iii) *and of Corollary 6.6 are those for which the neighbourhood named there is contained in W.*

For $W = \mathbb{T}$ these are the statements of the preceding subsections. By Lemma 6.19 the minimiser on a support of data which are regular on the whole torus is the restriction of the minimiser on the torus. The point of the theorem is that nothing is assumed outside W; this is the generality of [7], whose domain Ω_0 is the support.

7 The Small-Field Step

7.1 The Spaces of Pairs

The terms of an effective action are functions of the background field, and the induction needs them on complex neighbourhoods of the real backgrounds, as analytic functions with bounds. The neighbourhoods have to be large enough to contain every configuration at which a later step evaluates a term, and among these are the minimisers of the averages of a configuration: a step replaces the argument $\mathbf{U}$ of an old term by the configuration of least action with the averages of $\mathbf{U}$, on each of the scales below that of the term. The definition of the spaces therefore contains, besides bounds on the pair itself, bounds on these derived configurations [9, Definition, p. 262]. We first define the determining sets inside a domain and the derived configurations.

Definition 7.1 (The Minimal Sequences of a Domain)**.** Let M_c be the smallest power of L with $M_c \geq LR_2M_1$, with the numbers M_1 and R_2 of Theorem 6.5. Let $1 \leq j \leq k$ and let $X \subset \mathbb{T}$ be a union of cubes of the grid of side $M_c a_j$. We define $\Omega_1(X) \supset \cdots \supset \Omega_j(X)$ recursively: $\Omega_1(X)$ is the union of the cubes of the grid of side $M_c a_1$ which lie in X at an ℓ^∞-distance of at least $M_1\eta$ from the complement of X, and, for $1 \leq n < j$, $\Omega_{n+1}(X)$ is the union of the cubes of the grid of side $M_c a_{n+1}$ which lie in $\Omega_n(X)$ at an ℓ^∞-distance greater than $27M_c a_n$ from the complement of $\Omega_n(X)$. For $1 \leq n \leq j$ the *minimal sequence of X of the order n* is the sequence of domains

$$\mathbb{T} \supset \Omega_1(X) \supset \cdots \supset \Omega_n(X) \supset \emptyset \supset \cdots \supset \emptyset\,, \tag{7.1}$$

which has the block size M_c, the separation 27, and the support X; its determining set is denoted by $\mathbf{B}_n(X)$.

These are the determining sets $\mathbf{B}_j(\Omega)$ of [10, p. 256] and the sequences of the condition (iv) of [9, p. 262], with the block size and the separation for which Corollary 6.6 holds. The passage from X to $\Omega_1(X)$ removes a layer of the width $M_c a_1$, and the passage from $\Omega_n(X)$ to $\Omega_{n+1}(X)$ a layer of a width of at most $(27 + L)M_c a_n$, so that $\Omega_j(X)$ contains every point of X at a distance of at least $2(27 + L)M_c a_{j-1}$ from the complement of X.

Definition 7.2 (The Minimiser of the Averages)**.** Let X, j, and $1 \le n \le j$ be as in Definition 7.1. Let U be a gauge field on the bonds of X whose averages on the layers of $\mathbf{B}_n(X)$, the data V_0 with $V_0(c) \coloneqq M^i(U)(c)$ for the bonds c of the layer i, are regular with a constant ε_1 for which Theorem 6.17 holds on the support X, and let U_0 be the minimal configuration for V_0 on X which is in the axial gauge. Let $\mathbf{U}$ be a configuration on the bonds of X such that

$$B(c) \coloneqq \frac{1}{i} \operatorname{Log}(M^i(\mathbf{U})(c)\, V_0(c)^{-1}) \tag{7.2}$$

is defined for the bonds c of the layers and $|B(c)| \le C_{\mathrm{d}}\varepsilon_1$. Then

$$U_{n,X}(\mathbf{U};U) \coloneqq U(B)\,, \qquad J_{n,X}(\mathbf{U};U) \coloneqq J(U(B))\,, \tag{7.3}$$

with the configuration (6.22) of Theorem 6.17*(iii)* at the base U_0, and with the current (1.10).

For a gauge field $\mathbf{U}$ the configuration $U_{n,X}(\mathbf{U};U)$ is the minimal configuration with the averages of $\mathbf{U}$ which is in the axial gauge, by Theorem 6.17(iii), and it does not depend on U. For a complex $\mathbf{U}$ it is the value of the analytic continuation of the minimiser, as a function of the data, from the real data V_0.

Definition 7.3 (The Space of Pairs of a Scale)**.** Let $1 \le j \le k$, let $M \ge M_{\mathrm{c}}$ be the block size of the localisation domains, let $X \in \mathbf{D}_j$, and let $\alpha_0, \alpha_1, \gamma_0 > 0$ and $B_{\mathrm{r}} \ge 1$. The set $\mathfrak{U}^{\mathrm{c}}_j(X;\alpha_0,\alpha_1,\gamma_0)$ consists of the pairs $(\mathbf{U}^u, R(u)\mathbf{J})$, where u is a gauge transformation of the sites of X with values in G^c and $(\mathbf{U},\mathbf{J})$ is a pair on the bonds of X of the form $\mathbf{U} = e^{i\eta A'}U$ which satisfies the following four conditions.

(i) U is a gauge field with $|U(\partial p) - \mathbf{1}| \le \alpha_0(\eta/a_j)^2$ for the plaquettes p of X, and for every cube $\square \subset X$ which is a union of cubes of π_j and has a side $s_\square \le 20LMa_j$ there is a gauge transformation v on $\square$ with values in G such that $U^v = e^{i\eta A}$ on the bonds of $\square$ with

$$a_j|A| \le \frac{s_\square}{a_j}\, B_{\mathrm{r}}\alpha_0\,, \qquad a_j^2|\nabla_{\mathbf{1}} A| \le \frac{s_\square}{a_j}\, B_{\mathrm{r}}\alpha_0\,. \tag{7.4}$$

(ii) A' is a 1-form with values in $\mathfrak{g}^c$, and $a_j|A'| \le \alpha_1$, $a_j^2|\nabla_U A'| \le \alpha_1$ on X.

(iii) $|\mathbf{U}(\partial p) - \mathbf{1}| \le \alpha_0(\eta/a_j)^2$ for the plaquettes p of X, and $a_j^3|\mathbf{J}(b)| \le \gamma_0$ for the bonds b of X.

(iv) For $1 \le n \le j$ the configurations $U_{n,X}(\mathbf{U};U)$ and $U_{n,X}(U;U)$ of Definition 7.2 are defined, and each of them, written $\mathbf{U}_n$, satisfies

$$|\mathbf{U}_n(\partial p) - \mathbf{1}| \le \alpha_0\Big(\frac{\eta}{a_j}\Big)^2\,, \qquad a_n^3|J(\mathbf{U}_n)(b)| \le \alpha_0\Big(\frac{a_n}{a_j}\Big)^2 \tag{7.5}$$

for the plaquettes p and the bonds b of $X^{\sim -2}$, where the operation $^\sim$ is that of π_j.

We write $\mathfrak{U}^{\mathrm{c}}_j(X;\alpha_0,\alpha_1) \coloneqq \mathfrak{U}^{\mathrm{c}}_j(X;\alpha_0,\alpha_1,\alpha_0)$.

This is the Definition of [9, p. 262] in the units of the lattice $T^{(k)}$: the lattice of spacing $\xi = L^{-j}$ there is T_η in the units of the scale j, so that $\xi = \eta/a_j$, a field A' and its derivative there are a_jA' and $a_j^2\nabla A'$ here, and a current there is $a_j^3\mathbf{J}$ here; the current of the condition (iv) is taken there in the units of the scale n. The constant B_{r} is the constant B of [9, (1.12)], which is fixed with the constants of the construction. By definition the set is invariant under the gauge transformations with values in G^c, acting on pairs by [9, (1.10)], and it increases

with each of α_0, α_1, γ_0 when the conditions on the regularity of the data in Definition 7.2 are kept fixed.

The condition (iv) is not explicit, and the spaces are used through simpler sets which they contain [9, p. 263].

Proposition 7.4 (A Criterion for Membership)**.** *There are numbers $c_F \in (0,1]$ and $a_6 > 0$, which depend on L, G, M, and B_{r} only, with the following property. Let $\alpha_0, \alpha_1 \le a_6$, $X \in \mathbf{D}_j$, and let $(\mathbf{U}, \mathbf{J})$ be a pair on the bonds of X which satisfies the conditions* (i) *and* (iii) *of Definition 7.3 with $c_F\alpha_0$ in the place of α_0, the condition* (ii)*, and $a_j^3|\mathbf{J}| \le \gamma_0$. Then $(\mathbf{U}, \mathbf{J}) \in \mathfrak{U}_j^{\mathrm{c}}(X; \alpha_0, \alpha_1, \gamma_0)$. In particular, if U is a gauge field on X which satisfies* (i) *with $c_F\alpha_0$ and $a_j^3|J(U)| \le \alpha_0$, then $(U, J(U)) \in \mathfrak{U}_j^{\mathrm{c}}(X; \alpha_0, \alpha_1)$.*

In [9, (1.17), p. 263] this is derived in one line from the regularity of the minimisers, applied to the averages of the complex configuration, and it is said to be obvious that the condition (iv) follows. The statement is used with the full radius α_1 of the complex part, and only the plaquettes are required to be smaller.

A term which is localised near a region of large fields is a function of a background whose determining set has several layers in the domain of the term, and its radii of analyticity depend on the layer. For such terms the spaces are defined with respect to the sequence of domains of the background [10, (2.34)–(2.39), p. 261]. The derived configurations are now the minimisers of determining sets which follow the given sequence up to a scale and are minimal near the boundary of the domain.

Definition 7.5 (The Localised Sequences)**.** Let a sequence of domains $\{\Omega_n\}$ with a block size $M \ge M_{\mathrm{c}}$ and a separation $R \ge 27$ be given, let $1 \le p \le j \le k$, and let $X \in \mathbf{D}_j$. The *sequence localised to X, of the order p*, is

$$\mathbb{T} \supset \Omega_1 \cap \Omega_1(X) \supset \cdots \supset \Omega_p \cap \Omega_p(X) \supset \emptyset \supset \cdots \supset \emptyset \,, \tag{7.6}$$

with the domains $\Omega_n(X)$ of Definition 7.1. It is a sequence of domains with the block size M_{c}, the separation 27, and the support X, and its determining set is denoted by $\mathbf{B}_p(X; \{\Omega_n\})$. For a gauge field U and a configuration $\mathbf{U}$ on the bonds of X, the configuration $U_{p,X}(\mathbf{U}; U)$ and its current are defined as in Definition 7.2, with this determining set in the place of $\mathbf{B}_n(X)$.

The separation of (7.6) is that of the two sequences which are intersected: a point of $\Omega_{n+1} \cap \Omega_{n+1}(X)$ is at a distance greater than $27M_{\mathrm{c}}a_n$ from the complement of Ω_n and from that of $\Omega_n(X)$. The determining set is $\mathbf{B}_p(X) \cup \{\Gamma_i\}_{i<p}$ in the notation of [10, p. 261]: in the interior of X a point of the layer $i < p$ of the given sequence remains in the layer i, the points of the layers $i \ge p$ are in the layer p, and near the boundary of X the layers are those of the minimal sequence.

Definition 7.6 (The Space of Pairs of Several Scales)**.** Let the sequence of domains, j, and X be as in Definition 7.5, let $\tilde{\alpha}_0 = (\alpha_{0,n})_{0 \le n \le j}$ and $\tilde{\alpha}_1 = (\alpha_{1,n})_{0 \le n \le j}$ be sequences of positive numbers, let $0 < \beta < 1$ and $B_{\mathrm{r}} \ge 1$, and put

$$c_{j,n} \coloneqq 1 - \beta(1 - 2^{-(j-n)}) \,, \qquad X_n \coloneqq X \cap (\Omega_n \setminus \Omega_{n+1}) \quad (n < j) \,, \qquad X_j \coloneqq X \cap \Omega_j \,. \tag{7.7}$$

The set $\tilde{\mathfrak{U}}_j^{\mathrm{c}}(X; \tilde{\alpha}_0, \tilde{\alpha}_1)$ consists of the pairs $(\mathbf{U}, \mathbf{J})$ on the bonds of X of the form $\mathbf{U} = e^{i\eta A'}U$, with a gauge field U and a 1-form A' with values in $\mathfrak{g}^c$, which satisfy the following conditions for $0 \le n \le j$.

(i) On the plaquettes and the bonds of X_n,

$$|U(\partial p)-\mathbf{1}|,\ |\mathbf{U}(\partial p)-\mathbf{1}|\le c_{j,n}\alpha_{0,n}\Big(\frac{\eta}{a_n}\Big)^2\,,\qquad a_n^3|\mathbf{J}(b)|\le c_{j,n}\alpha_{0,n}\,. \tag{7.8}$$

(ii) For $1\le p\le j$ the configurations $U_{p,X}(\mathbf{U};U)$ and $U_{p,X}(U;U)$ of Definition 7.5 are defined, and each of them, written $\mathbf{U}_p$, satisfies on the plaquettes and the bonds of X_n

$$|\mathbf{U}_p(\partial p)-\mathbf{1}|\le c_{j,n}\alpha_{0,n}\Big(\frac{\eta}{a_n}\Big)^2\,,\qquad a_p^3|J(\mathbf{U}_p)(b)|\le c_{j,n}\alpha_{0,n}\Big(\frac{a_p}{a_n}\Big)^e\,, \tag{7.9}$$

with $e=3$ for $n\le p$ and $e=2$ for $n>p$.

(iii) For every cube $\square$ which, for $n<j$, is a cube of the class of the sequence of the index n (Definition 4.3(i)), and, for $n=j$, is a union of cubes of π_j contained in Ω_j with a side $10Ma_j\le s_\square\le 20Ma_j$, there is a gauge transformation v on $\square\cap X$ with values in G such that $U^v=e^{i\eta A}$ on the bonds of $\square\cap X$ with $a_n|A|\le(s_\square/a_n)B_{\mathrm{r}}\alpha_{0,n}$ and $a_n^2|\nabla_{\mathbf{1}}A|\le(s_\square/a_n)B_{\mathrm{r}}\alpha_{0,n}$.

(iv) $a_n|A'|\le c_{j,n}\alpha_{1,n}$ and $a_n^2|\nabla_U A'|\le c_{j,n}\alpha_{1,n}$ on X_n.

These are the conditions [10, (2.34)–(2.39), p. 261]; the exponents in (7.9) are those of [10, (2.37)], where the current of $\mathbf{U}_p$ is taken in the units of the scale p, and for $n>p$ they agree with (7.5). The conditions are stated there for the layers $n\ge 1$. We include $n=0$, the part of X outside Ω_1, because the derived configurations of (ii) are defined only if the data of U are regular on the whole support X. The set is invariant under the gauge transformations with values in G, and not under those with values in G^c. The factor $c_{j,n}$ decreases from 1 at $n=j$ to more than $1-\beta$: a term is analytic in a domain which shrinks slightly on the layers below its own scale, and this is the room in which the operations of a step move the configurations at which they evaluate the older terms.

7.2 The Terms of an Effective Action

An effective action is a sum of local terms, each a function of the background in a localisation domain, and what the induction carries from step to step is a short list of properties of these functions. We state the list for the terms which a step creates in the region of small fields [9, (1.7)–(1.19), pp. 261–263], in the form with a base point which [10, (2.26), (2.27), p. 259] introduces. Since the sets of Definition 7.3 are closed, we call a function on such a set *holomorphic* if it is the restriction of a holomorphic function on an open subset of the complex manifold of the pairs on the bonds of X, that is, of $(G^c)^{\mathsf{b}}\times(\mathfrak{g}^c)^{\mathsf{b}}$ with b the set of the positive bonds of X, which contains the set.

Definition 7.7 (Regular Terms)**.** Let $1\le j\le k$, and let α_0, α_1, E_0, and κ be positive numbers. A *family of regular terms of the scale j* with these constants is a family of functions $\mathbf{E}(X,\cdot\,,z)$, one for each $X\in\mathbf{D}_j$ and each point $z\in T^{(j)}\cap X$, with the following properties.

(i) *Locality and analyticity.* $\mathbf{E}(X,\cdot\,,z)$ is a holomorphic function on $\mathfrak{U}_j^c(X;\alpha_0,\alpha_1)$, a set of pairs on the bonds of X.

(ii) *Gauge invariance.* $\mathbf{E}(X,(\mathbf{U}^u,R(u)\mathbf{J}),z)=\mathbf{E}(X,(\mathbf{U},\mathbf{J}),z)$ for all gauge transformations u of the sites of X with values in G^c.

(iii) *The bound.* $|\mathbf{E}(X,(\mathbf{U},\mathbf{J}),z)|\le E_0e^{-\kappa d_j(X)}$ on $\mathfrak{U}_j^c(X;\alpha_0,\alpha_1)$.

(iv) *Covariance.* Let r be an isometry of the torus which maps $T^{(j)}$ onto itself, let U be a gauge field on T_η, and let rU be the gauge field with $(rU)(rb) \coloneqq U(b)$ for the oriented bonds b of T_η. If, for U and for rU in the place of U, the pair $(U|_X, J(U)|_X)$ lies in $\mathfrak{U}_j^c(X;\alpha_0,\alpha_1)$ for every $X \in \mathbf{D}_j$, then

$$\mathbf{E}(rU, rz) = \mathbf{E}(U, z)\ , \qquad \mathbf{E}(U, z) \coloneqq \sum_{X\in\mathbf{D}_j,\ X\ni z} \mathbf{E}(X, (U|_X, J(U)|_X), z)\ . \tag{7.10}$$

For a gauge field U, a domain $\Lambda \in \mathbf{D}_j$, and a point $z \in T^{(j)} \cap \Lambda$ we write

$$\mathbf{E}(\Lambda, U, z) \coloneqq \sum_{X\in\mathbf{D}_j,\ z\in X\subset\Lambda} \mathbf{E}(X, (U|_X, J(U)|_X), z)\ . \tag{7.11}$$

The current in (7.10) and (7.11) is the current of U on the torus, restricted to the bonds of X; at a bond near the boundary of X it depends on U outside X, and the pair is a point of the space, at which the term is evaluated. The sums are finite, and by (iii) and Lemma 1.18 they are bounded uniformly in the torus for κ large. The isometries of (iv) are generated by the translations by the vectors of $a_j\mathbb{Z}^4$, the permutations of the coordinates, and the reflections; they map T_η onto itself, but not the grid π_j, and for this reason the covariance is a property of the sum over all domains and not of its terms [10, (2.29), p. 260].

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